

Exhibit 96
Part 1 of 2

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart

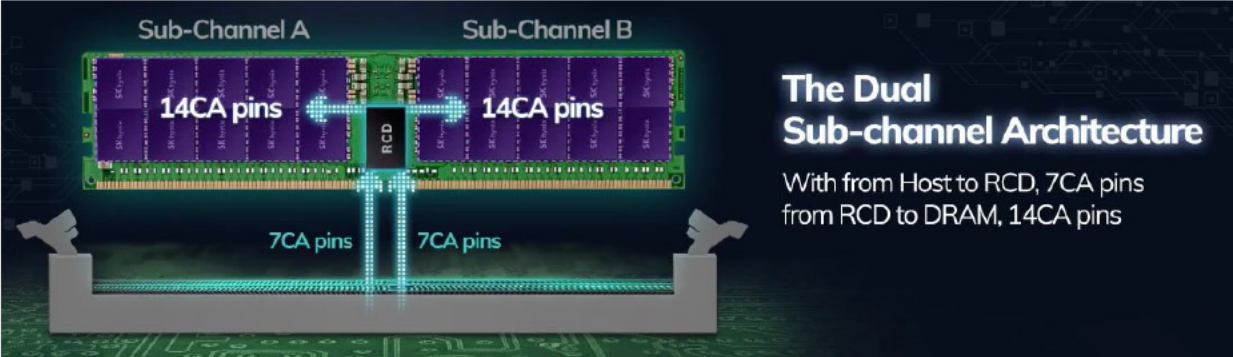
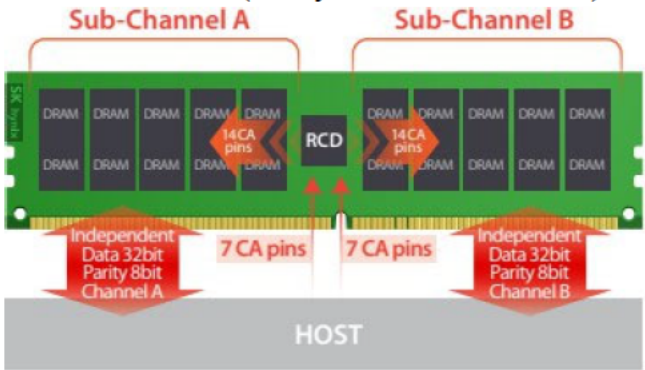
EXHIBIT 96 (Part 1)

Domestic Industry Claim Chart for Claims of U.S. Patent No. 12,373,366 (the “’366 Patent”) Based on Practicing ’366 Products –
Part I of II

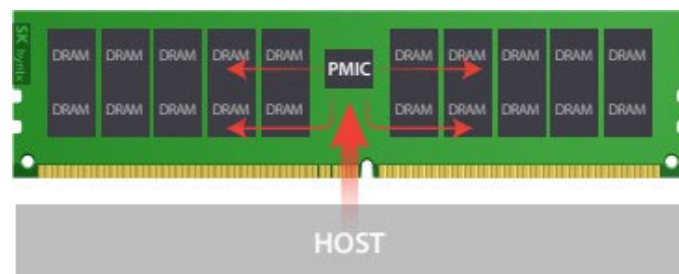
As shown in the chart below, the Practicing ’366 Products, which include Netlist DDR5 VLP RDIMMs, Netlist Lightning DDR5 RDIMMs, Netlist DDR5 LP MRDIMMs, [REDACTED], and SK hynix RDIMMs and MRDIMMs, practice at least claim 1 of the ’366 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Practicing ’366 Products do not practice the ’366 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Practicing ’366 Products, such differences are insubstantial and the Practicing ’366 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Practicing ’366 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Practicing ’366 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

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Claim 1	Evidence of Use
[1.pre] A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:	Each of the Practicing '366 Products, including Netlist's DDR5 Very Low Profile Registered DIMMs ("VLP RDIMMs"), Netlist's OverClock DDR5 Registered DIMMs and Ultra Low Latency DDR5 Registered DIMMs (collectively, "Lightning RDIMMs"), SK hynix's Registered DIMMs ("RDIMMs"), and SK hynix's MRDIMMs, is a dual in-line memory module (DIMM). Each of the Practicing '366 Products is configured to fit into a memory slot connector of a system board of a computer system. For example, SK hynix's web site shows that SK hynix DDR5 RDIMMs and DDR5 MRDIMMs are configured to fit into a memory slot corresponding to a host computer. <i>See, e.g.,</i> Ex. CC- 41 (RDIMM SK hynix) at 4:  The Dual Sub-channel Architecture With from Host to RCD, 7CA pins from RCD to DRAM, 14CA pins <i>See also</i> Ex. CC-40 (SK hynix A New Standard) at 5–6:  New Sub-channel DRAM Architecture The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

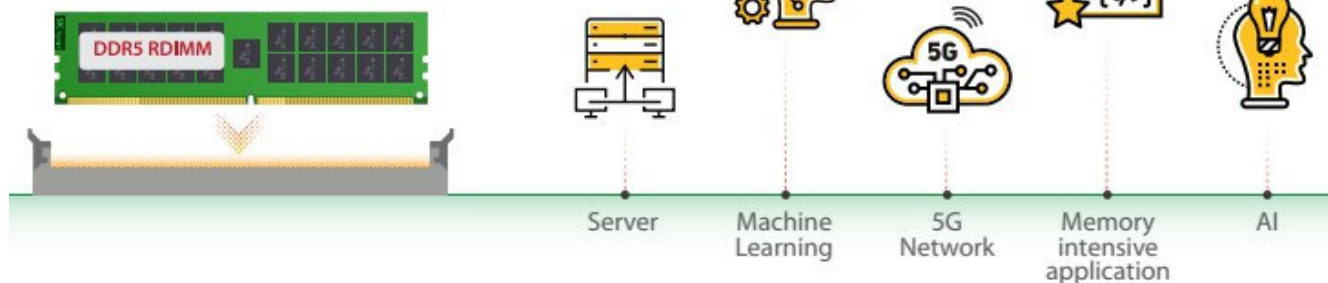
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| Stable Power Management

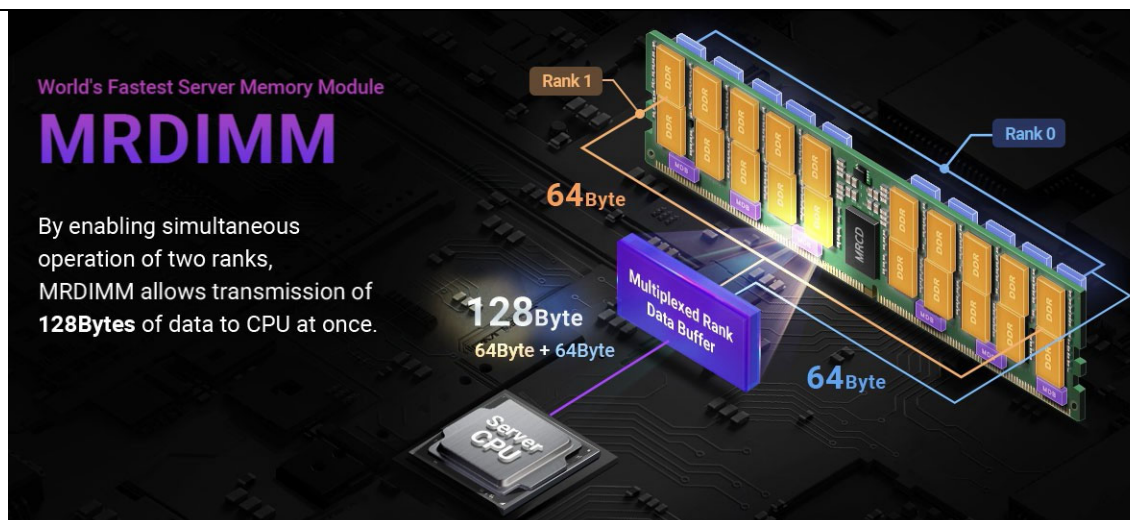
On-board power management IC (PMIC) eliminates the need for external voltage regulation and ensures more stable power provision to the DRAM.

Use Cases



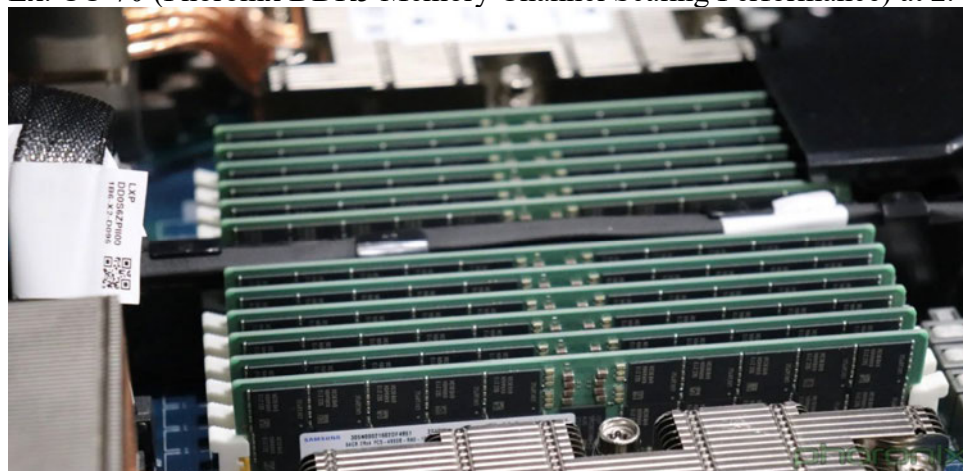
1. See also Ex. CC-42 (SK hynix MRDIMM) at 4:

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Furthermore, each of the Practicing '366 Products is configured to fit into a memory slot connector of a system board of a computer system.

Ex. CC-70 (Phoronix DDR5 Memory Channel Scaling Performance) at 2:



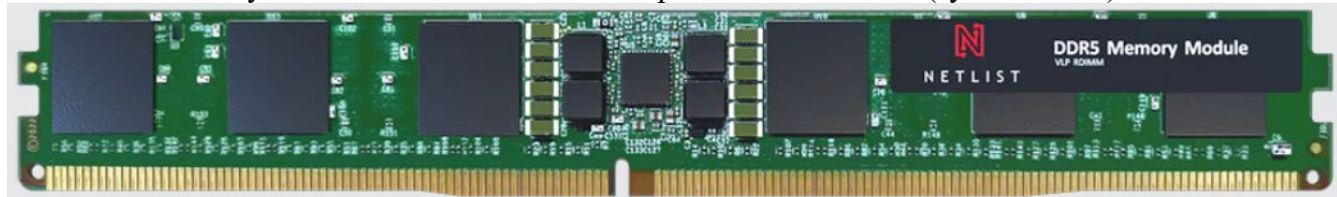
For example, Netlist's web site advertises various dual in-line memory modules for various applications, including Client PC, WS/Server, and HPC workload applications. Ex. CC-33 (Netlist

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DDR5 DIMMs). For instance, Netlist advertises RDIMMs, including ones with ECC capability, and MRDIMMs as suitable for server applications. *Id.* Additionally, Netlist's product brief advertises Netlist Lightning RDIMMs as configured for Servers and Workstations. *See* Ex. CC-56 (Netlist Ultra Low Latency DDR5 RDIMM Product Brief) at 1 (stating that Netlist DDR5 Ultra Low Latency RDIMMs are used for Server and Workstation applications); Ex. CC-57 (Lightning DDR5 High Performance Module Product Brief) at 1 (stating that Netlist DDR5 Overclock RDIMMs are used for Server and Workstation applications).

See also, e.g., infra at (1.a.4).

Each of the Practicing '366 Products includes edge connections with gold fingers that are configured to fit into a memory slot connector of a server computer's motherboard (system board).



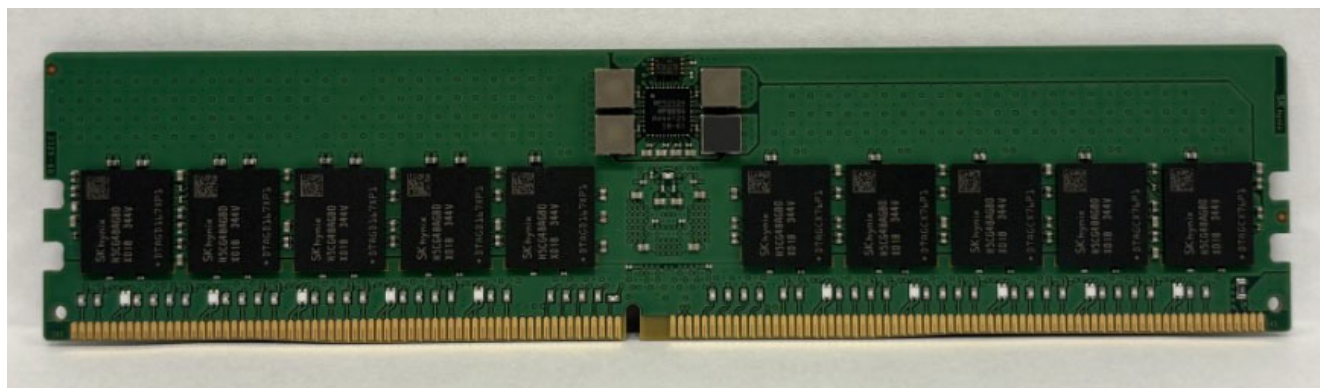
VLP RDIMM

Ex. CC-33 (Netlist DDR5 DIMMs) at 2.

See also images of Netlist RDIMMs (e.g., SK hynix 32GB DDR5-5600 RDIMM PC5-44800R Dual

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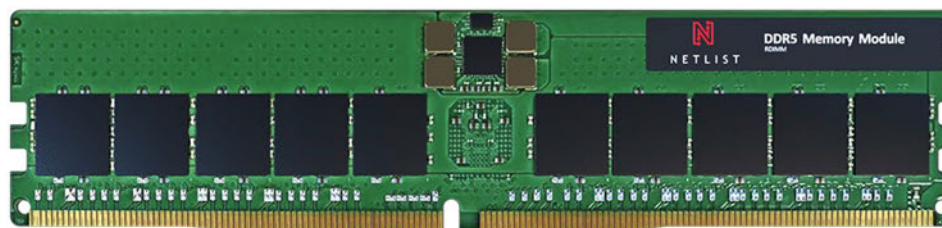
Rank x8 Module):



Ex. CC-36 (Product Images) at 10.



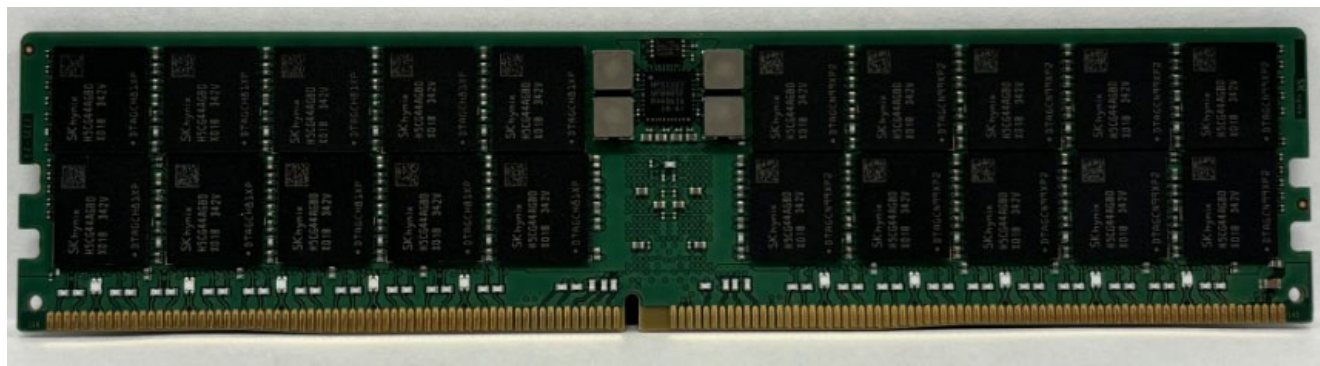
Ex. CC-36 (Product Images) at 7



Ex. CC-33 (Netlist DDR5 DIMMs) at 2.

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See also images of Netlist Overclock RDIMMs (Netlist 64GB 2Rx4 PC5-6000OC6-R):



Ex. CC-36 (Product Images) at 3.

See also images of Netlist and SK hynix MRDIMMs (e.g., SK hynix 32GB DDR5-8800 MCR DIMM Dual Rank x8 Module):

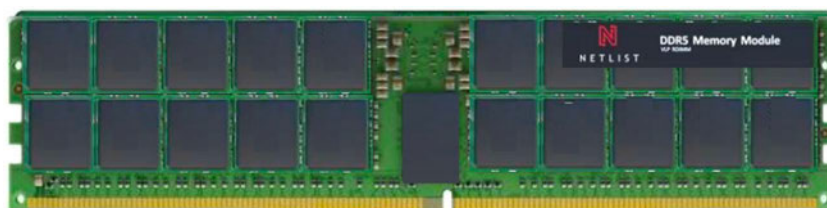


Ex. CC-36 (Product Images) at 16.

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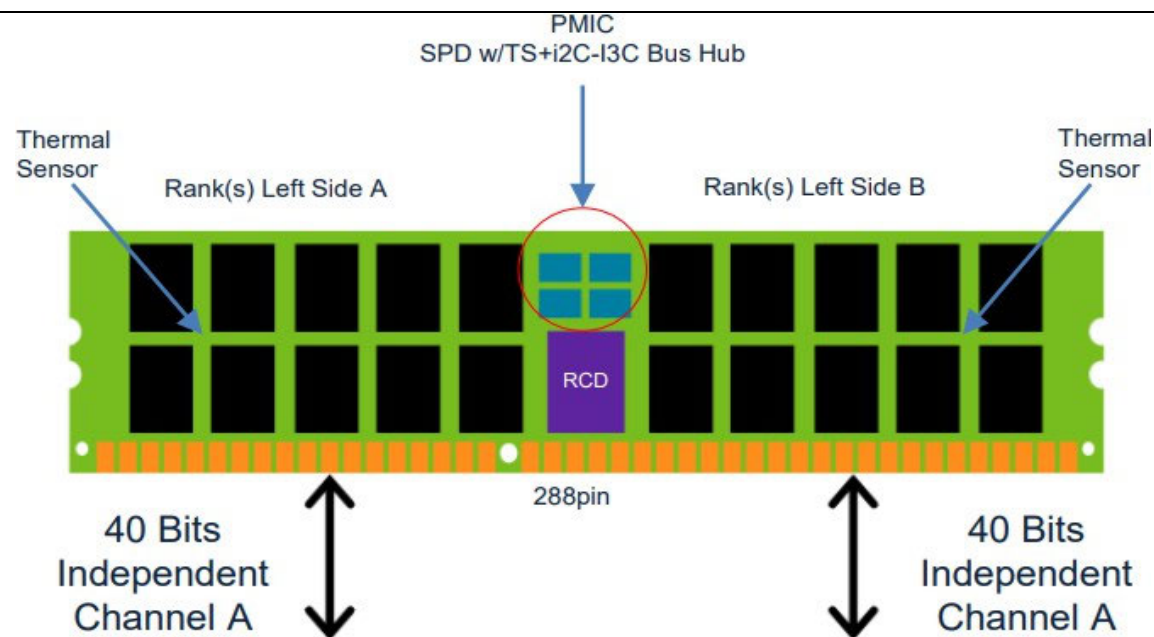
Ex. CC-36 (Product Images) at 13.



Ex. CC-33 (Netlist DDR5 DIMMs) at 2.

See also Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1:

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See also Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 2 (annotated) (advertising the connector and physical layout for Netlist DDR5 MRDIMMs are the same as Netlist DDR5 RDIMMs, therefore Netlist DDR5 RDIMMs' connectors and physical layouts are representative of Netlist DDR5 MRDIMMs):

DDR5 MRDIMM Features

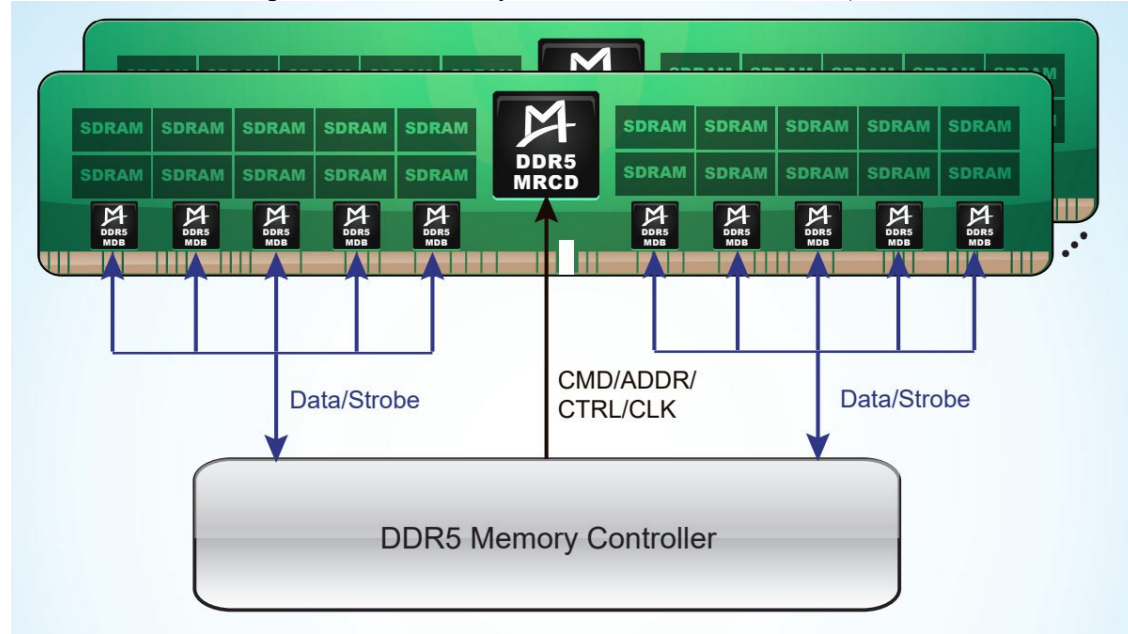
Features	Description
Bandwidth	Up to 8.8K MT/s, providing a 39% bandwidth upgrade
Latency	Up to 40% lower latency compared to traditional DDR5
Capacity	Available in sizes from 32GB to 128GB
Parallel DRAM Activation	Enables pairs of DRAM ranks to be activated and accessed in parallel
Multiplexed Data Streams	Doubles the data rate per signal pin
Power Management	Includes PMIC 5030 for efficient power management
Compatibility	Optimized for Intel Xeon 6 processors. They maintain the same connector and physical layout as RDIMMs, allowing easy integration into existing systems without modifications
Operating Voltage	1.1 volts
Operating Temperature	0°C to +70°C

See also Ex. CC-33 (Netlist DDR5 DIMMs) at 6 (advertising that Netlist's DDR5 MRDIMMs

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support “Backward Compatibility: Compatible with existing DDR5 RDIMMs, providing flexibility in system configurations”).

See also Ex. CC-12 (M88MR5RCD01 Flyer English) at 2 (showing data, control, and address signals sent between example DDR5 memory controller and MRDIMM):

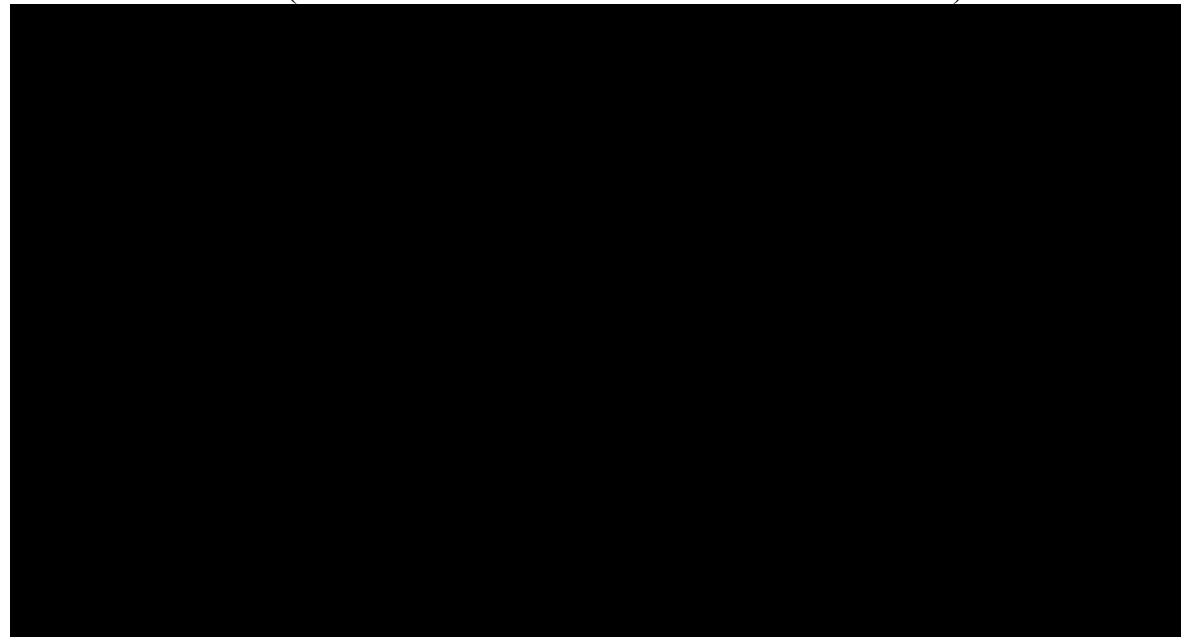


See also Ex. CC-36 (Product Images) at 14 (showing, on Netlist DDR5 MRDIMM, part number M88MR5RCD01, which corresponds to Montage DDR5 MRCD, as referenced above):

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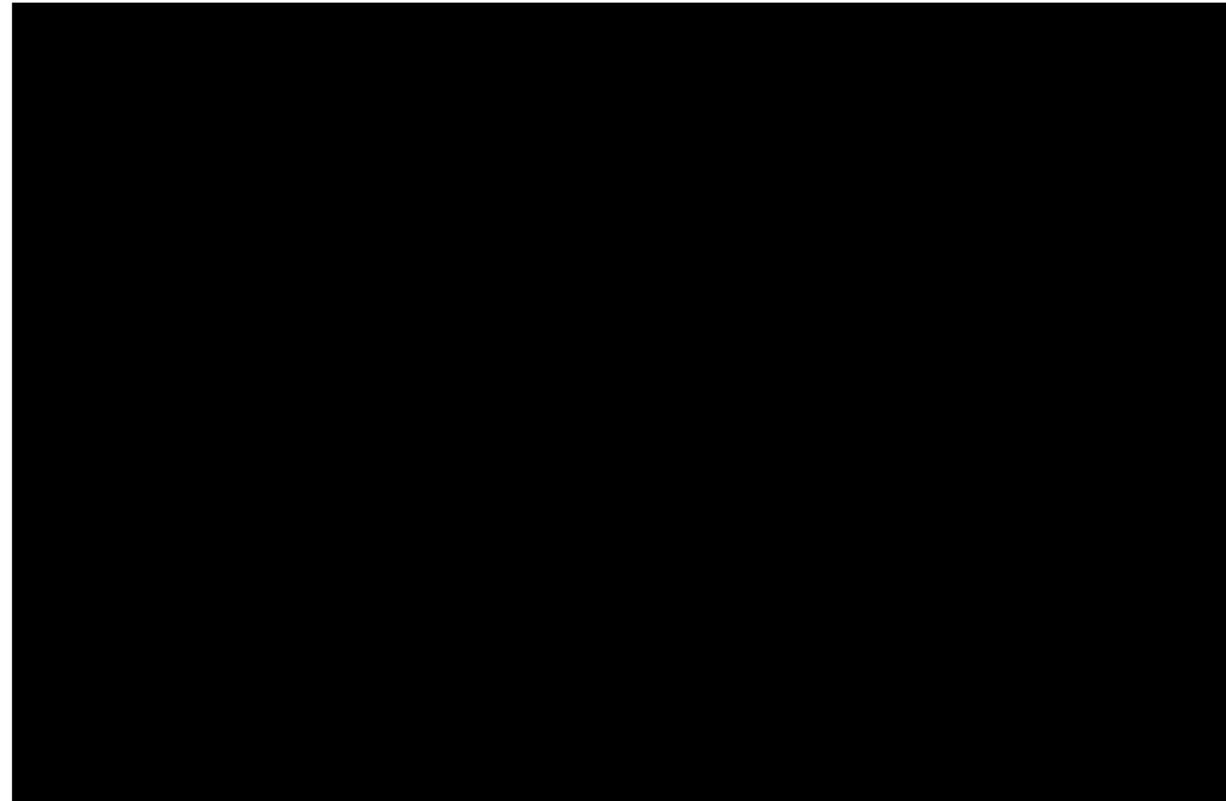


See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 45:



See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 25:

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The memory slot connector on the host system's motherboard includes a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits.

[Redacted text block]

Ex. CC-38C (Netlist DDR5 VLP RDIMM

Datasheet Version 1.1) at 17, 19, 22, 24:



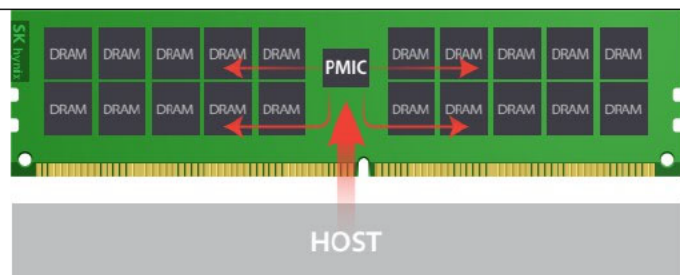
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	 <i>See also</i> Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10 [Redacted text block]
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**| Stable Power Management**

On-board power management IC (PMIC) eliminates the need for external voltage regulation and ensures more stable power provision to the DRAM.

See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 13

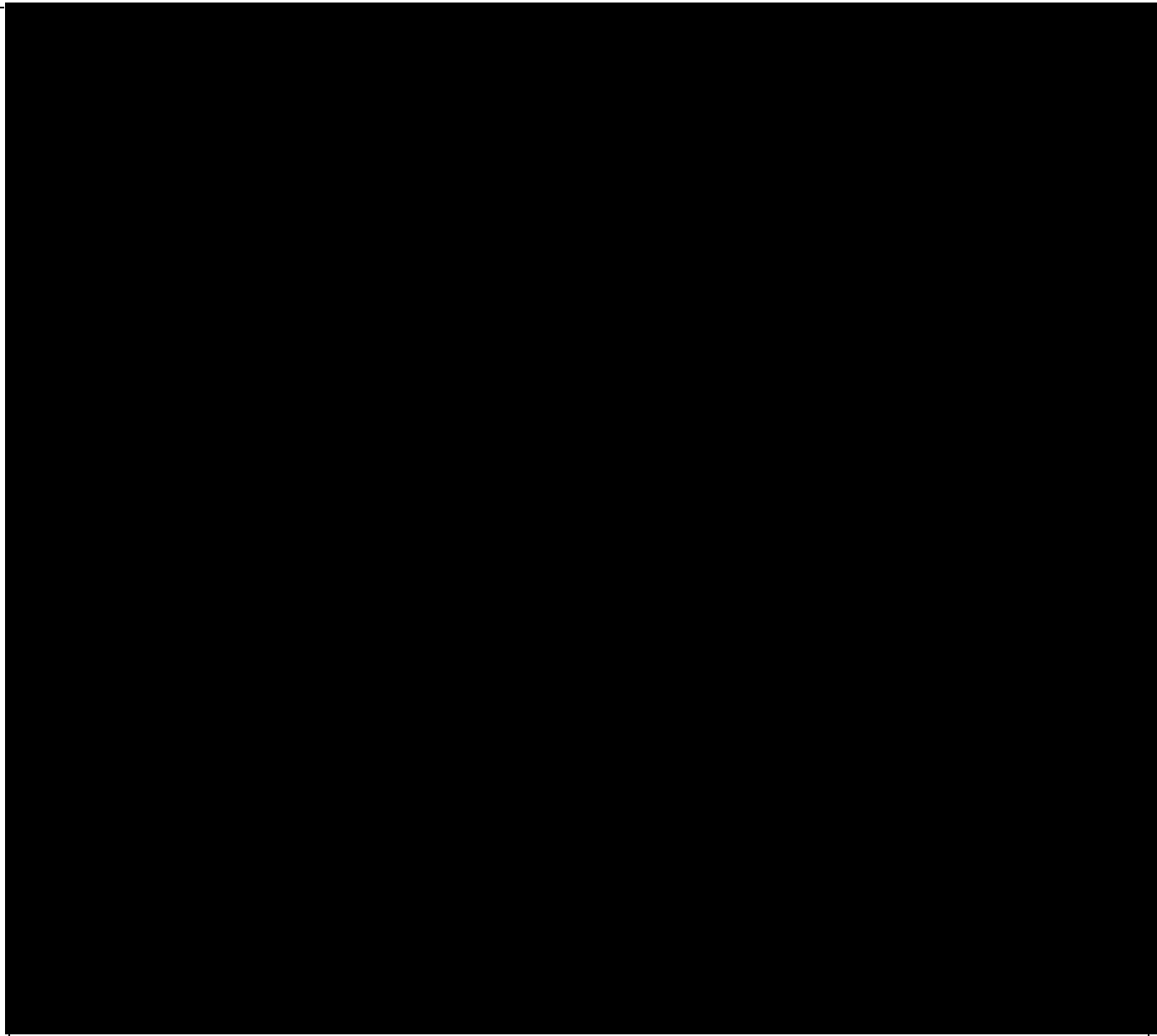
See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7–8 (annotated)



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See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–6 (annotated)

[REDACTED]

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	[REDACTED] [REDACTED] [REDACTED] : [REDACTED]
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Additionally, each of the Practicing '366 Products is "JEDEC [c]ompliant," (Ex. CC-33 (Netlist DDR5 DIMMs) at 2), or at least includes a JEDEC compliant edge connector, and is therefore configured to fit into a JEDEC-compliant memory slot, as illustrated below. *See, e.g.,* Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 1

; Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1

Ex. CC-37 (Lightning DDR5 DIMM - Netlist Inc.) at 2 (advertising, on Netlist's web site, that Netlist DDR5 Lightning RDIMMs are JEDEC compliant and "utiliz[e] JEDEC-standard PCB, DRAM, RCD, PMIC, and clock components"); Ex. CC-56 (Netlist Ultra Low Latency DDR5 RDIMM Product Brief) at 1–2 (advertising Netlist DDR5

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Ultra Low Latency RDIMMs as JEDEC DDR5 RDIMMs and describing various JEDEC compliant configurations); Ex. CC-57 (Lightning DDR5 High Performance Module Product Brief) at 1–2 (advertising various JEDEC compliant configurations of Netlist DDR5 Overclock RDIMMs); Ex. CC-72 (SK hynix DDR5) (stating, on SK hynix’s web site, that SK hynix DDR5 DRAMs, included in each of the Practicing ’366 Products, are JEDEC compliant).

Each of the Practicing ’366 Products includes SK hynix DDR5 DRAMs. *See, e.g.*, images of Netlist DDR5 RDIMM, Overclock RDIMM, MRDIMM:



Ex. CC-36 (Product Images) at 9 (showing SK hynix DRAM on Netlist DDR5 RDIMM).



Ex. CC-36 (Product Images) at 6 (showing SK hynix DRAM on Netlist DDR5 Overclock RDIMM).

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Ex. CC-36 (Product Images) at 13 (showing SK hynix DRAM on Netlist DDR5 MRDIMM).

See also Ex. CC-53C (0887-99A-1M6060 Prelim 02-08-22) at 3

Ex. CC-54C (0888-97A-1A6B61-X00) at 2

While the specific Netlist Overclock RDIMM referenced throughout in Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0)

Furthermore, XMP profiles, such as the Netlist DDR5 Overclock RDIMM described in the datasheet, are built on top of JEDEC-compliant RDIMMs, and therefore include at least mandatory JEDEC functionality as well as additional features.

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Figure 1 DDR5 socket and module (SMT)

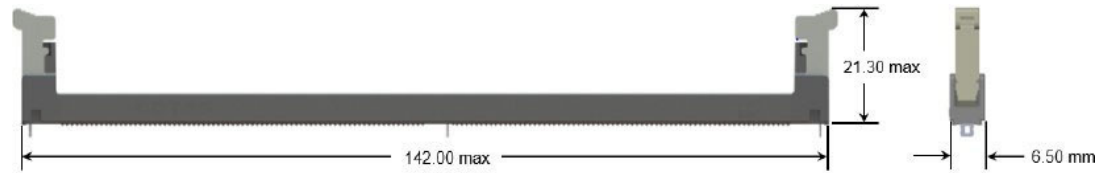
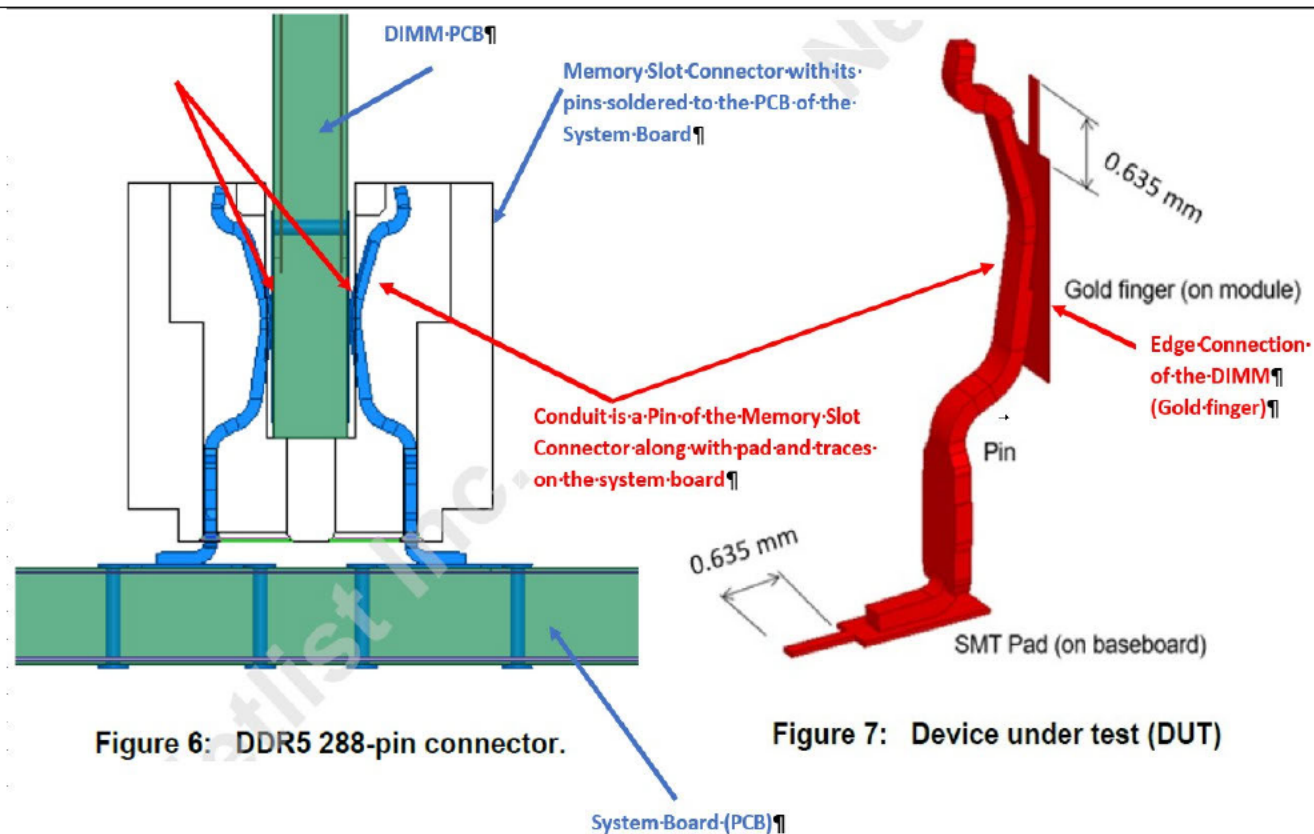


Figure 2 DDR5 U/R/LR DIMM Surface Mount (SMT) Connector Socket Outline

Ex. CC-73 (JEDEC PS-005B) at 6.

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Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on the DIMM being electrically connected with a corresponding pin on the memory slot connector).

The memory slot connector of the system board comprises conduits as illustrated above that correspond to Netlist DDR5 RDIMM pin assignments, such as



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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 5–6
See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 7–9

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4 (annotated). <i>See also</i> Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4 (annotated):
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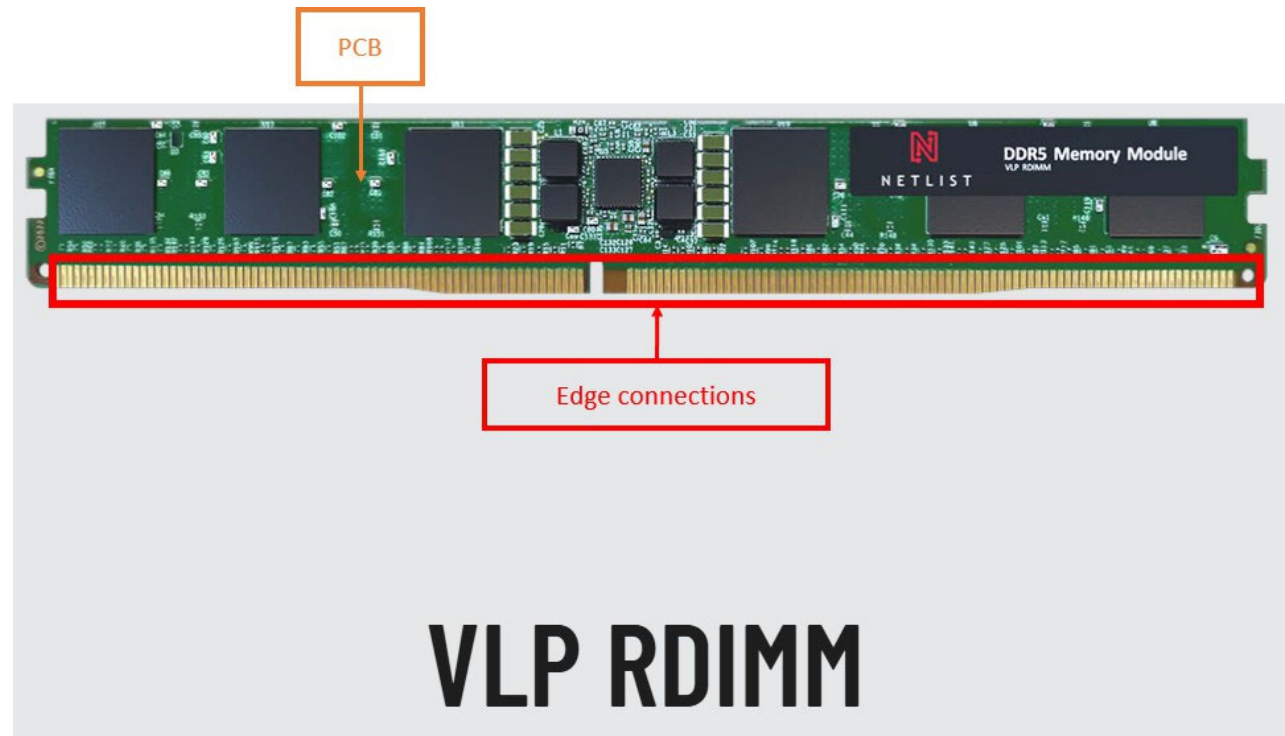
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[1.a] a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:

Each of the Practicing '366 Products comprises a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board.

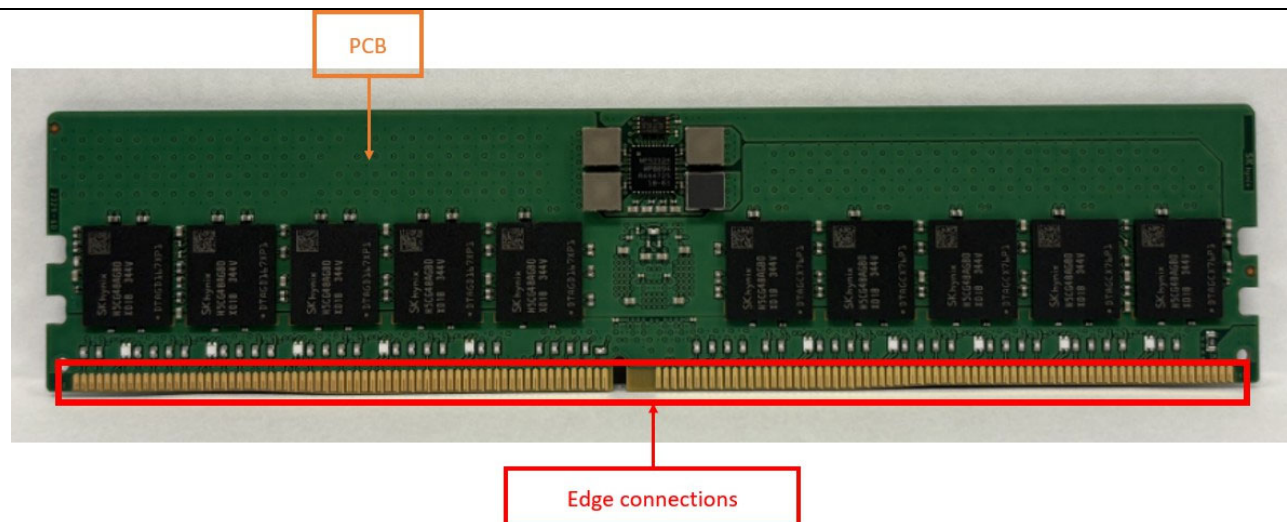
For example, each of the Practicing '366 Products comprises a printed circuit board (PCB) that comprises edge connections, as shown below:



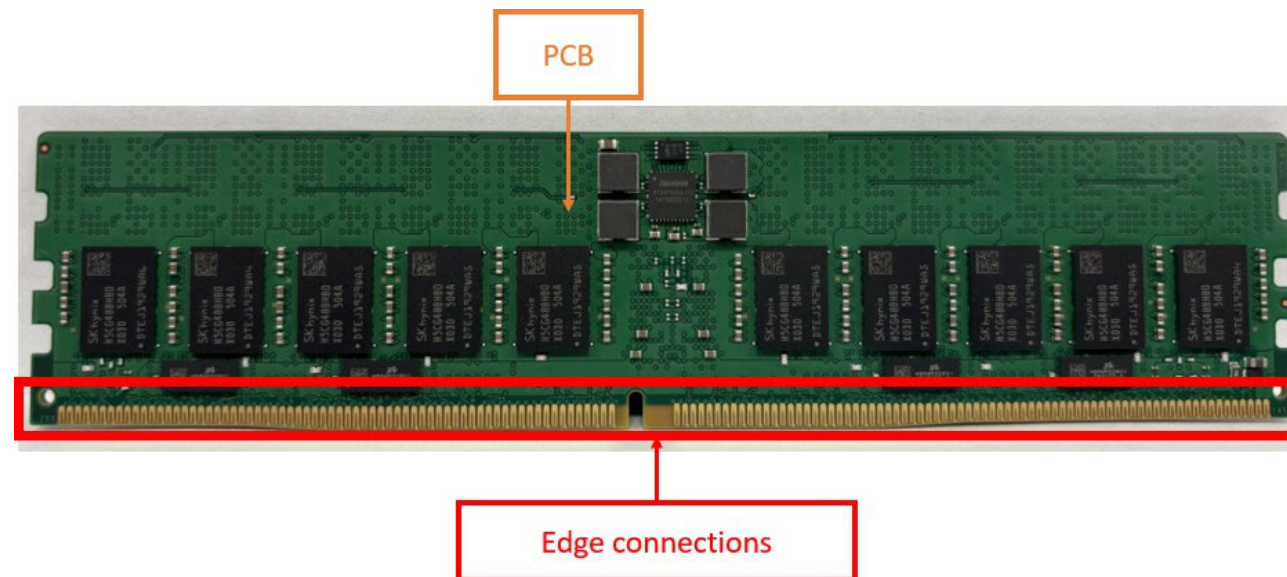
Ex. CC-33 (Netlist DDR5 DIMMs) at 2 (annotated).

See also images of Netlist DDR5 RDIMM, DDR5 Overclock RDIMM, and DDR5 MRDIMM:

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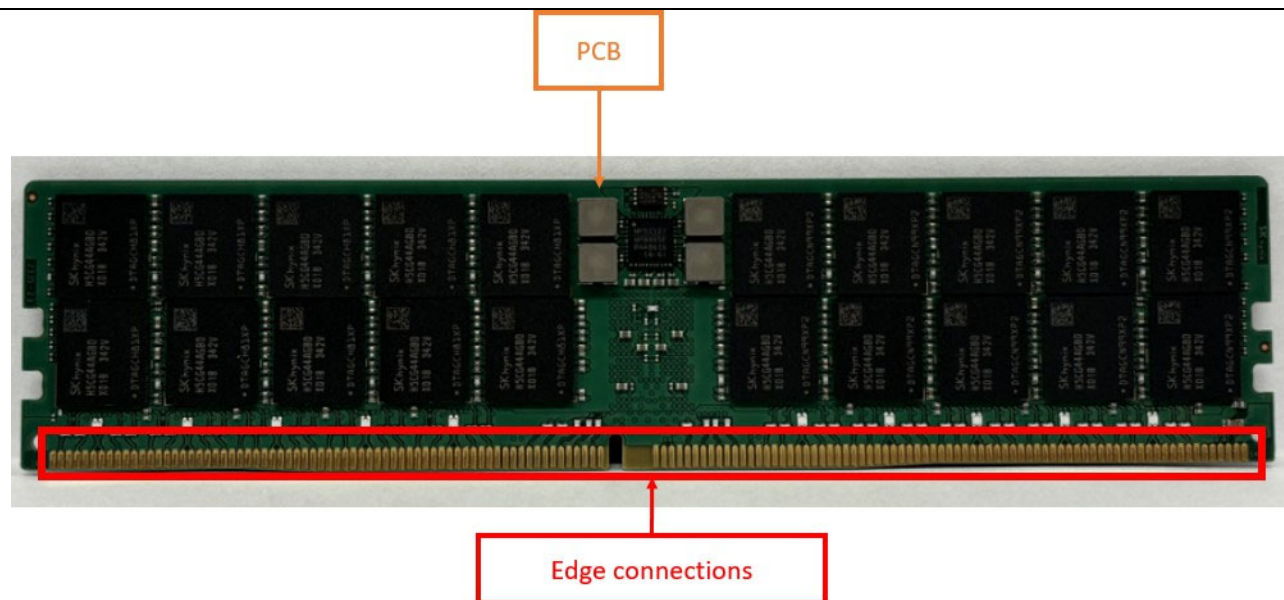


Ex. CC-36 (Product Images) at 10 (annotated).



Ex. CC-36 (Product Images) at 16 (annotated).

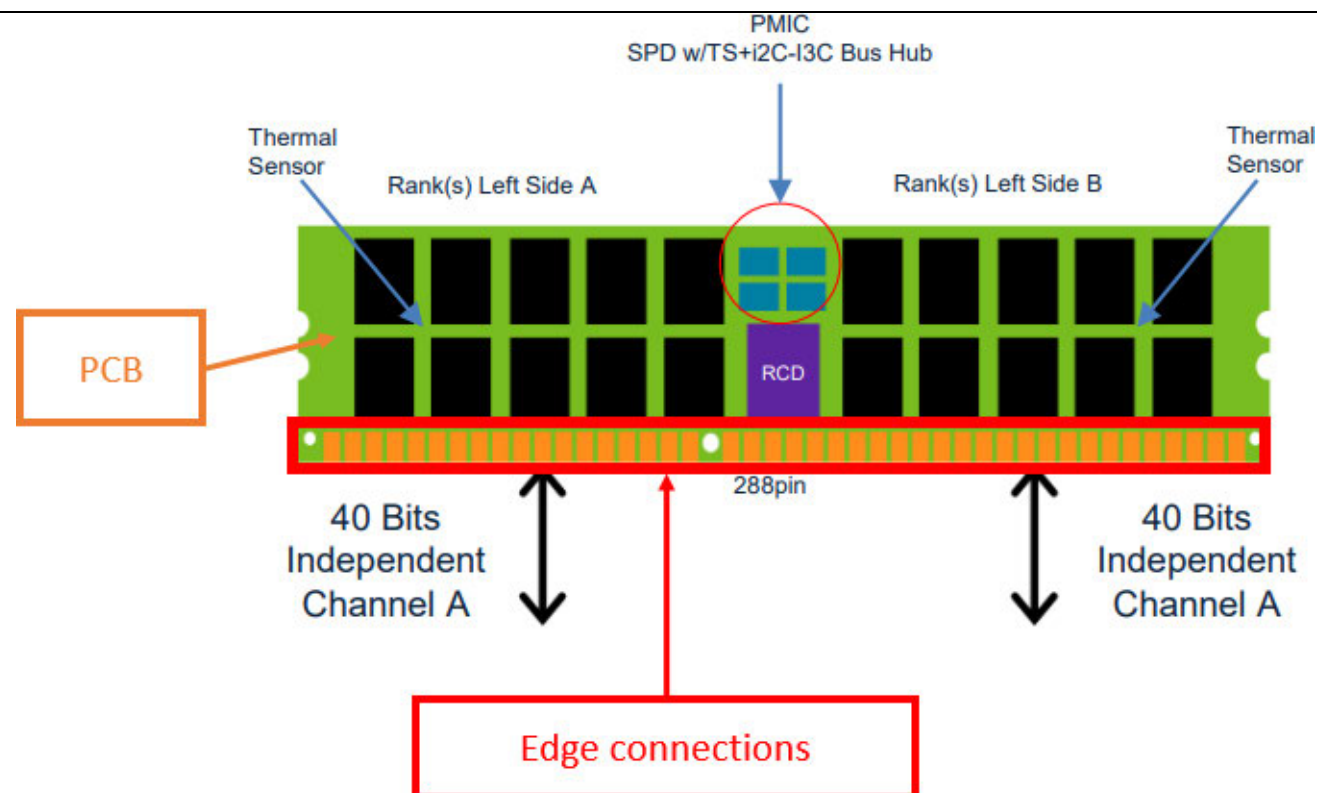
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Ex. CC-36 (Product Images) at 3 (Netlist DDR5 Overclock RDIMM) (annotated).

See also Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1 (annotated) (showing a representative Netlist DIMM, including a PCB and edge connections):

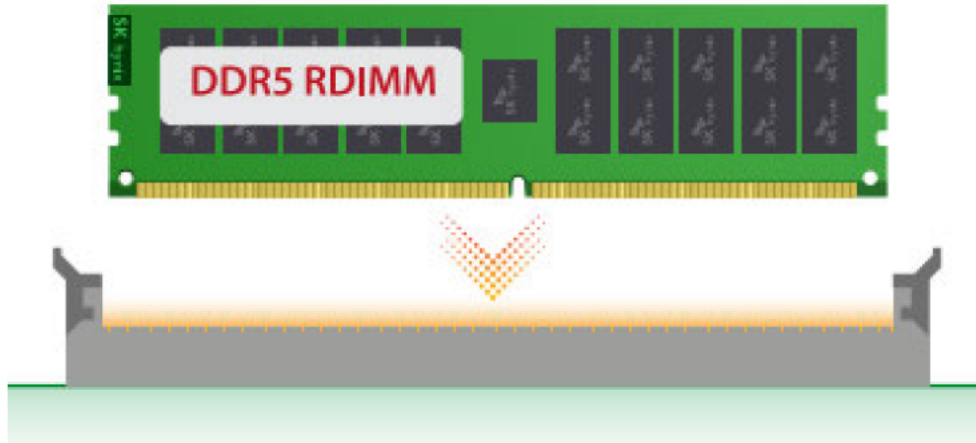
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As explained in 1[pre], the edge connections of the PCB of each of the Practicing '366 Products are configured to fit into the memory slot connector of the system board, for example, as illustrated below where a Netlist DDR5 DIMM was inserted into a corresponding memory slot.

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Use Cases



Ex. CC-40 (SK hynix A New Standard) at 6.

See also 1[pre].

Additionally, as explained in 1[pre], each of the Practicing '366 Products is “JEDEC [c]ompliant,” (Ex. CC-33 (Netlist DDR5 DIMMs) at 2), or at least is backwards compatible with JEDEC compliant RDIMMs (e.g., Netlist’s DDR5 MRDIMMs), and is therefore configured to fit into a JEDEC-compliant memory slot, as illustrated below. *See also* Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 1 [REDACTED]

[REDACTED] Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1 [REDACTED]

[REDACTED] Ex. CC-37 (Lightning DDR5 DIMM - Netlist Inc.) at 2 (advertising, on Netlist’s web site, that Netlist DDR5 Lightning RDIMMs are JEDEC compliant and “utiliz[e] JEDEC-standard PCB, DRAM, RCD, PMIC, and clock components”); Ex. CC-56 (Netlist Ultra Low Latency DDR5 RDIMM Product Brief) at 1–2 (advertising Netlist DDR5 Ultra Low Latency RDIMMs as JEDEC DDR5 RDIMMs and describing various JEDEC compliant configurations); Ex. CC-57 (Lightning DDR5 High

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Performance Module Product Brief) at 1–2 (advertising various JEDEC compliant configurations of Netlist DDR5 Overclock RDIMMs); Ex. CC-72 (SK hynix DDR5) (stating, on SK hynix’s web site, that SK hynix DDR5 DRAMs, included in each of the Practicing ’366 Products, are JEDEC compliant).

See also 1[pre].



Figure 1 DDR5 socket and module (SMT)

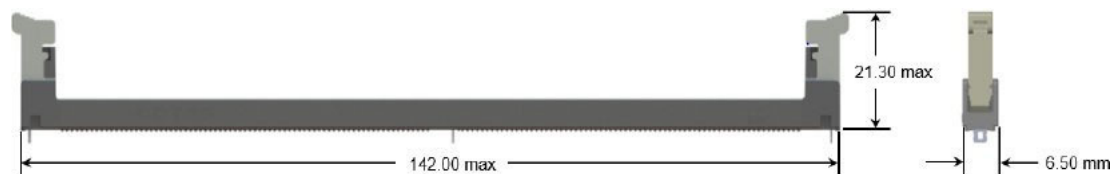


Figure 2 DDR5 U/R/LR DIMM Surface Mount (SMT) Connector Socket Outline

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Ex. CC-73 (JEDEC PS-005B) at 6.

See also *id.* at 13, Fig. 7 (illustrating an edge connection on the DIMM being electrically connected with a corresponding pin on the memory slot connector):

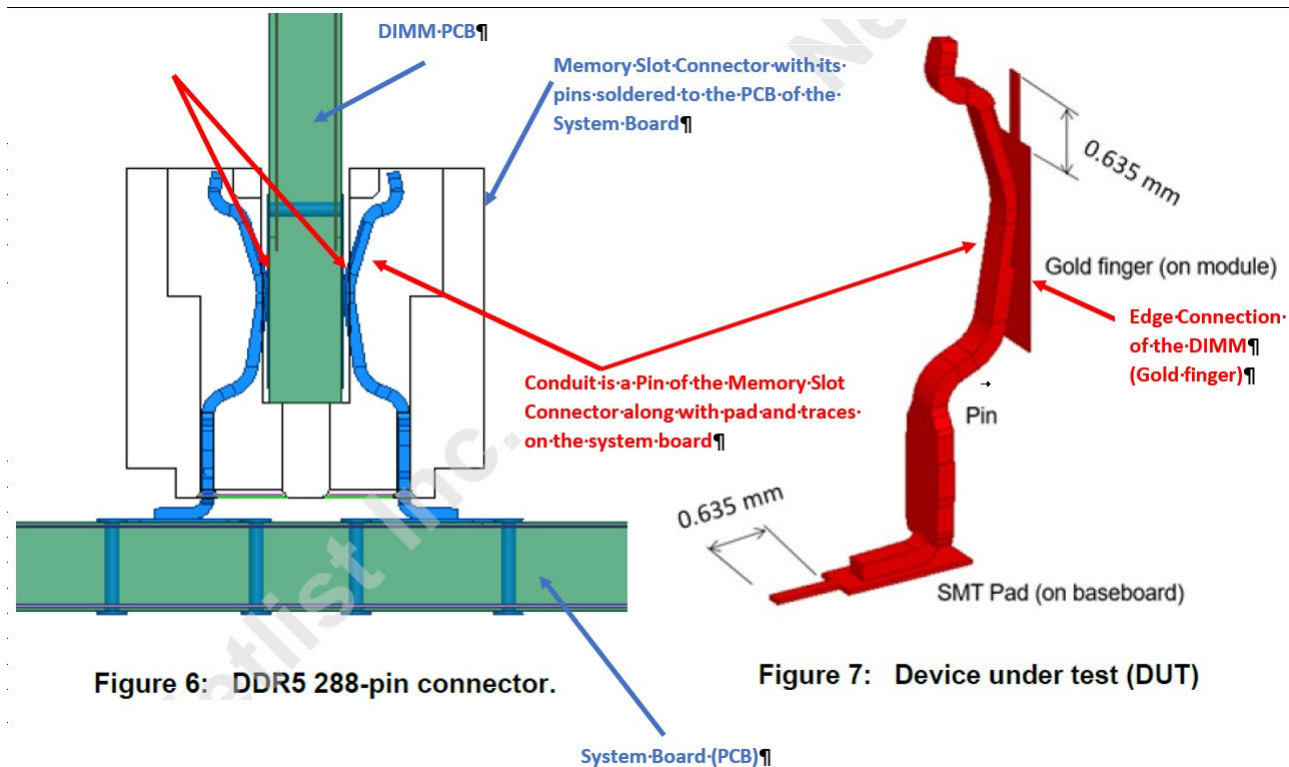


Figure 6: DDR5 288-pin connector.

Figure 7: Device under test (DUT)

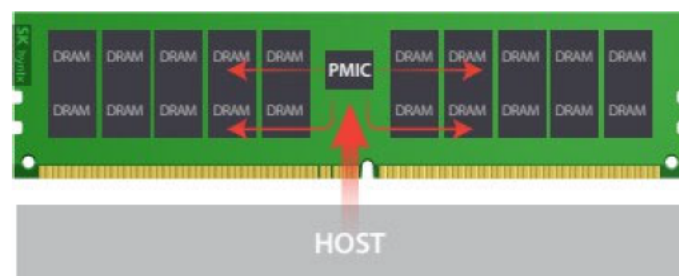
Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated).

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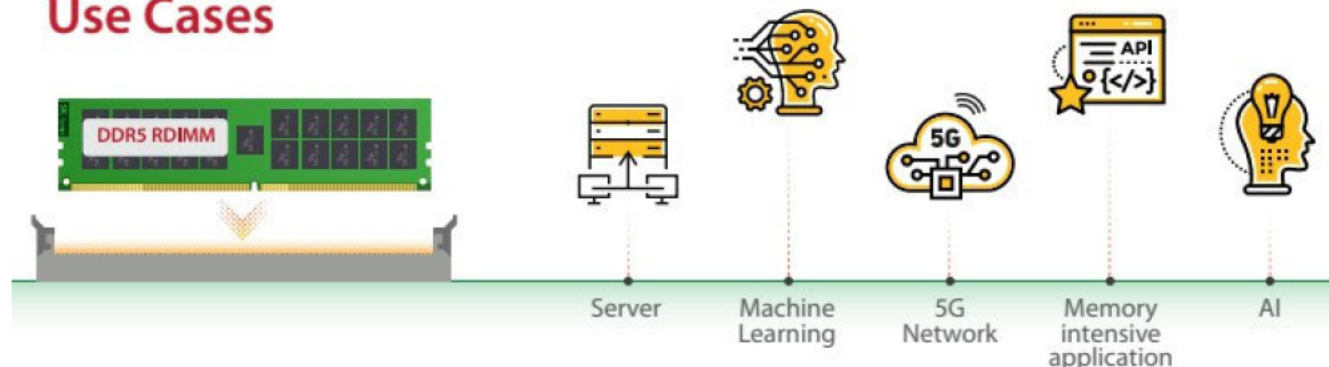
[1.a.1] a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

The printed circuit board of each of the Practicing '366 Products includes a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits. Additionally, the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM.

For example, SK hynix's website shows that SK hynix DDR5 RDIMMs are configured to receive power via power rails from a slot of a host computer, and this is the only power received by the RDIMM. *See, e.g., Ex. CC-40 (SK hynix A New Standard) at 5-6:*



Use Cases



As yet another example, as both Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet) and Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) show, [REDACTED]



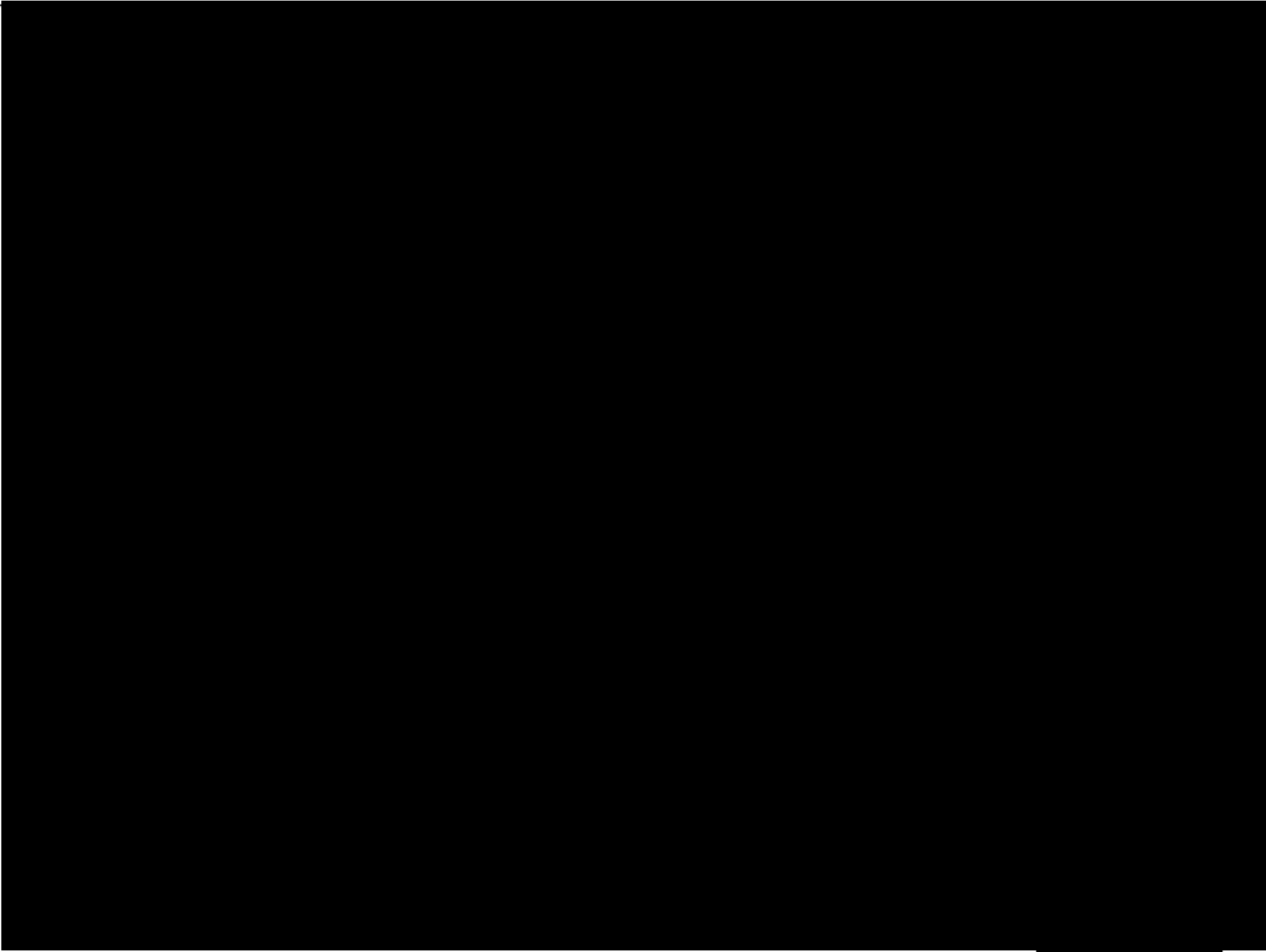
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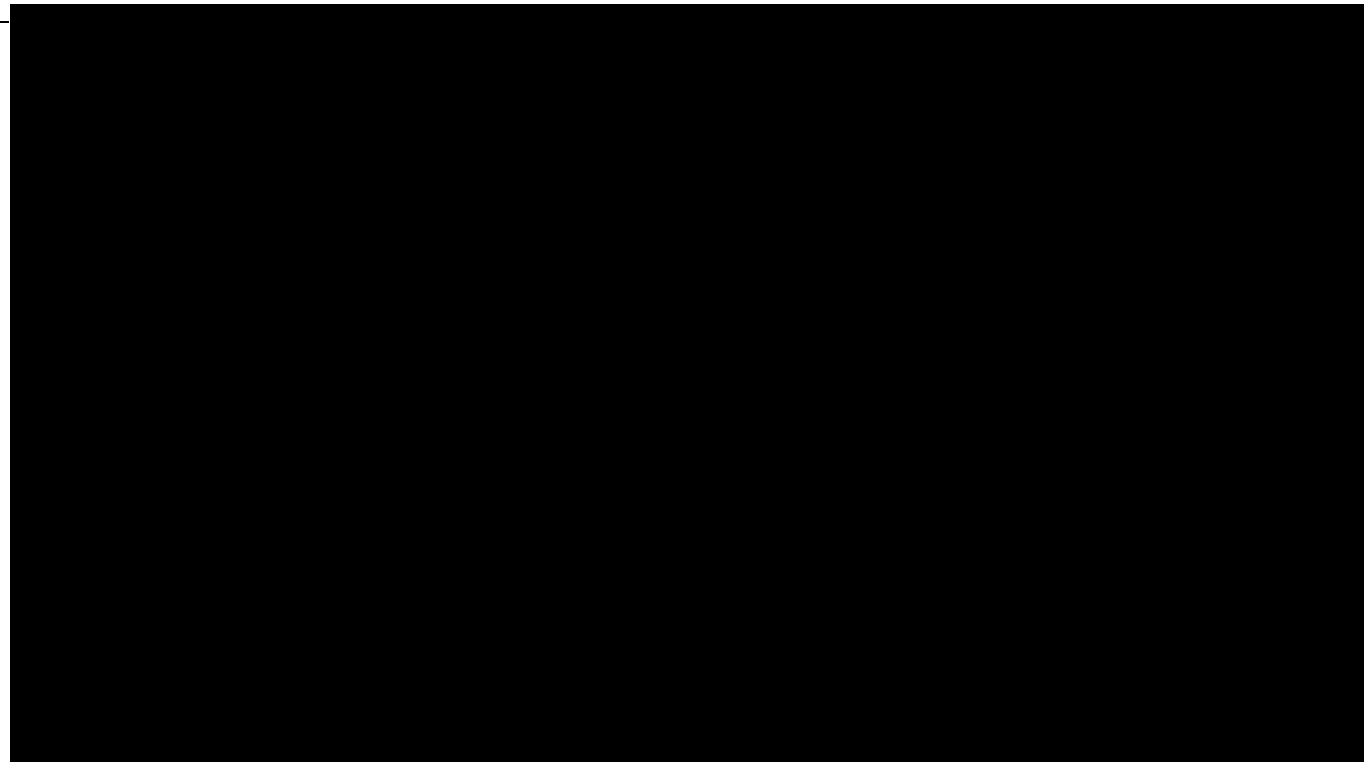
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 5–6 (annotated).

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	 Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4 (annotated) ().
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 8 (annotated) ([REDACTED]).

See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–9 (annotated).



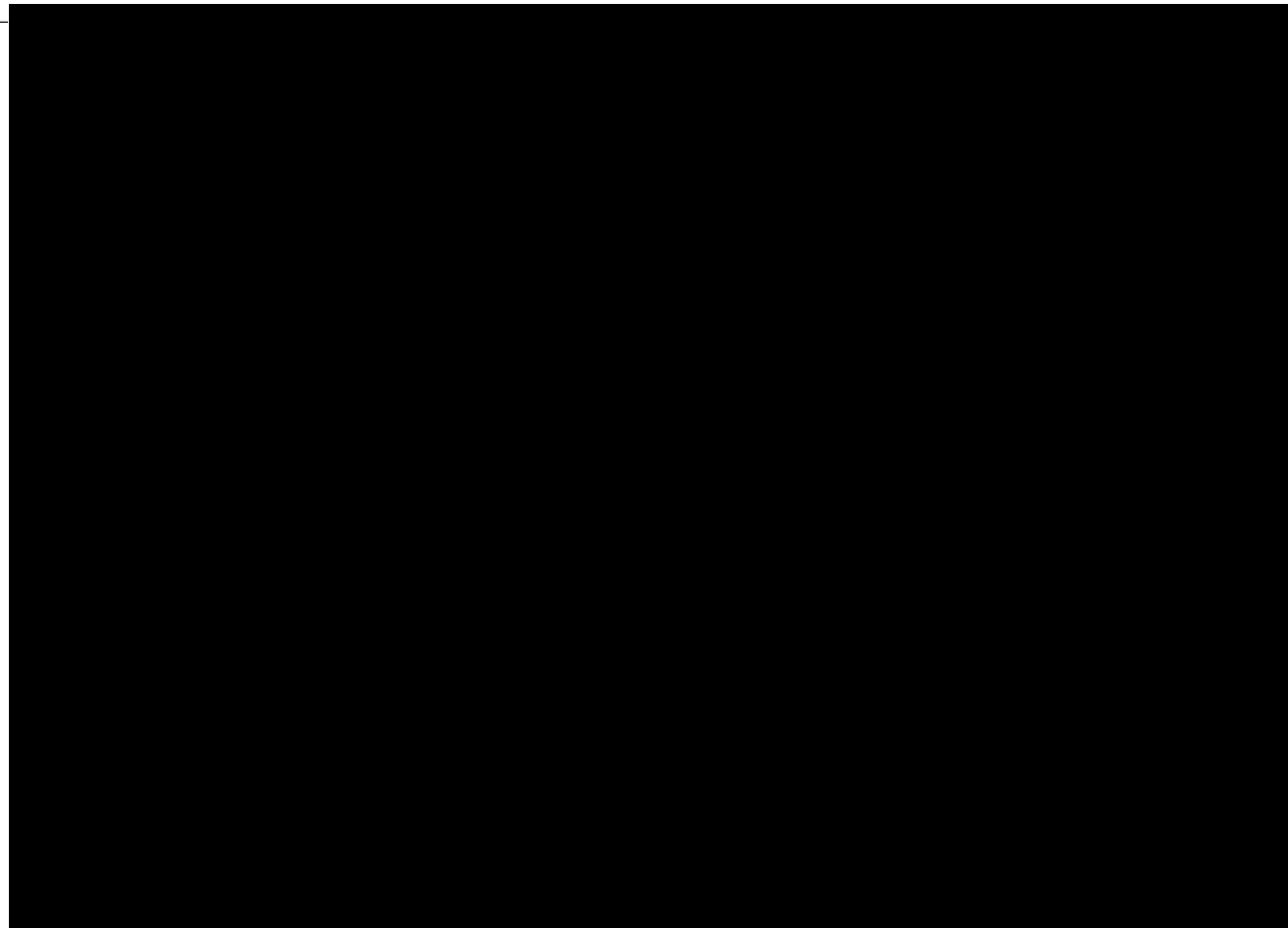
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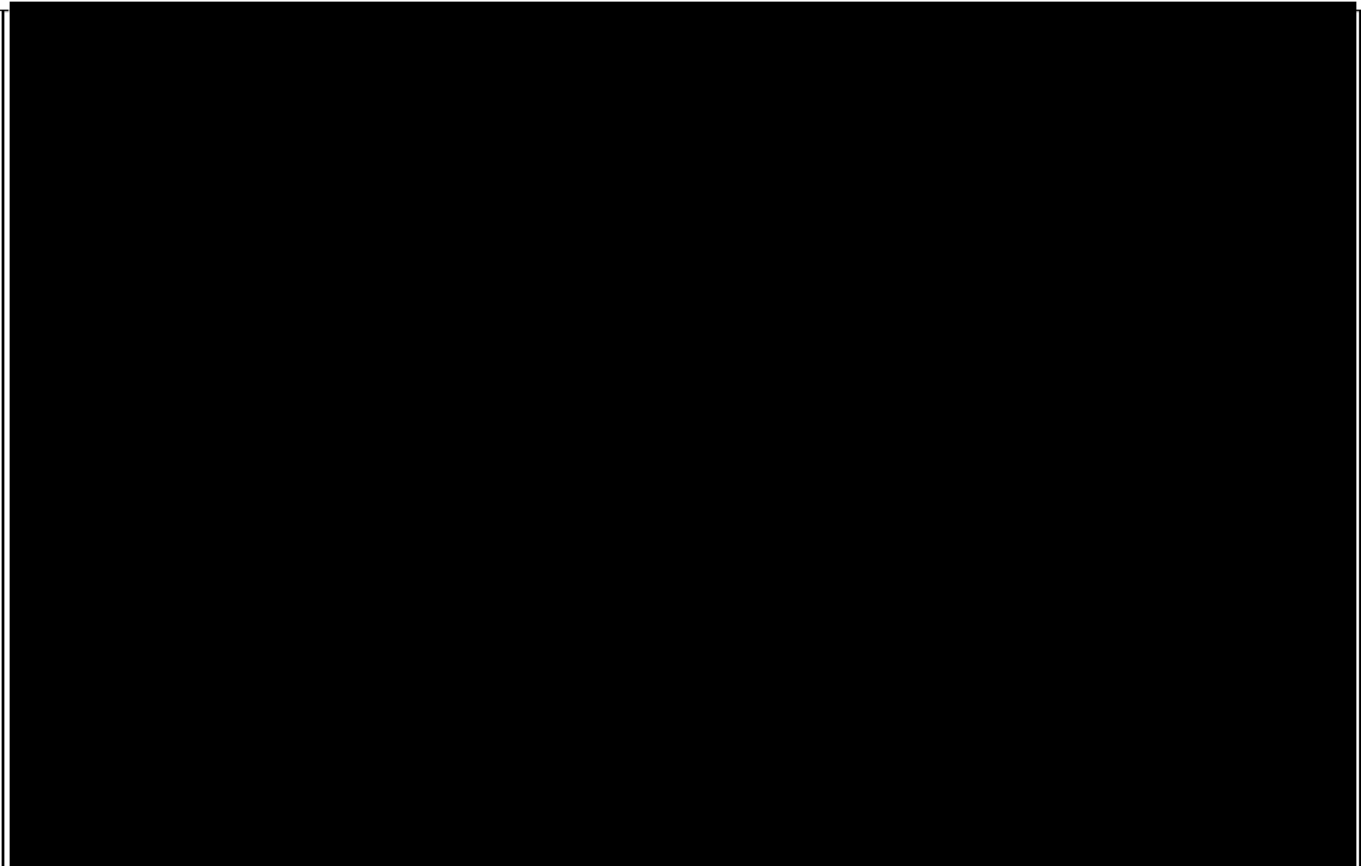
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U.S. Patent No. 12,373,366 Domestic Industry Claim Chart



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4 (annotated) (

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 6 (annotated) ([REDACTED]).

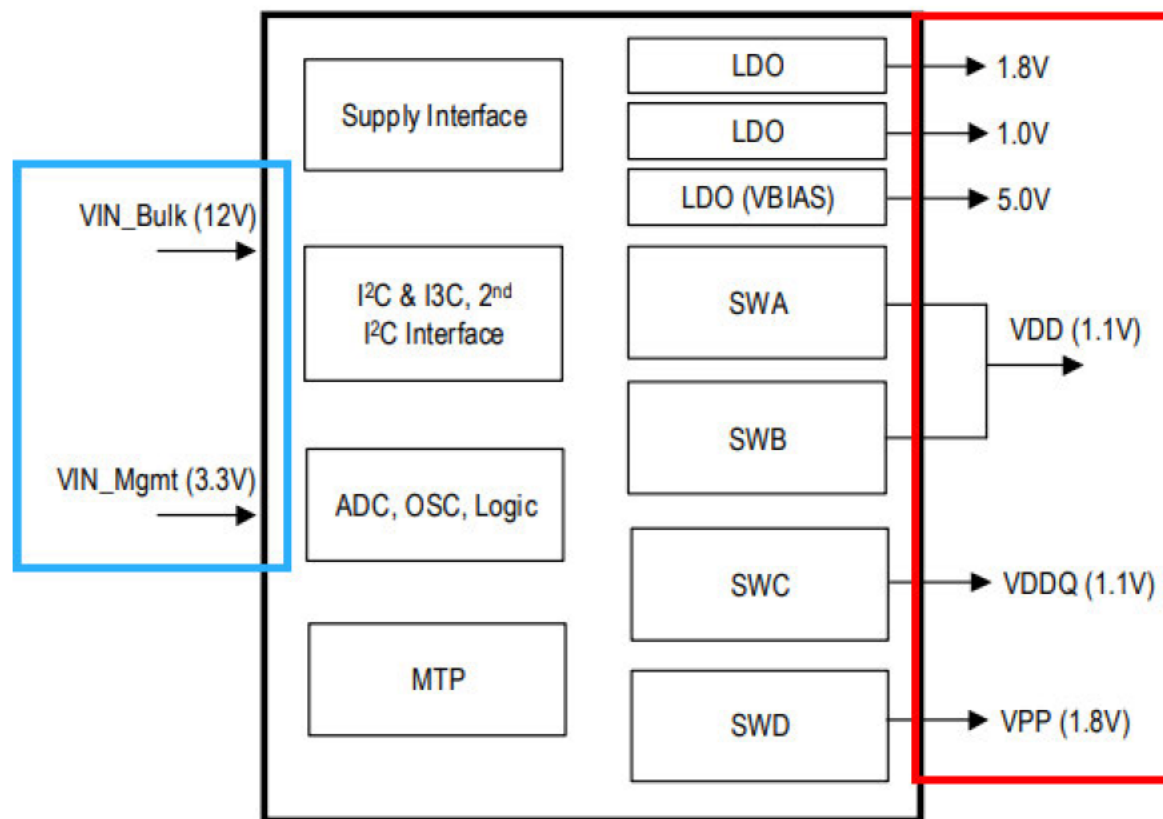
See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 13 [REDACTED]

[REDACTED]

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	For another example, <i>ee</i> Ex. CC-53C (0887-99A-1M6060 Prelim 02-08-22) at 3; Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1. And, as shown below, Renesas PMIC 5010 is configured to receive VIN_Bulk and VIN_Mgmt inputs to send out various voltage outputs.
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Block Diagram



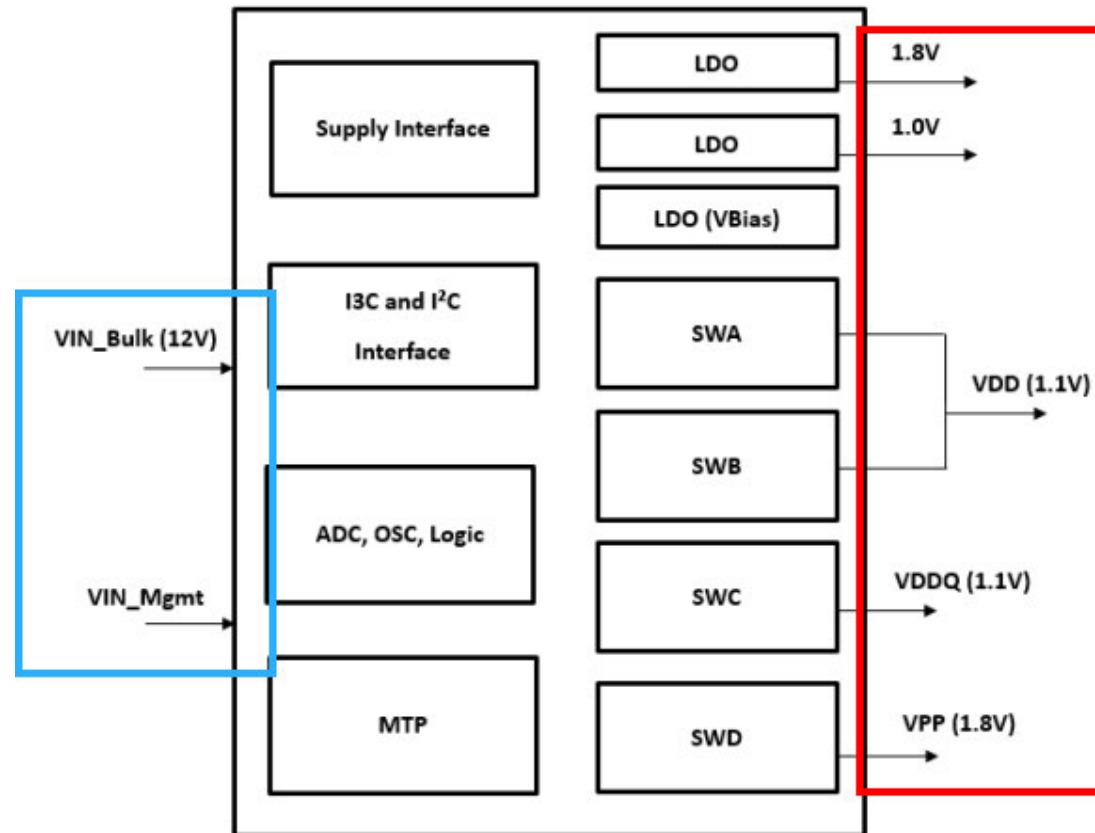
Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 (annotated) (showing a block diagram of inputs in blue and outputs in red for Renesas PMIC 5010).

Additionally,

See Ex. CC-54C (0888-97A-1A6B61-X00) at 3; Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1. And, as shown below, Renesas PMIC 5000 is configured to receive VIN_Bulk and VIN_Mgmt inputs to send out various voltage outputs. See Ex. CC-75 (Renesas P8900 Short Form

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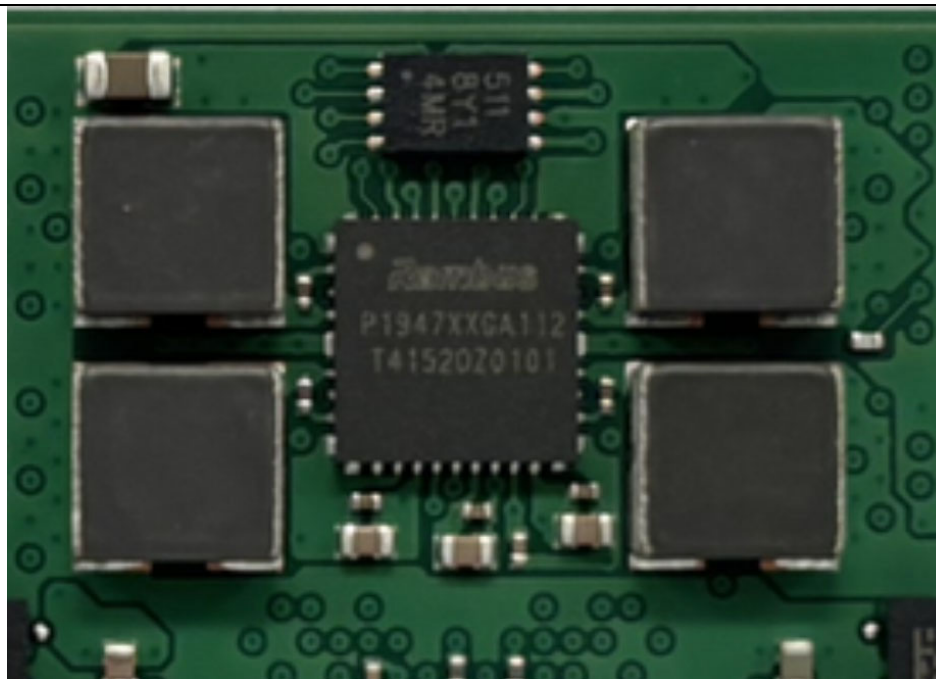
Datasheet) at 1:

Block Diagram

Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1 (annotated) (showing a block diagram of inputs in blue and outputs in red for Renesas PMIC 5000).

Concerning Netlist DDR5 MRDIMMs, Rambus's DDR5 PMICs, specifically "PMIC 5020 Extreme Current Power Management IC," is included in at least Netlist DDR5 MRDIMMs based on physical images and speed specifications. *See* Ex. CC-36 (Product Images) at 16 (showing Rambus part number P1947XXGA112 corresponding to Rambus's PMIC 5020):

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See also Ex. CC-76 (Rambus DDR5 PMICs) at 1 (showing part numbers corresponding to PMIC 50x0 versions and describing that “the PMIC5020 supports MRDIMM 8800, and the PMIC5030 will be used in MRDIMM 12800 [] modules”):

Description	Part Number	Product Brief	Applications
PMIC5000 High Current Power Management IC	P1945XXGxx	[Link]	Server RDIMM 4800 – 7200
PMIC5010 Low Current Power Management IC	P1943XXGxx	[Link]	Server RDIMM 4800 – 7200
PMIC5020 Extreme Current Power Management IC	P1947XXGxx	[Link]	4800 – 7200, MRDIMM 8800
PMIC5030 Gen2 Server Power Management IC	P2965XXGxx		RDIMM 8000, MRDIMM 12800

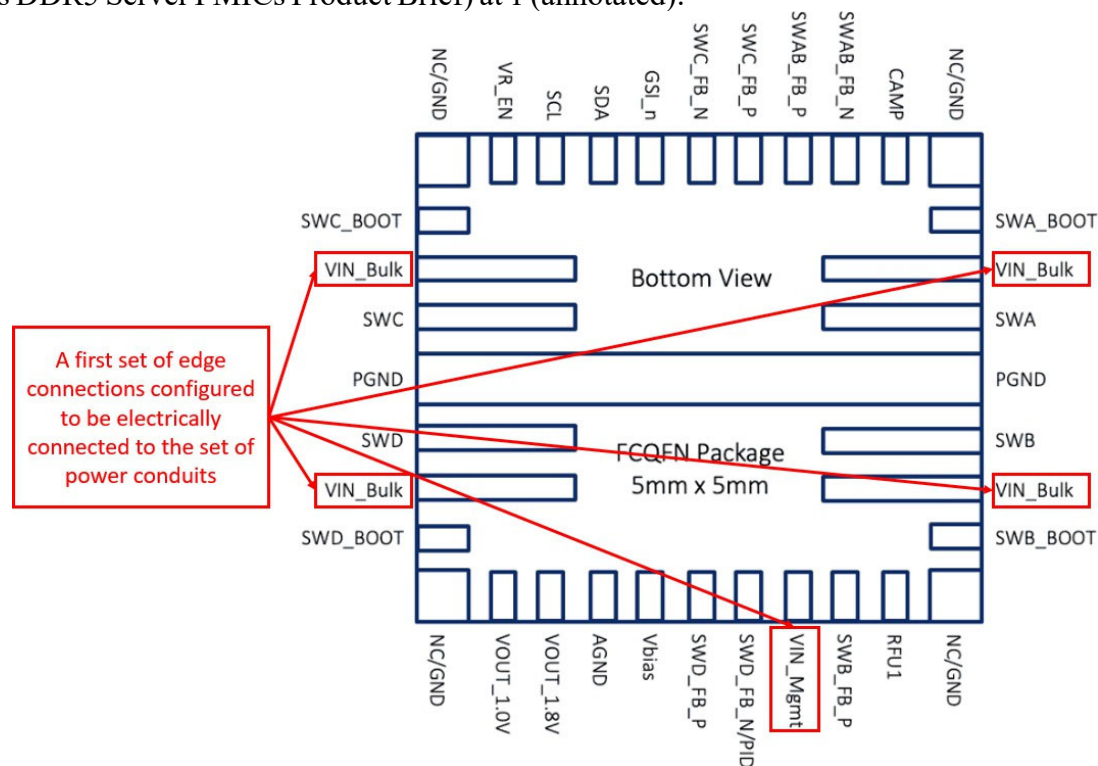
See also Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 2 (showing all part numbers for available

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MRDIMMs having speeds of 8800MT/s):

Density	Organization	Speed	Latency	Part Number
32GB	1Rx4	8800MT/s	36-36-36 @ 1.1V	NLM5G0054G0388200HC
32GB	2Rx8	8800MT/s	36-36-36 @ 1.1V	NLM5G0054G0788200HC
64GB	2Rx4	8800MT/s	36-36-36 @ 1.1V	NLM6G0054G0388200HC
64GB	4Rx8	8800MT/s	36-36-36 @ 1.1V	NLM6G0054G0788200HC
128GB	2Rx4	8800MT/s	36-36-36 @ 1.1V	NLM7G0055G0388200HM

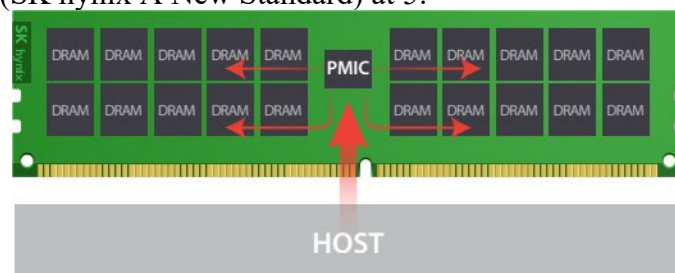
Therefore, Netlist DDR5 MRDIMMs include JEDEC-compliant Rambus PMIC 5020, which is configured to receive VIN_BULK and VIN_MGMT inputs to send out various voltage outputs. *See* Ex. CC-77 (Rambus DDR5 Server PMICs Product Brief) at 1 (annotated):



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Additionally, as described in 1[pre], the Netlist DDR5 RDIMM connector and physical layout are representative of the Netlist DDR5 MRDIMM connector and physical layout. *See, e.g.,* Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 2.

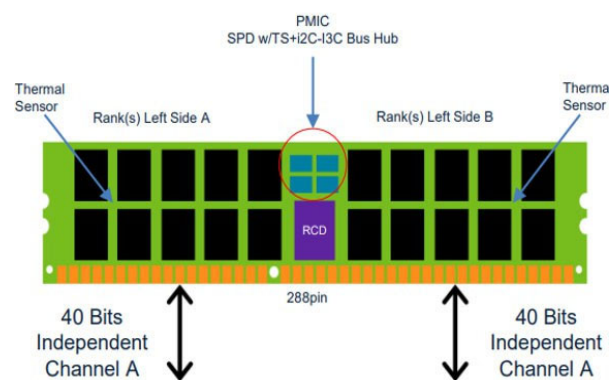
Furthermore, as advertised by SK hynix, DDR5 RDIMMs include an on-board PMIC that “eliminates the need for external voltage regulation and ensures more stable power provision to the DRAM.” Ex. CC-40 (SK hynix A New Standard) at 5:



| Stable Power Management

On-board power management IC (PMIC) eliminates the need for external voltage regulation and ensures more stable power provision to the DRAM.

See also Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1 (annotated):



DIMM Changes from DDR4 to DDR5

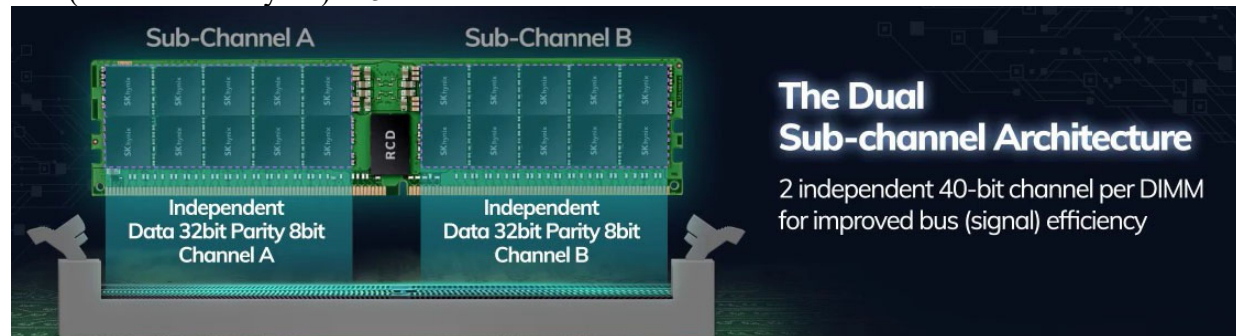
Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data+ 8 ECC 2 channel per DIMM
DQE Support	DB Only	RCD, DB, and DRAM

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[1.a.2] a second set of edge connections configured to be electrically connected to the set of data conduits, and

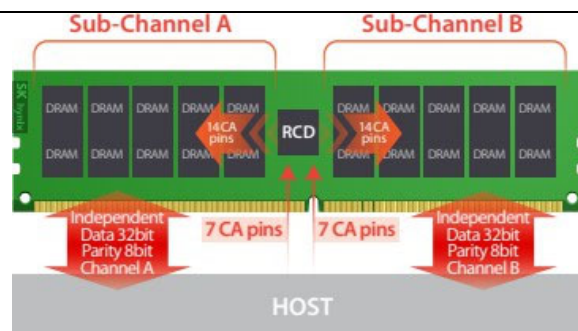
Each of the Practicing '366 Products comprises a printed circuit board (PCB) having a second set of edge connections, such as DQ[31:0]_A, DQ[31:0]_B, CB[7:0]_A, and CB[7:0]_B, configured to be electrically connected to the set of data conduits.

For example, SK hynix advertises that SK hynix DDR5 RDIMMs receive 32 data bits and 8 parity bits for both Channel A and Channel B, which are received via a second set of edge connections. *See* Ex. CC- 41 (RDIMM SK hynix) at 5:



See also Ex. CC-40 (SK hynix A New Standard) at 5:

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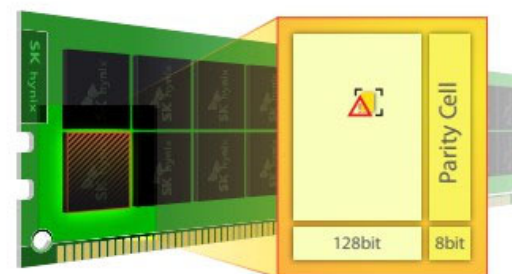


New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

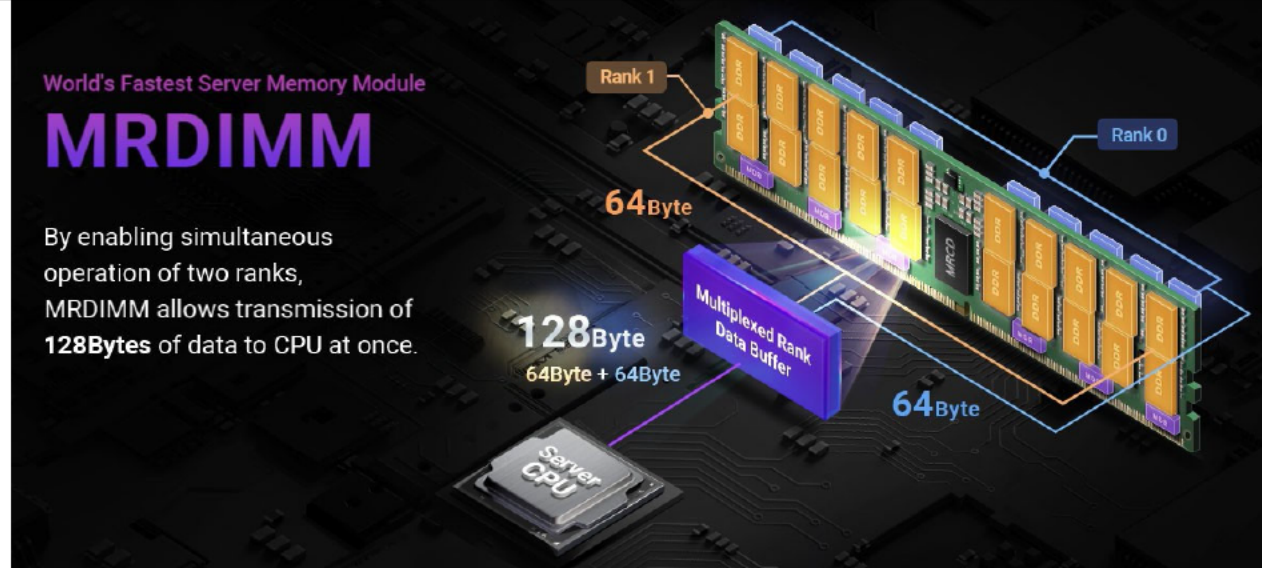
Enhanced Reliability

In-DRAM ECC on the DIMM offloads the system's error correction burden and considerably improves Reliability, Availability, and Serviceability (RAS).



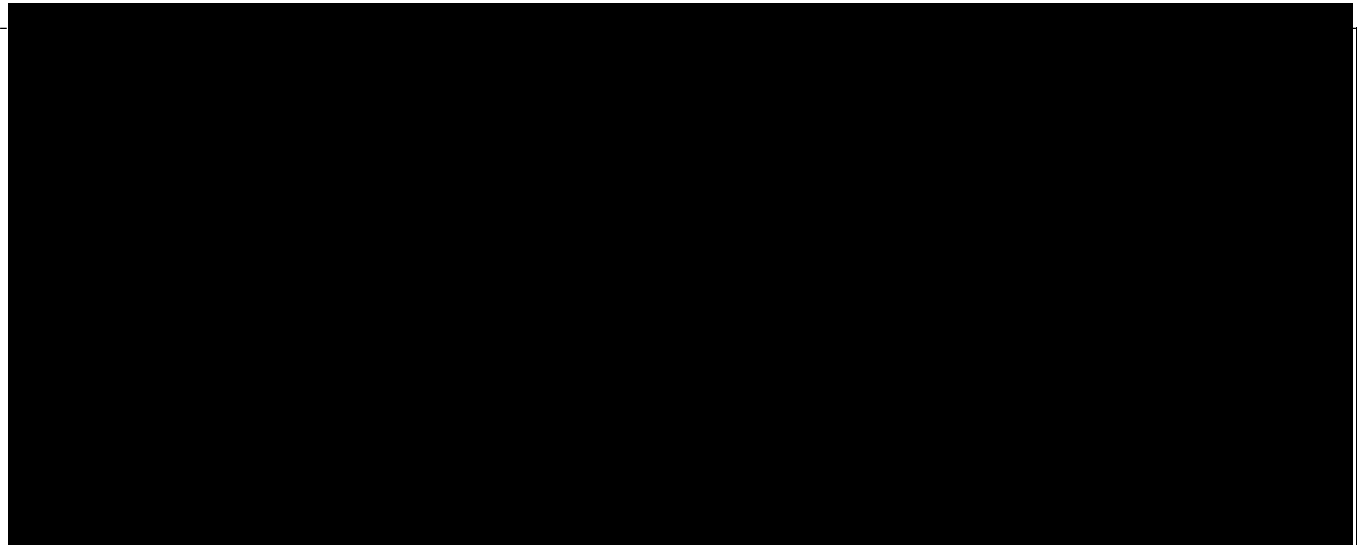
Additionally, SK hynix advertises transmission of 128 bytes of data between a host CPU and SK hynix's DDR5 MRDIMMs, which is performed via the second set of edge connections. *See Ex. CC-42 (SK hynix MRDIMM) at 4:*

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For another example, Netlist's DDR5 VLP RDIMM Datasheet shows

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7 (annotated).

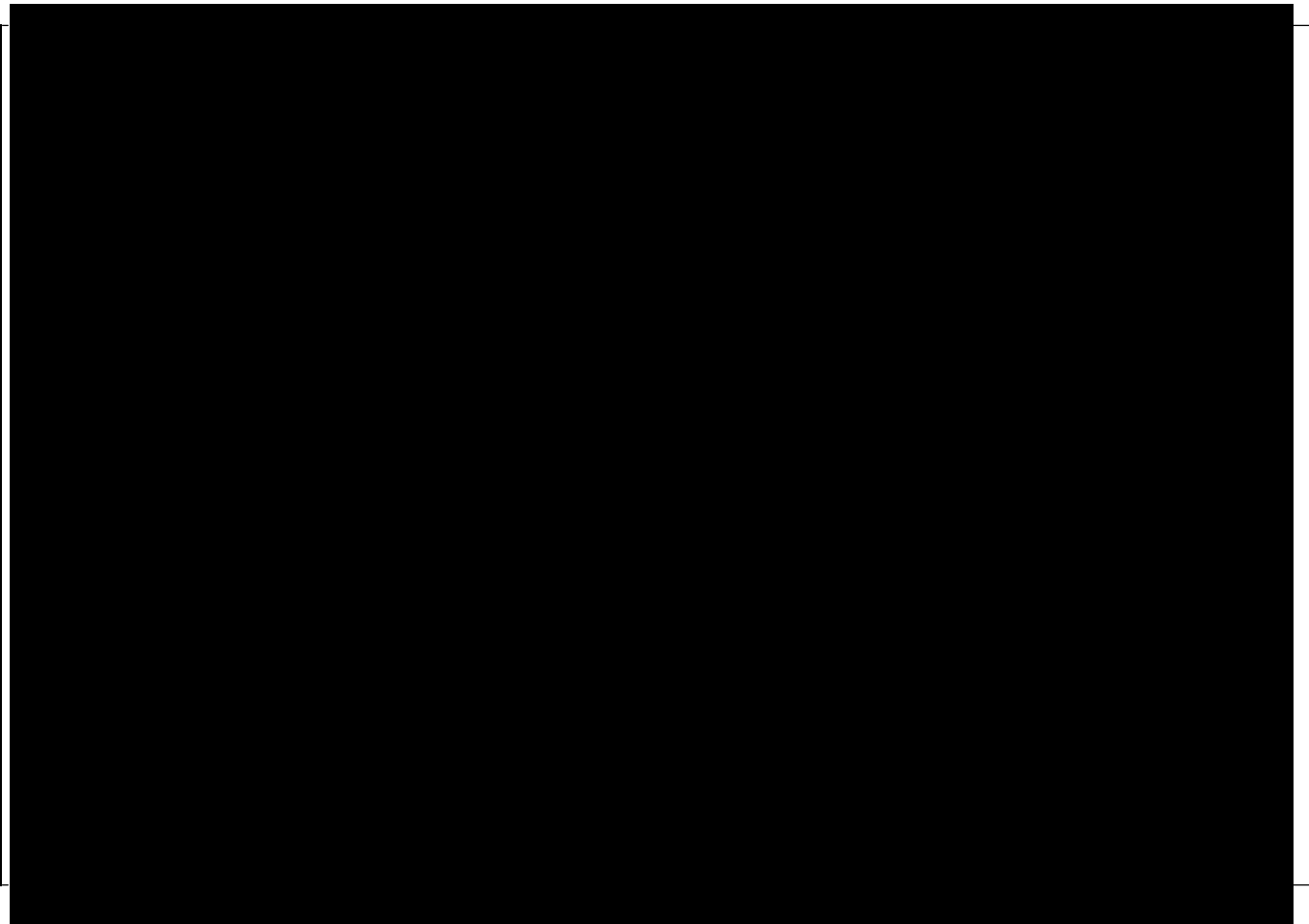
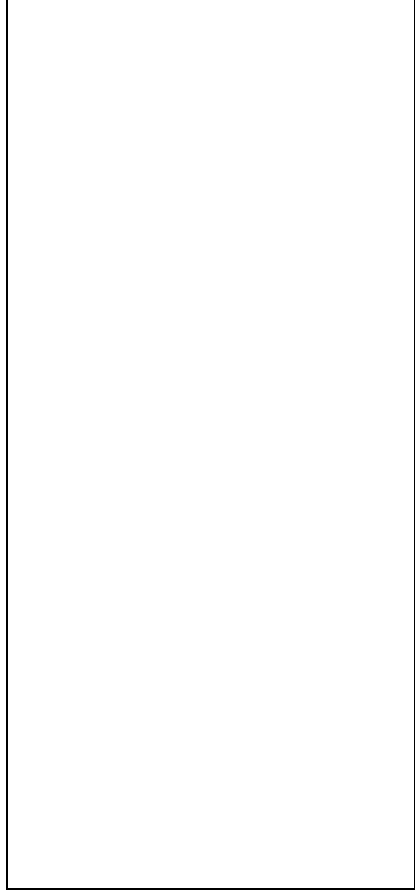
8

[illegible]

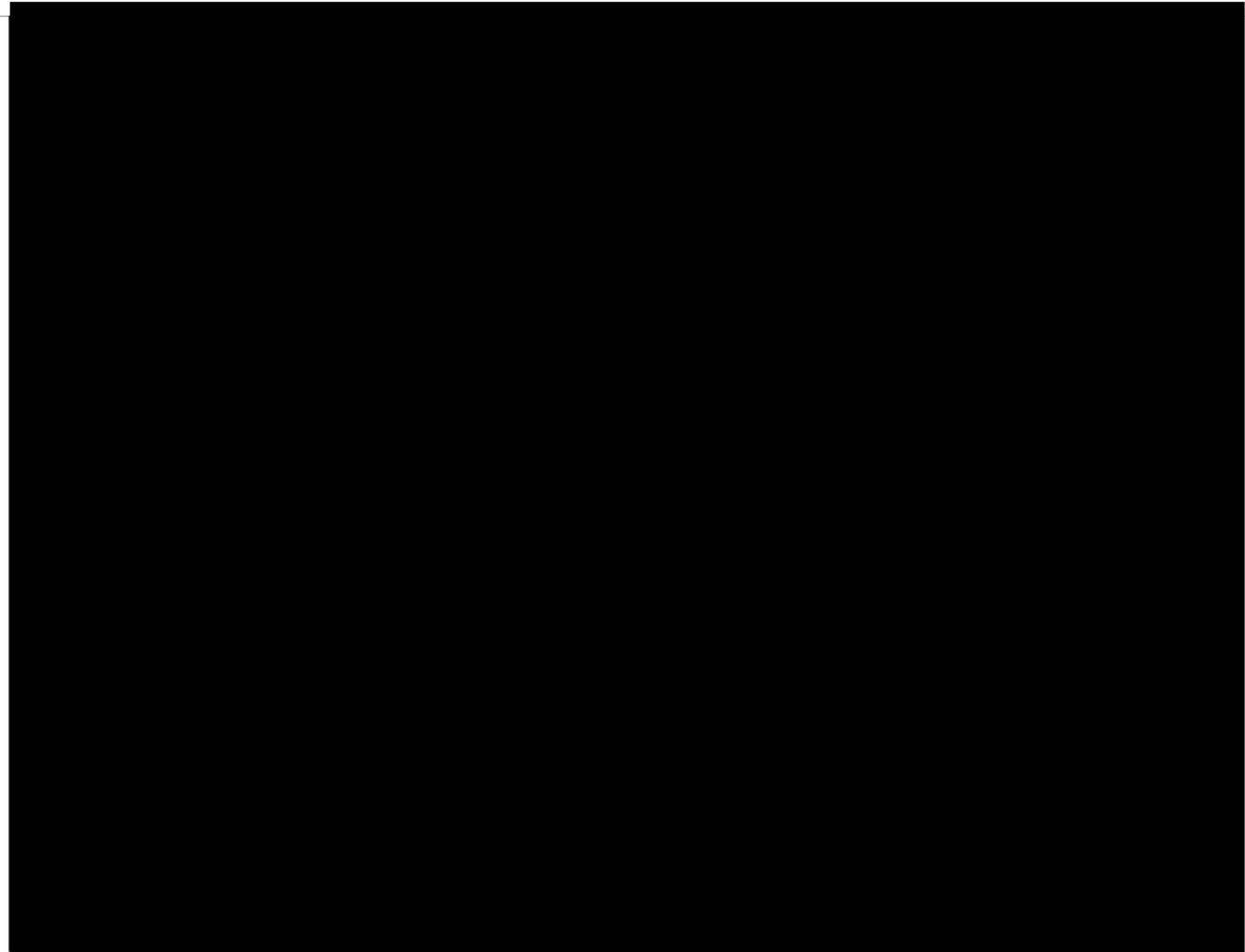
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 5–6.

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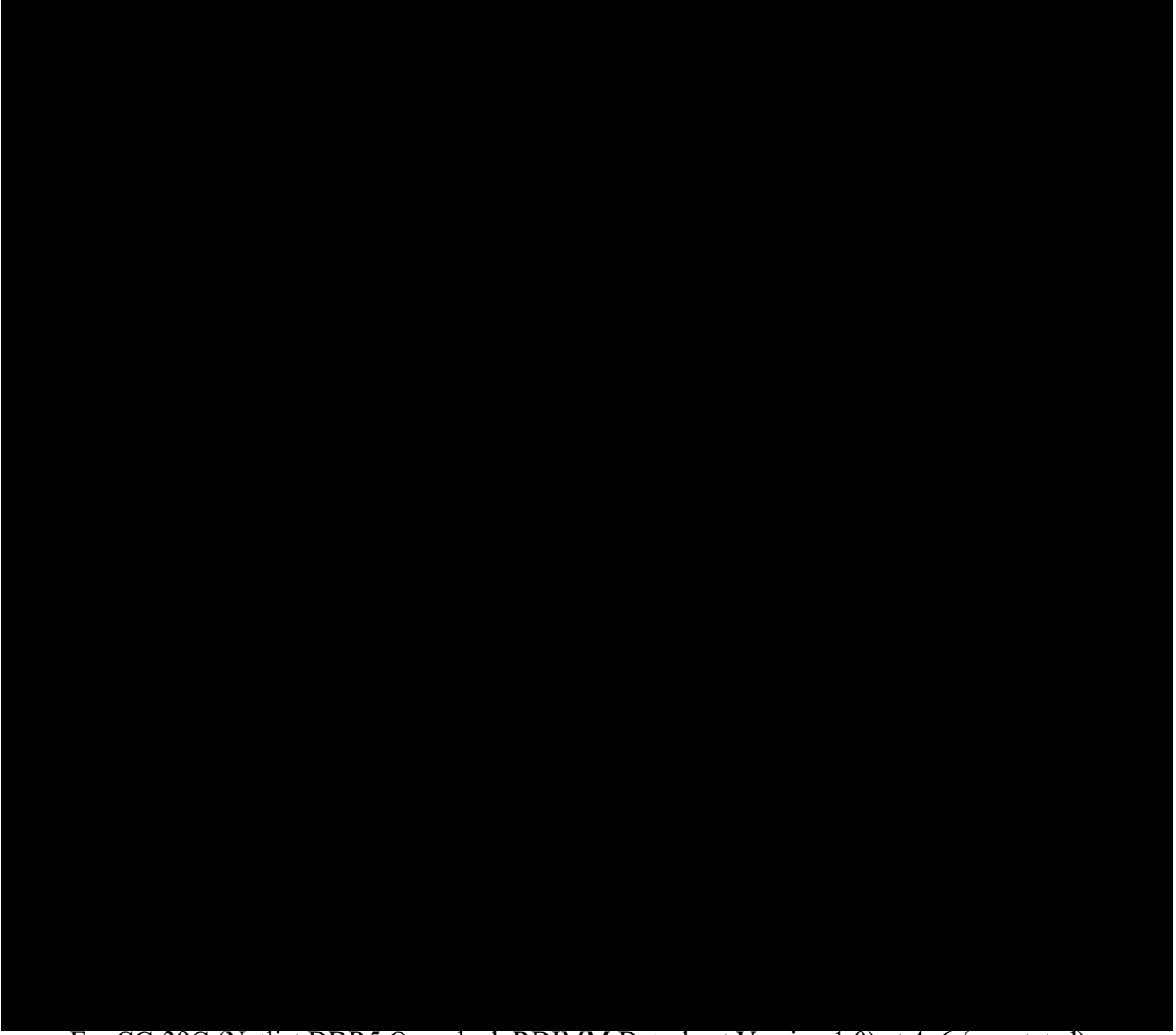
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart



Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 17, 19, 22, 24.
As another example, Netlist's DDR5 Overclock RDIMM Datasheet shows

[Redacted text block]

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	 Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–6 (annotated).
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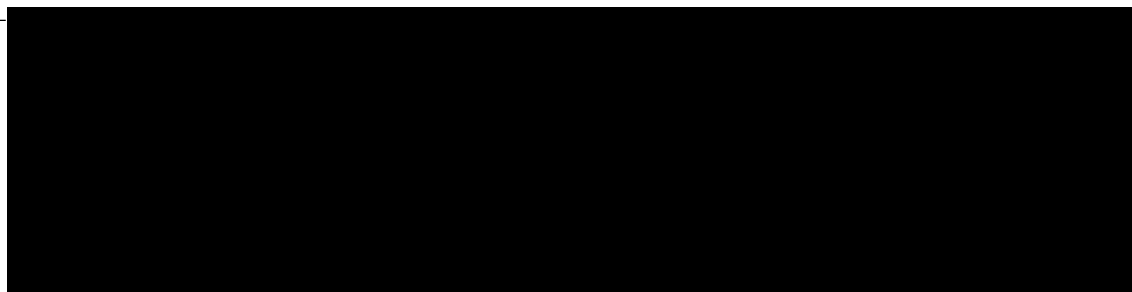
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U.S. Patent No. 12,373,366 Domestic Industry Claim Chart

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 7–9.

These data edge connections, similar to above with reference to Netlist DDR5 VLP RDIMMs, are configured to be electrically connected to the set of data conduits.

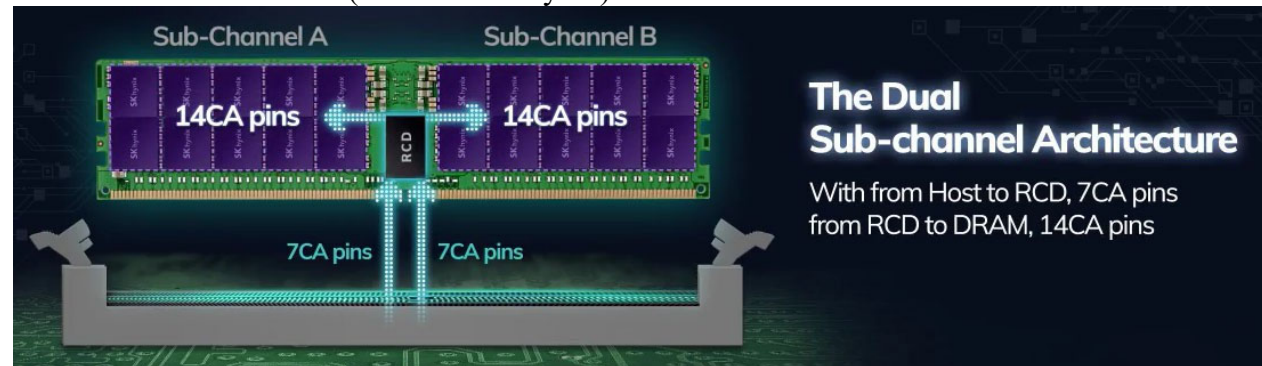
Furthermore, as described in 1[pre], the Netlist DDR5 RDIMM connector and physical layout are representative of the Netlist DDR5 MRDIMM connector and physical layout. *See, e.g.*, Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 2.

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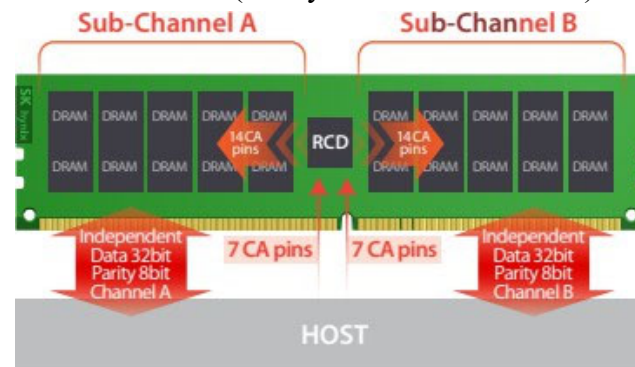
[1.a.3] a third set of edge connections configured to be electrically connected to the set of address and control conduits,

Each of the Practicing '366 Products comprises a printed circuit board (PCB) having a third set of edge connections, such as CA[6:0]_A, CA[6:0]_B, CS[1:0]_A_n, CS[1:0]_B_n configured to be electrically connected to the set of address and control conduits in the memory slot.

For example, SK hynix advertises that SK hynix DDR5 RDIMMs receive at least 7 CA pins for Channel A and 7 additional CA pins for Channel B, which would be performed via a third set of edge connections. *See* Ex. CC- 41 (RDIMM SK hynix) at 4:

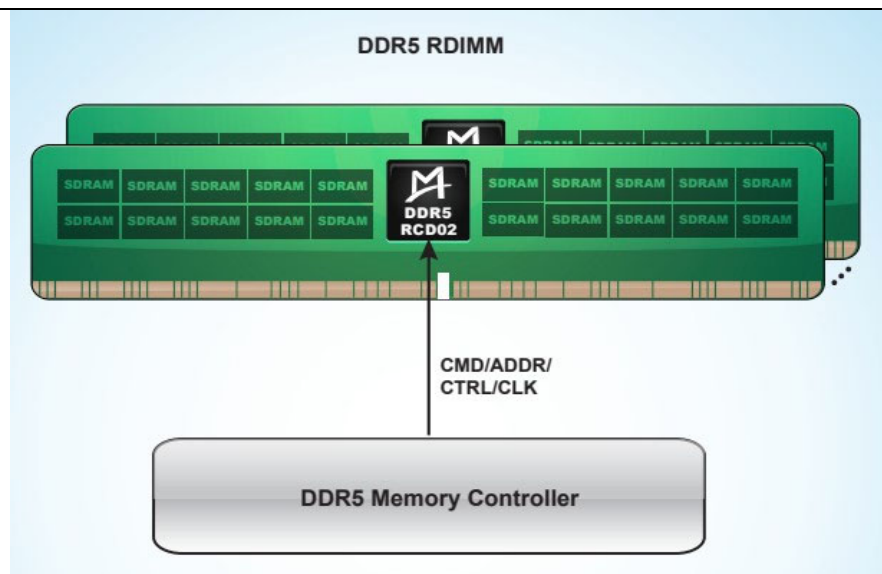


See also Ex. CC-40 (SK hynix A New Standard) at 5:



Additionally, both Netlist DDR5 RDIMMs and Netlist DDR5 Overclock RDIMMs include Montage M88DR5RCD02 RCD, which are advertised as configured to receive “Command/Address inputs and control signals” from a DDR5 memory controller (e.g., Host server or Host CPU). *See* Ex. CC-43 (M88DR5RCD02 Flyer English) at 1–2:

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See also Ex. CC-36 (Product Images) at 2 (showing M88DR5RCD02 included on Netlist's DDR5 Overclock RDIMM):



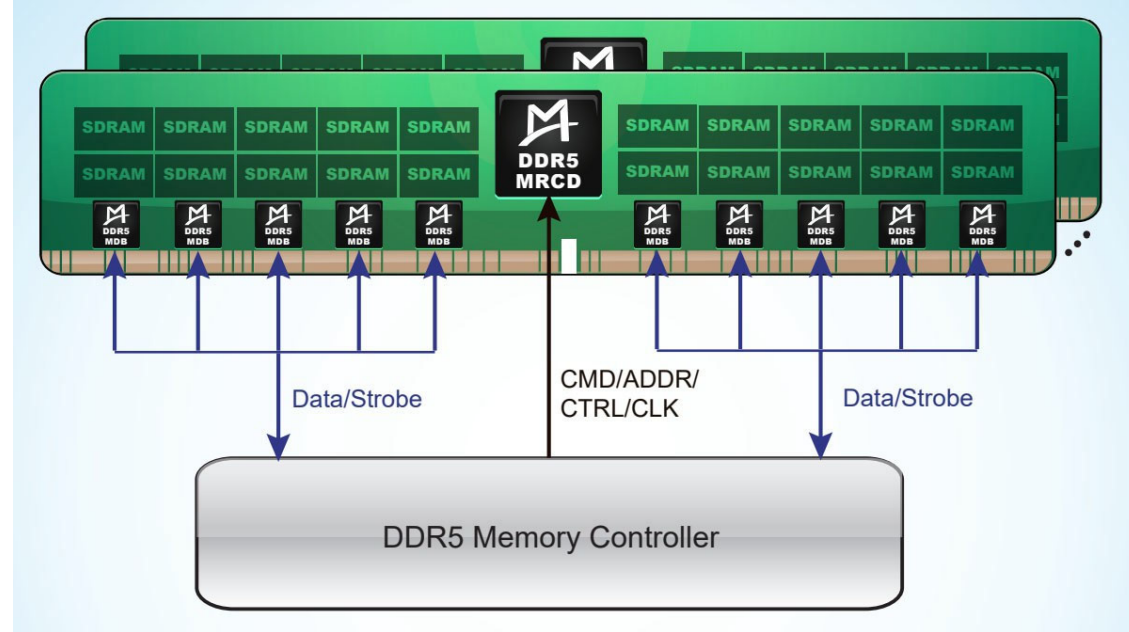
See also Ex. CC-36 (Product Images) at 9 (showing M88DR5RCD02 included on Netlist's DDR5 RDIMM):

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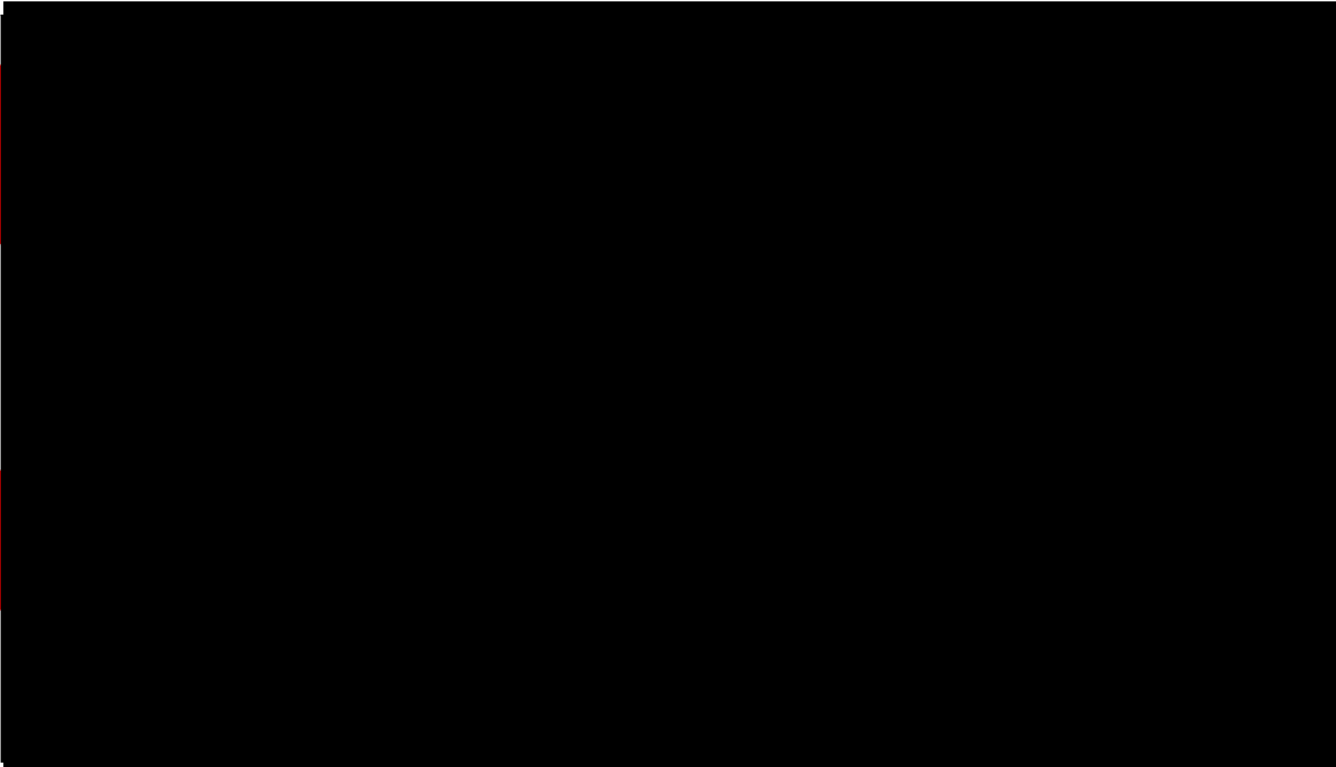
Furthermore, Netlist's DDR5 MRDIMMs, including a Montage M88MR5RCD01 MRCD, as shown in 1[pre], are configured to receive command, address, and select signals from a DDR5 memory controller (e.g., host server or host CPU). *See also* Ex. CC-12 (M88MR5RCD01 Flyer English) at 1–2 (showing and describing data, control, and address signals sent between example DDR5 memory controller and MRDIMM):

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For another example, the Netlist DDR5 VLP RDIMM Datasheet shows pin assignment for edge connections, which include pins for data transmission, such as CA[6:0]_A, CA[6:0]_B, CS[1:0]_A_n, CS[1:0]_B_n, configured to be connected to data conduits:

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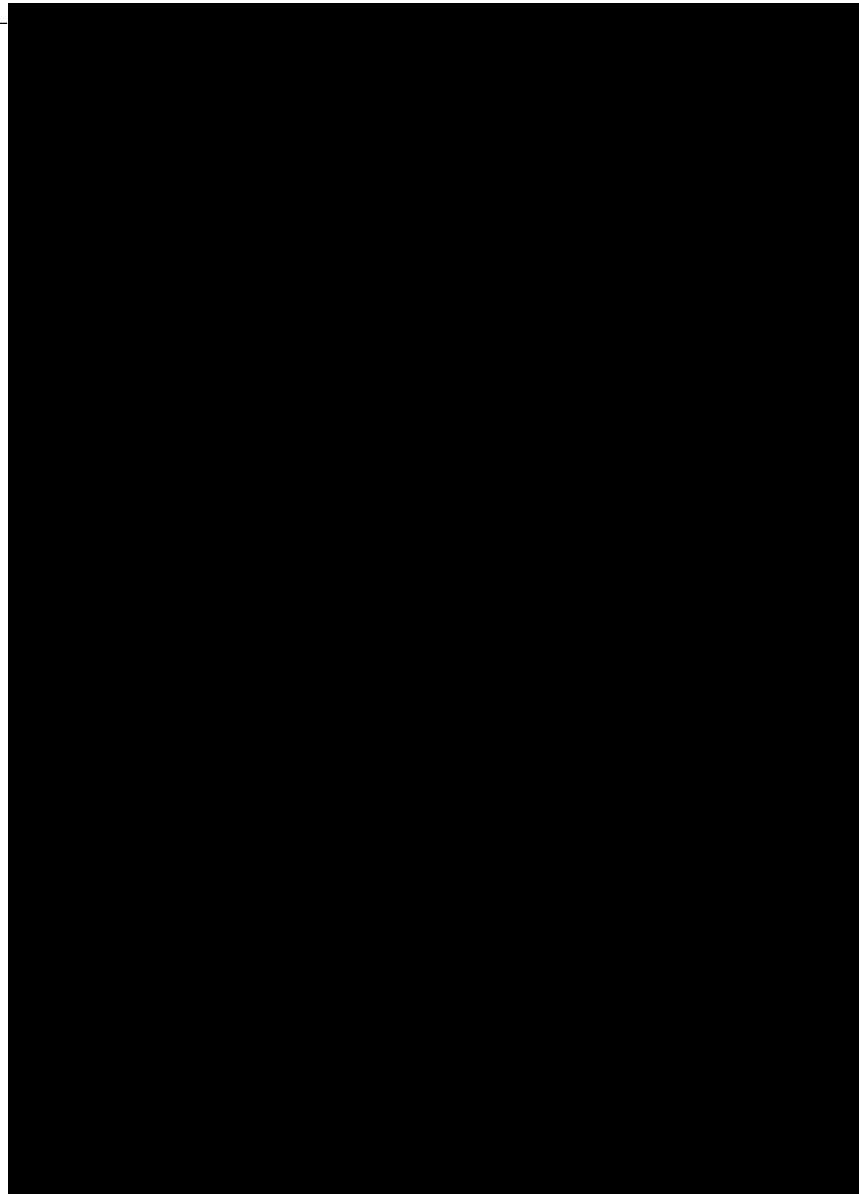
	 Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7 (annotated).
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 5–6.

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart

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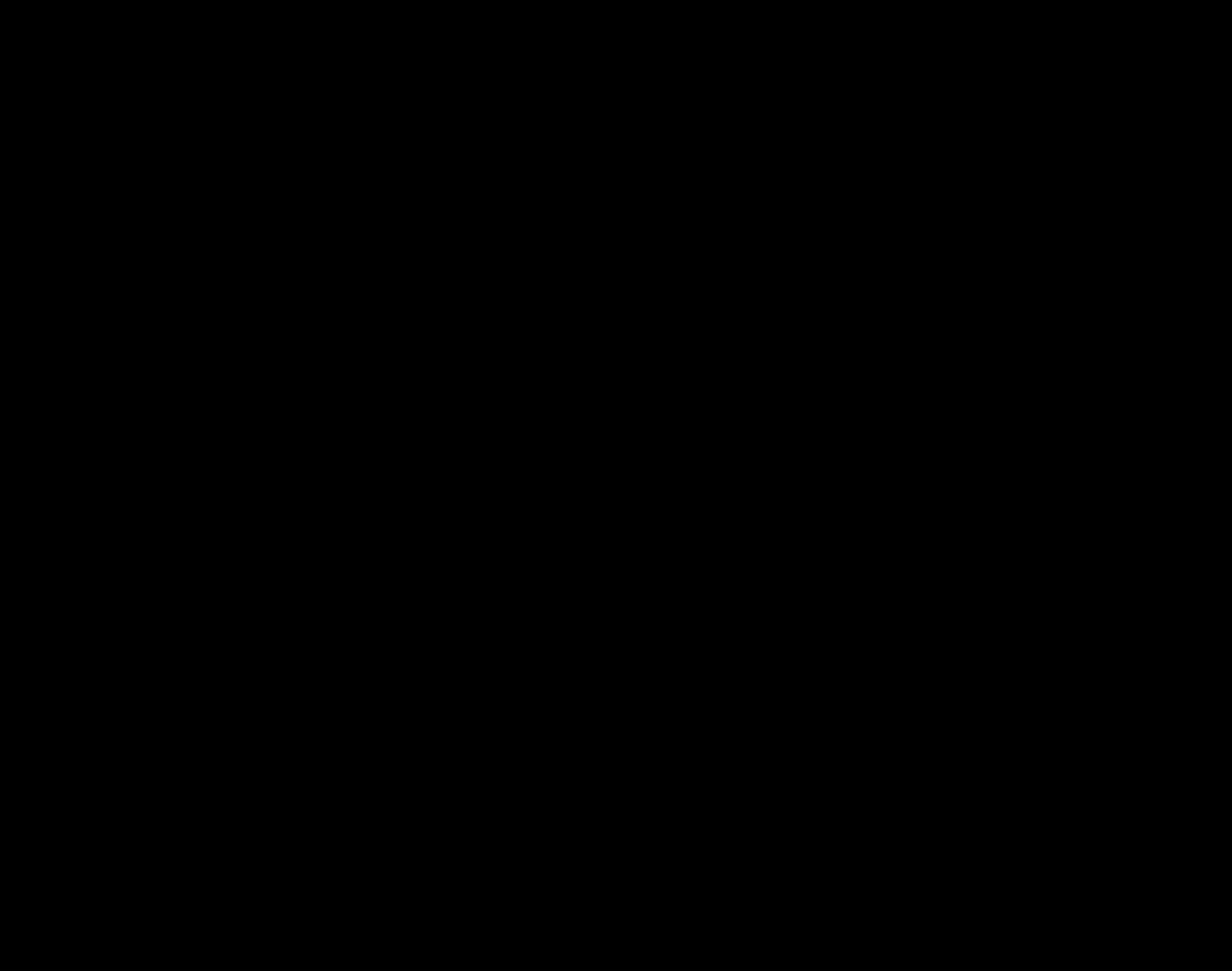
Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9 (

As another example, Netlist's DDR5 Overclock RDIMM Datasheet shows

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	 Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–5 (annotated).
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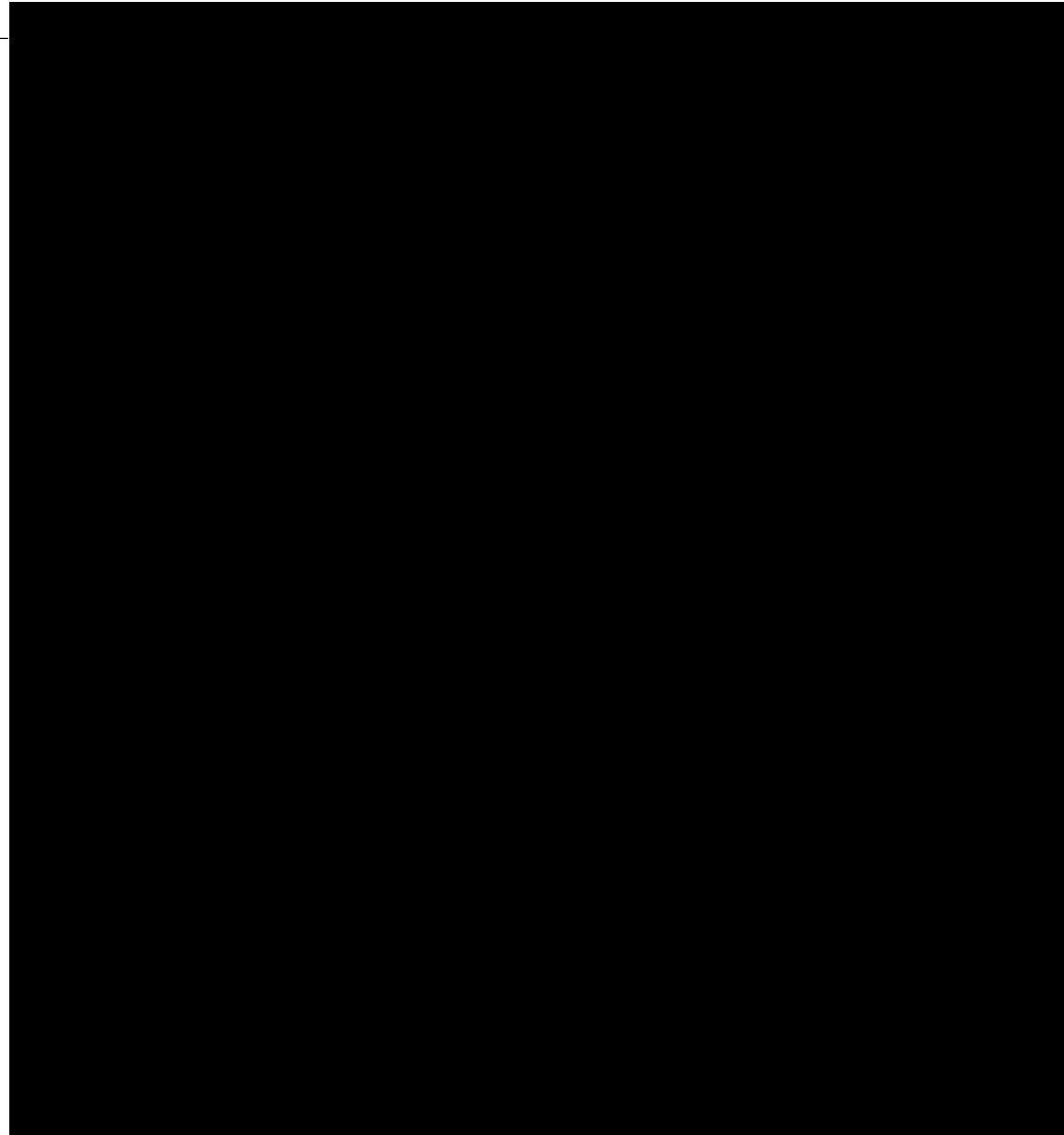
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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 7–9.

See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10

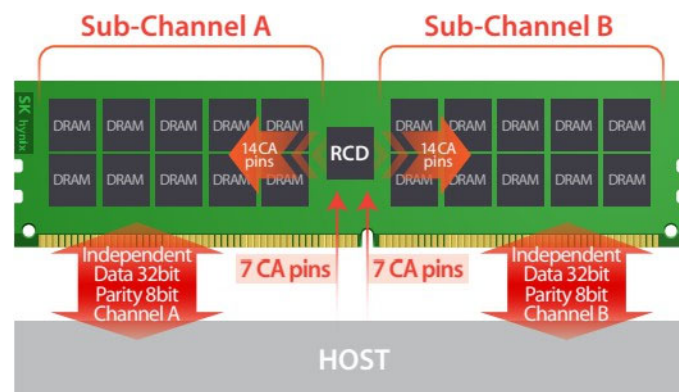
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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10.

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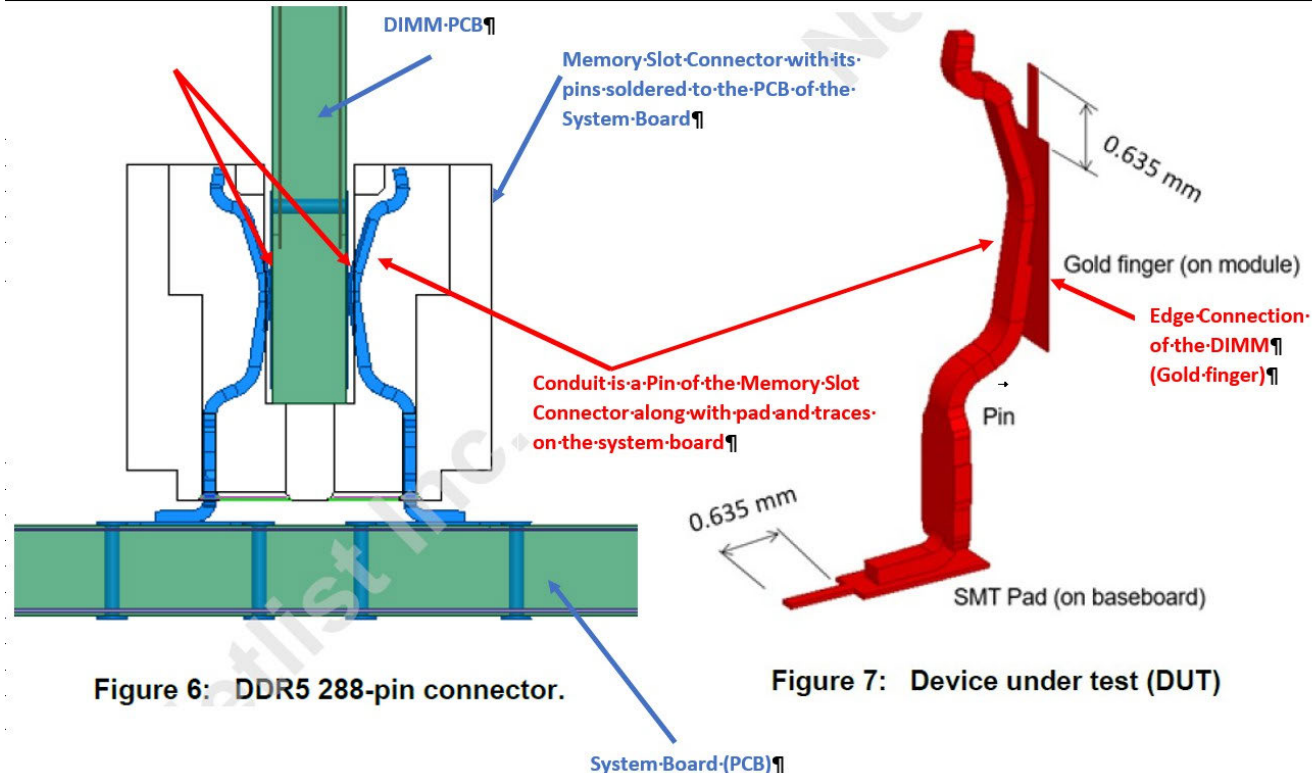
See also Ex. CC-40 (SK hynix A New Standard) at 5 (showing control and address signals from edge connections to RCD).



New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

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Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated).

Furthermore, as described in 1[pre], the Netlist DDR5 RDIMM connector and physical layout are representative of the Netlist DDR5 MRDIMM connector and physical layout. *See, e.g.,* Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 2.

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[1.a.4] a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

Each of the Practicing '366 Products comprises a printed circuit board (PCB) that has a set of voltage supply lines electrically connected to the first set of edge connections.

Furthermore, as described in 1[pre], each of the Practicing '366 Products is configured to fit into a memory slot connector of a system board of a computer system, and as illustrated below in an infrastructure diagram for DDR5, DDR5 DIMMs receive VIN_BULK and VIN_MGMT power from the power rails of the system board by way of the DIMM memory slot.

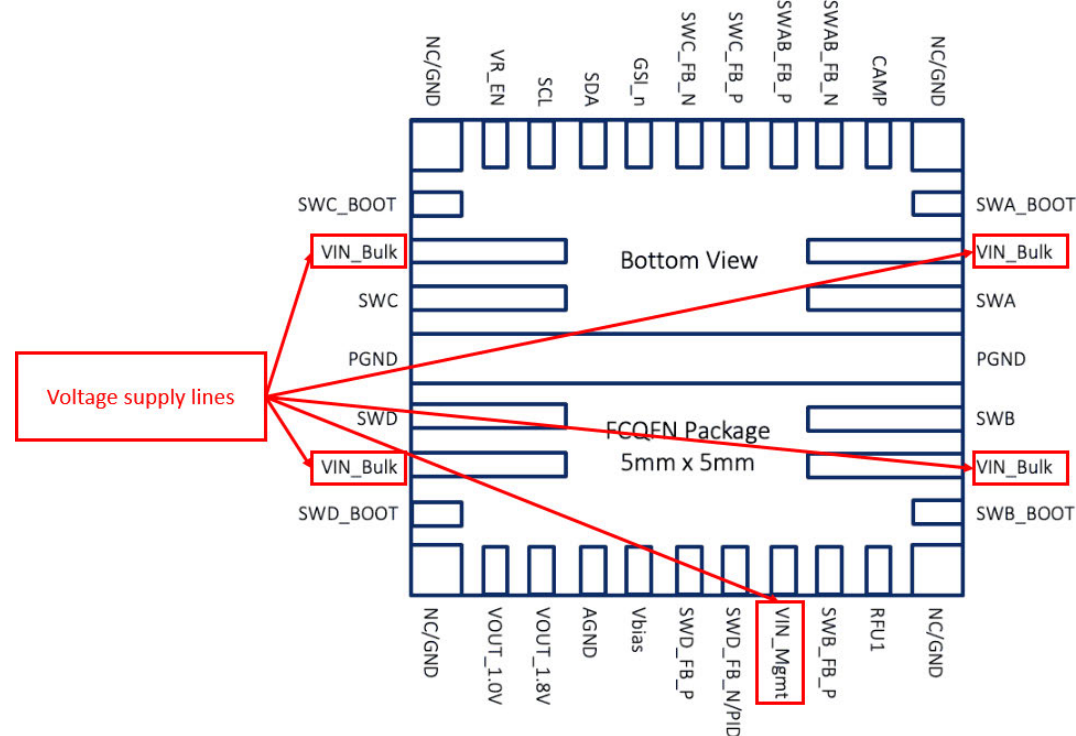
For example, as illustrated in Netlist's DDR5 Overclock RDIMM Datasheet, there is a set of voltage supply lines electrically connecting the VIN_BULK and VIN_MGMT edge connection on the DIMM with the corresponding power supply input pins on the PMIC (see below). As illustrated, the set of voltage supply lines is configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM (e.g., VIN_BULK and VIN_MGMT pins).



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet) at 13

As another example, Rambus, a supplier of PMIC 5020 for Netlist's DDR5 MRDIMMs, states that "[t]he DDR5 on-module PMIC receives a 12V input and generates five distinct voltage levels needed by the various components on the DDR5 module." Ex. CC-77 (Rambus DDR5 Server PMICs Product Brief) at 1.

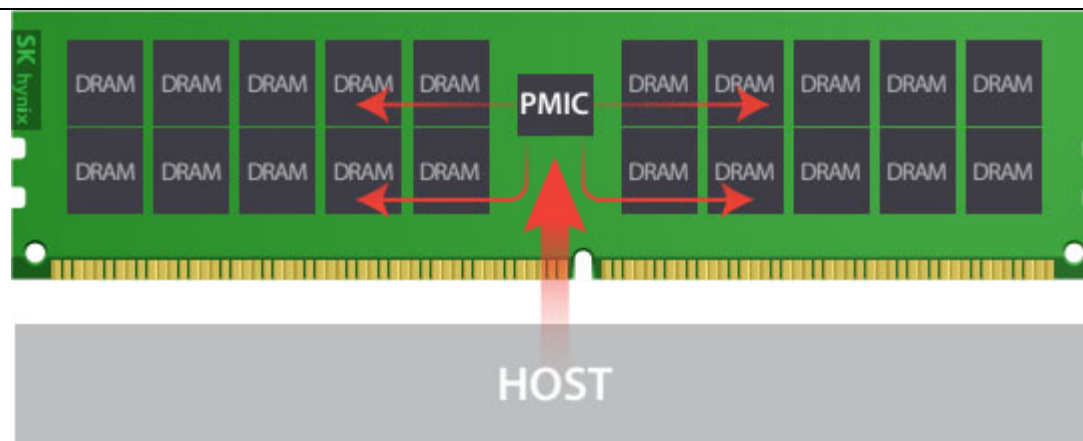
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Ex. CC-77 (Rambus DDR5 Server PMICs Product Brief) at 1 (annotated).

As yet another example, as shown in the below SK hynix illustration, power is supplied to the RDIMM from a power supply off the RDIMM, such as via DIMM slots through the edge connections. *See* Ex. CC-40 (SK hynix A New Standard) at 5.

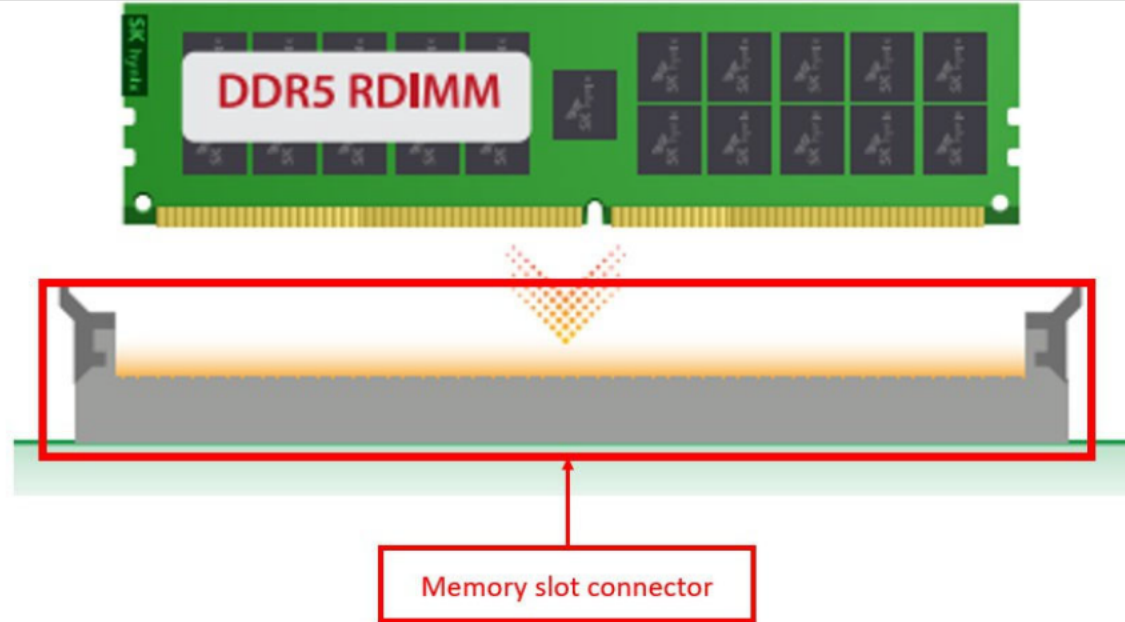
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See also, e.g., supra at (1.pre).

The external power supply is delivered from the power rails of the system board by way of the memory slot connector of the system board, as illustrated in Ex. CC-40 (SK hynix A New Standard) at 6 (annotated) (power lines, such as a 12V power line would be fed to the memory slot connector and from the slot connector to the power pins of the DIMM's edge connections):

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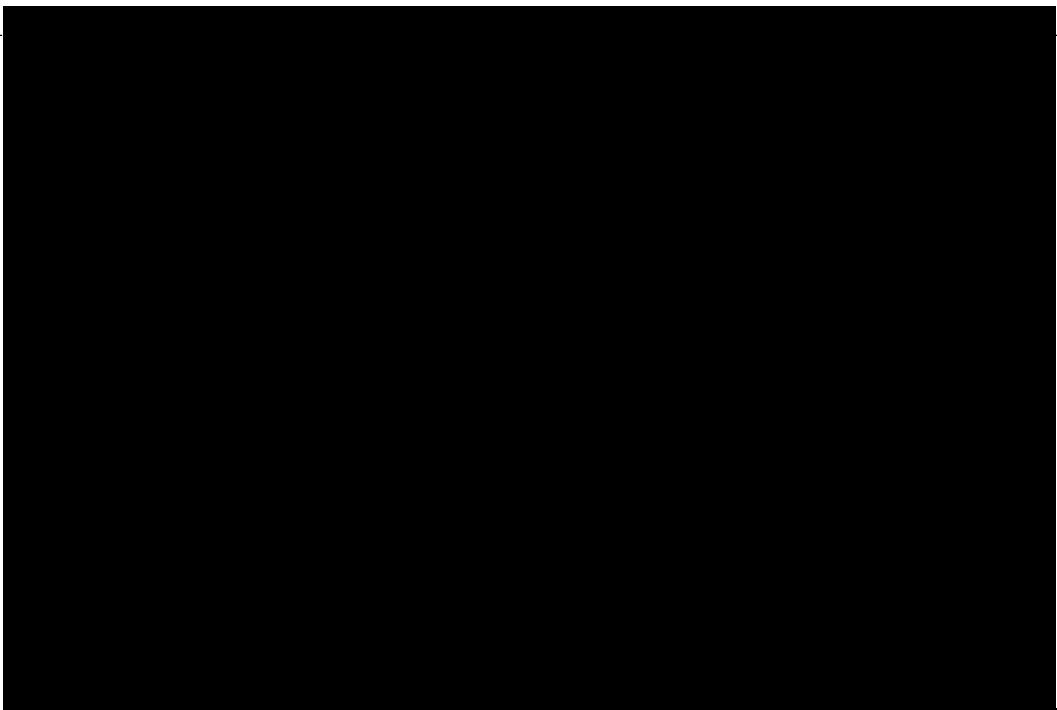
See also Ex. CC-78C (0887-06REVA-0114) at 2, 4



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	 <i>See also</i> Ex. CC-79C (0888-06REVA 0722) at 2, 4 (showing 
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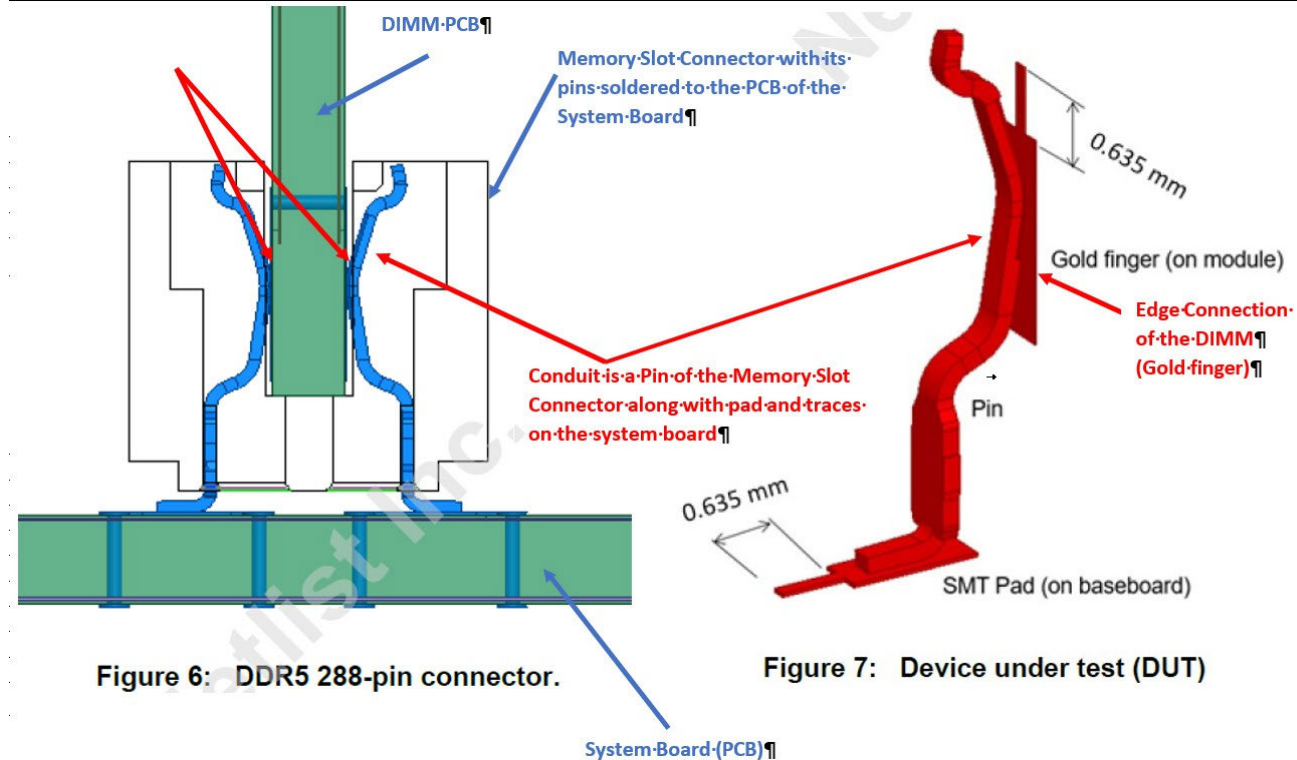
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Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated) (showing an electrical connection between a connector in a memory slot and a corresponding edge connection of the DIMM).

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[1.a.5] a set of data lines electrically connected to the second set of edge connections, and

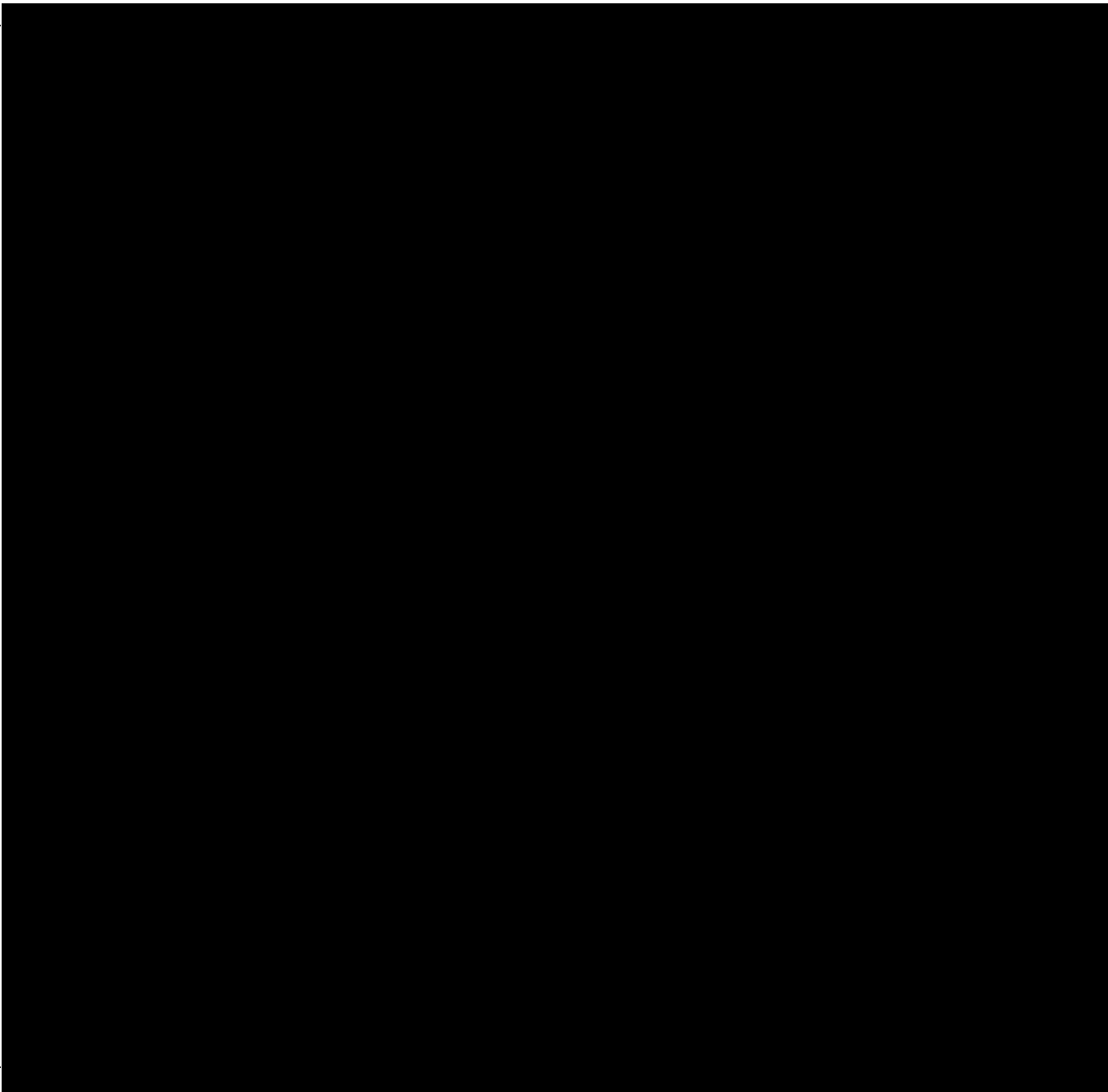
Each of the Practicing '366 Products comprises a printed circuit board (PCB) having a set of data lines electrically connected to the second set of edge connections (*e.g.*, connections with pin assignment DQ[31:0]_A, DQ[31:0]_B, CB[7:0]_A, and CB[7:0]_B).

For example, as illustrated by Netlist's DDR5 VLP RDIMM Datasheet.



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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 17, 19, 22, 24.

See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 10 (annotated)

See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 11–12 (annotated)

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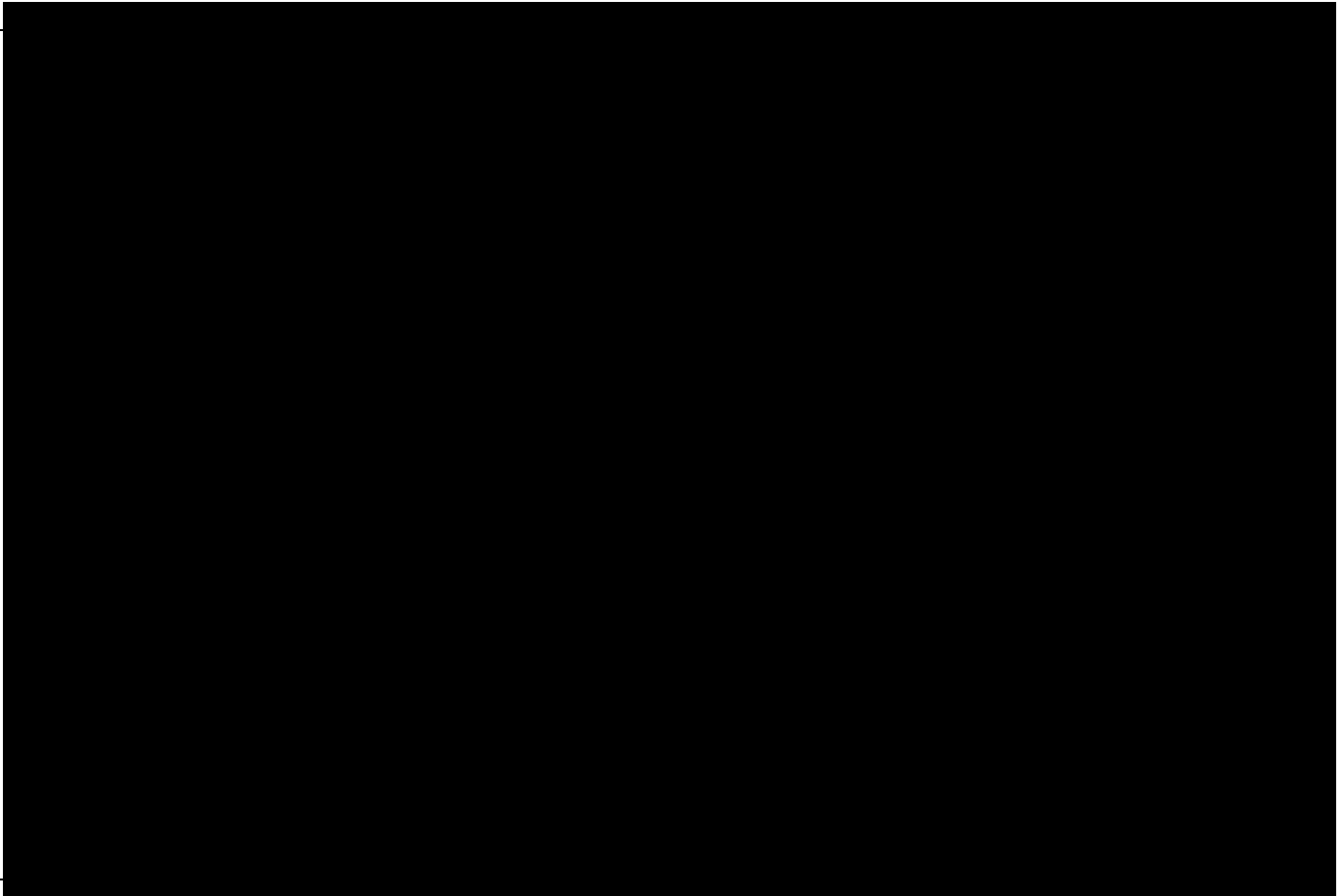
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U.S. Patent No. 12,373,366 Domestic Industry Claim Chart

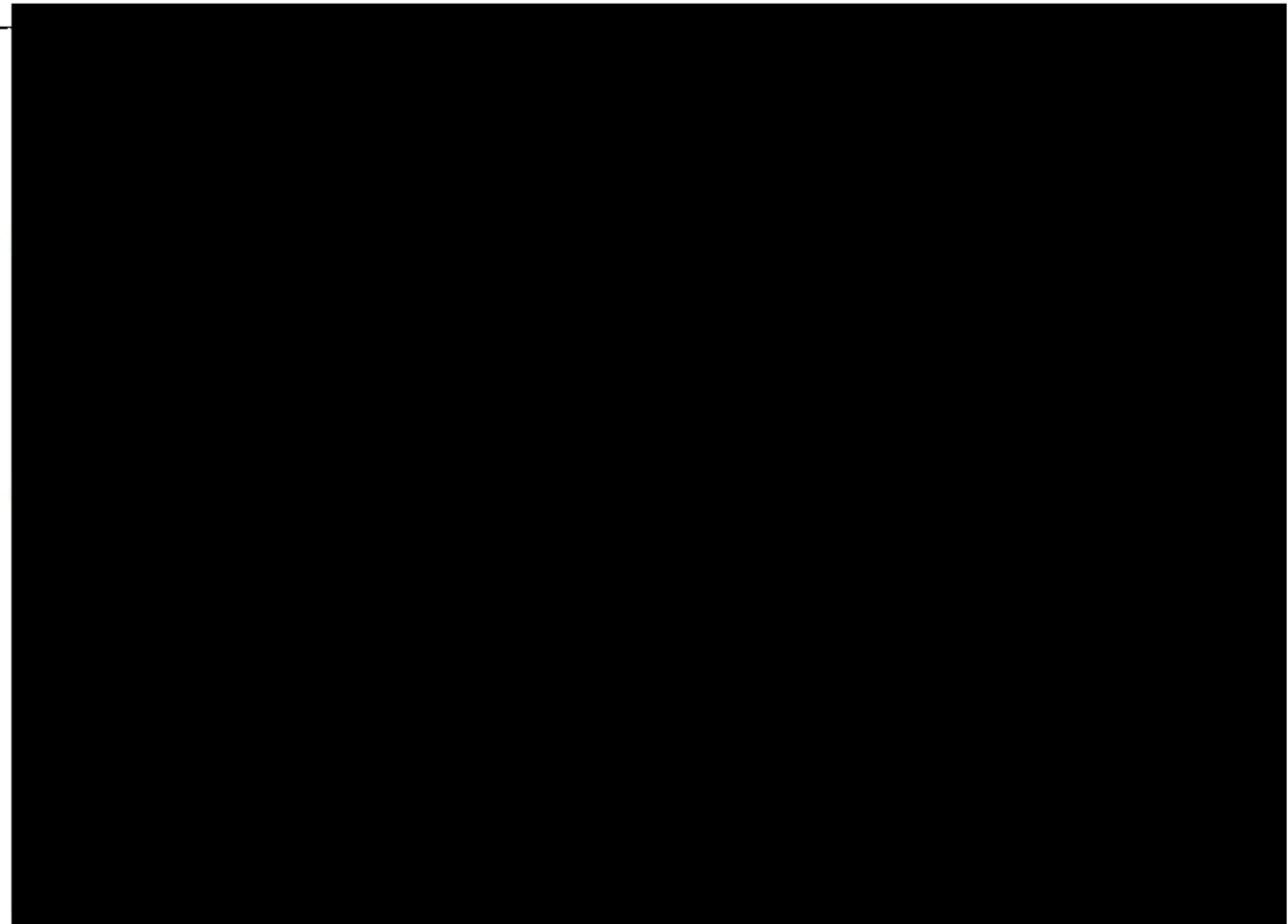
See also Ex. CC-78C (0887-06REVA-0114) at 2, 5–6 (annotated)

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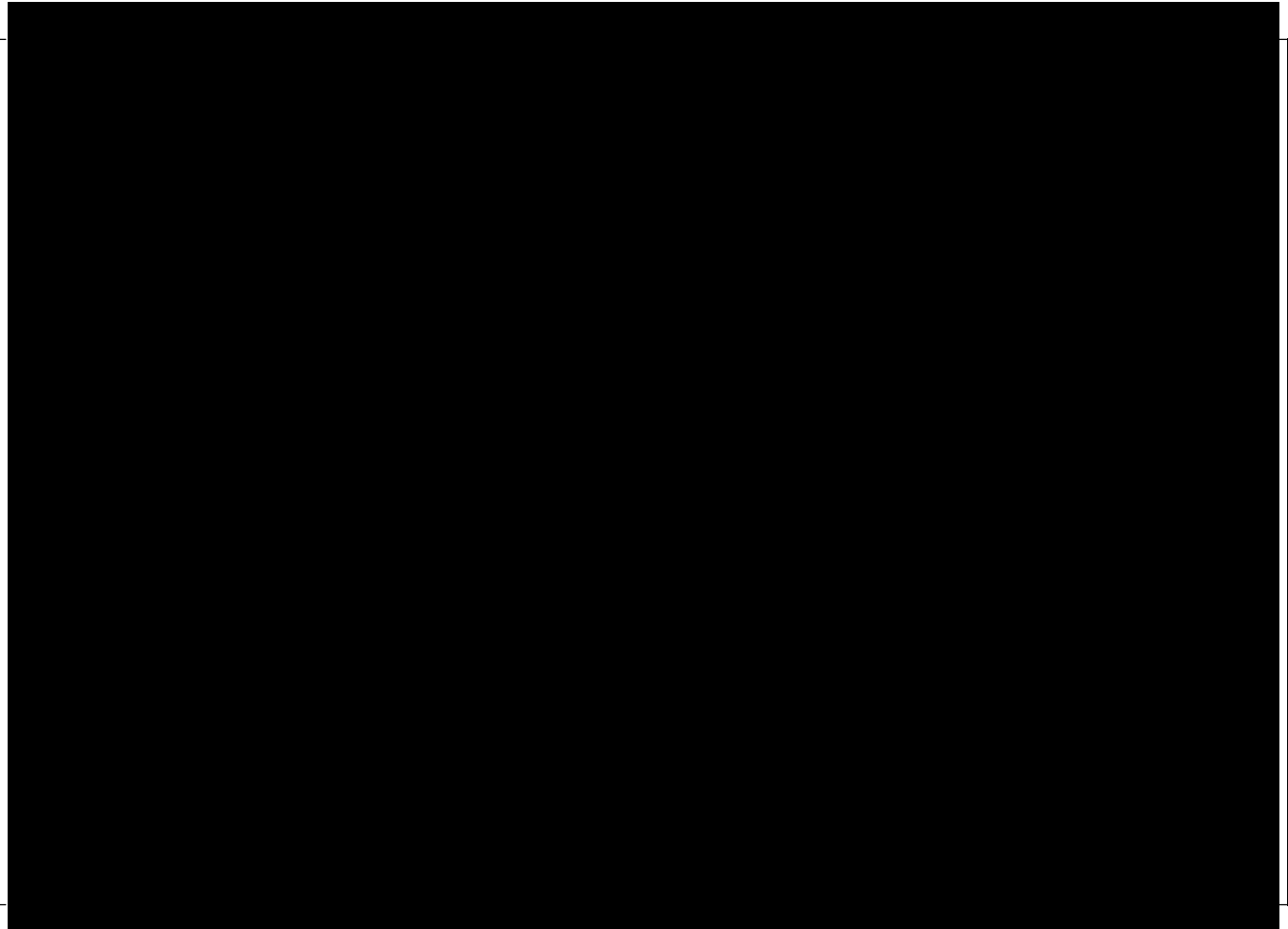
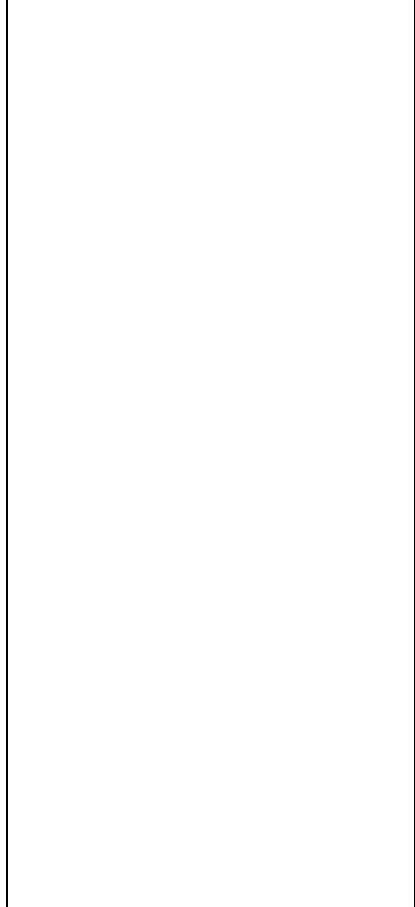
Ex. CC-78C (0887-06REVA-0114) at 6 (

[Redacted text block]

See also Ex. CC-79C (0888-06REVA 0722) at 2, 5–6 (annotated)

[Redacted text block]

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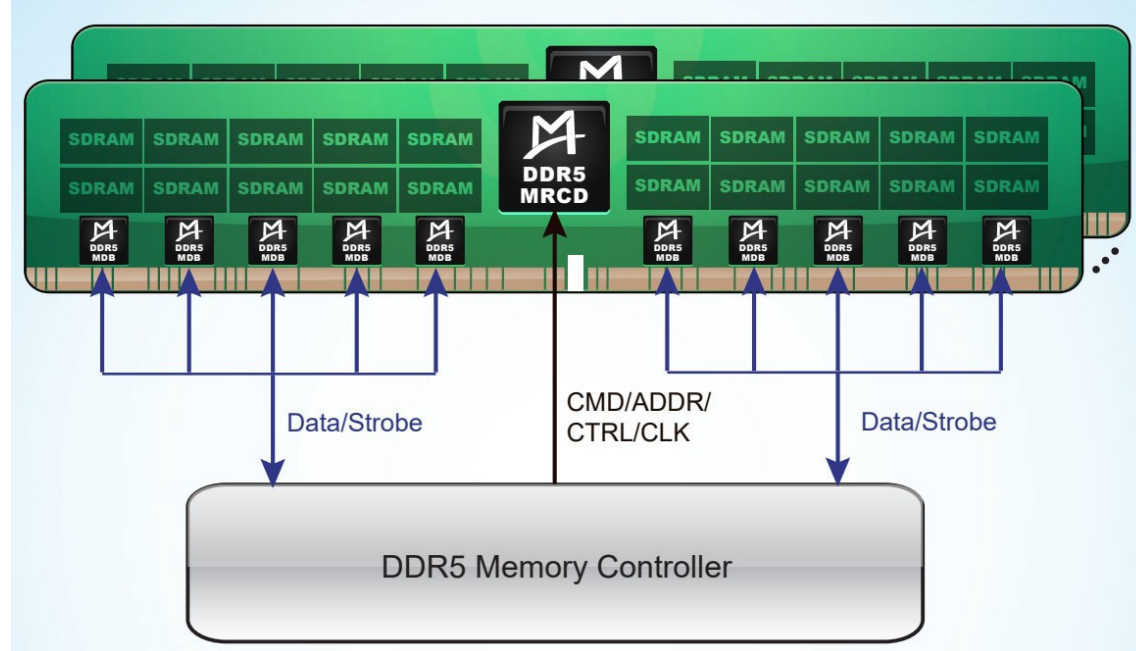


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Ex. CC-79C (0888-06REVA 0722) at 6 (

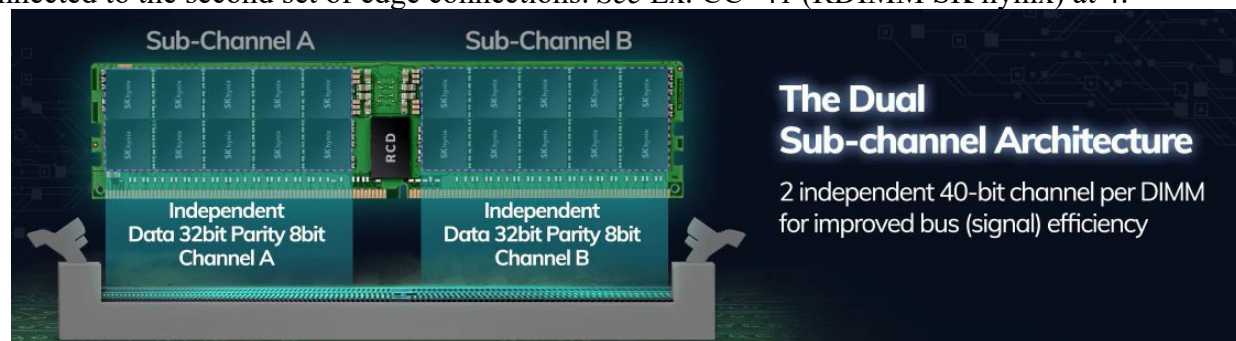
Furthermore, Montage DDR5 MRCD flyer depicts various data lines extending from the memory controller to the data buffers, which Montage DDR5 MDB flyer describes as facilitating transmission of data between the DRAMs and the memory controller. *See* Ex. CC-12 (M88MR5RCD01 Flyer English) at 2:

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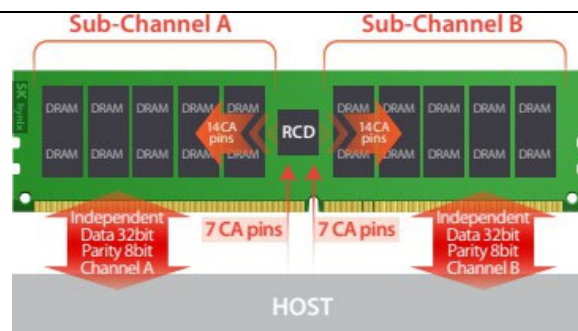
See also Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) (describing communication between DRAMs and memory controller via Montage DDR5 MDBs).

For another example, SK hynix advertises that SK hynix DDR5 RDIMMs receive 32 data bits and 8 parity bits for both Channel A and Channel B, which would be performed via a set of data lines connected to the second set of edge connections. See Ex. CC-41 (RDIMM SK hynix) at 4:



See also Ex. CC-40 (SK hynix A New Standard) at 5:

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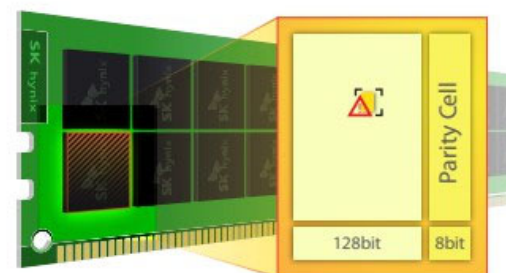


| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

| Enhanced Reliability

In-DRAM ECC on the DIMM offloads the system's error correction burden and considerably improves Reliability, Availability, and Serviceability (RAS).

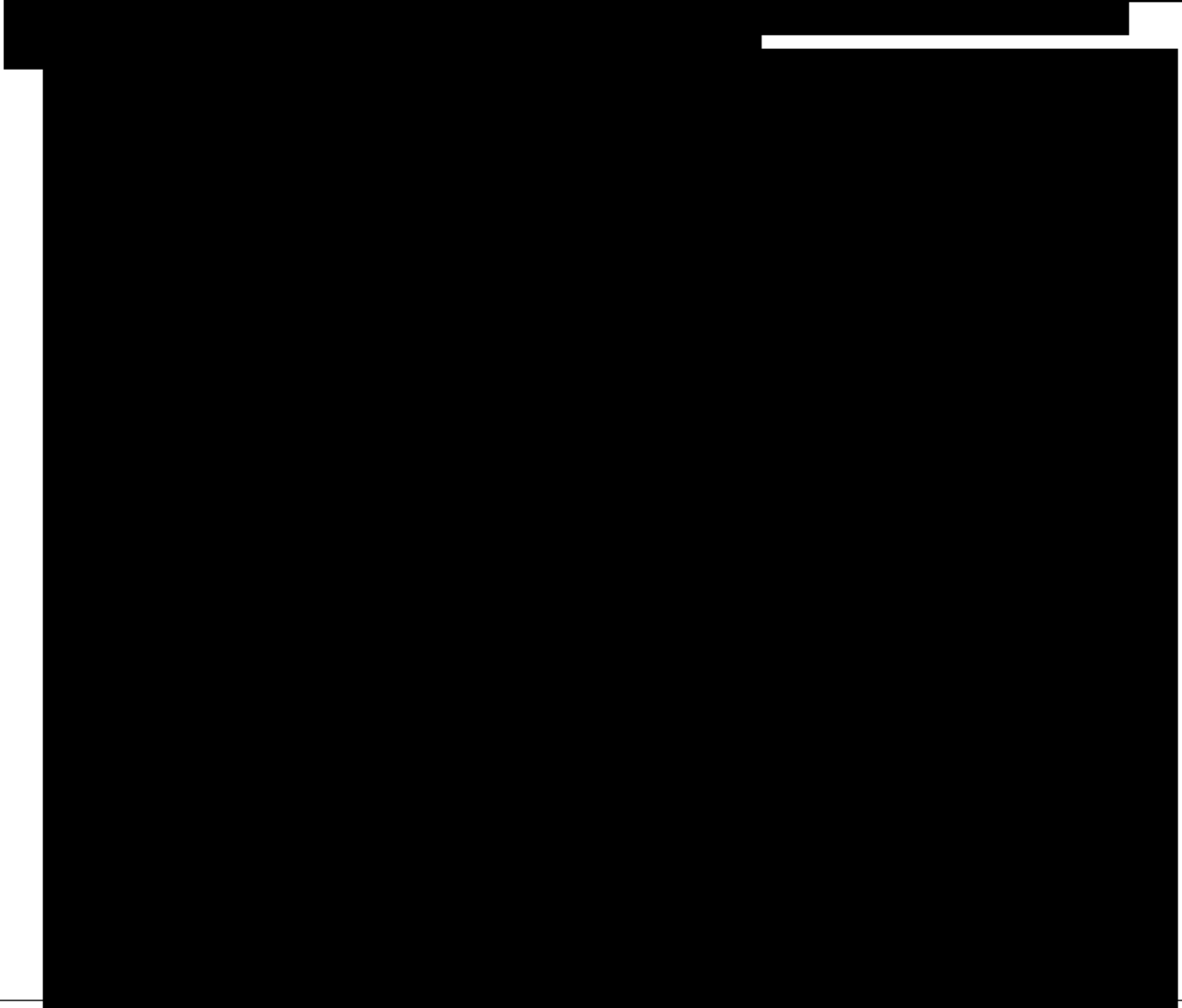


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[1.a.6] a set of address and control lines electrically connected to the third set of edge connections;

Each of the Practicing '366 Products comprises a printed circuit board (PCB) having a set of address and control lines electrically connected to the third set of edge connections (*e.g.*, connections with pin assignment CA[6:0]_A, CA[6:0]_B, CS[1:0]_A_n, CS[1:0]_B_n).

For example, as illustrated by Netlist's DDR5 VLP RDIMM Datasheet, [REDACTED]

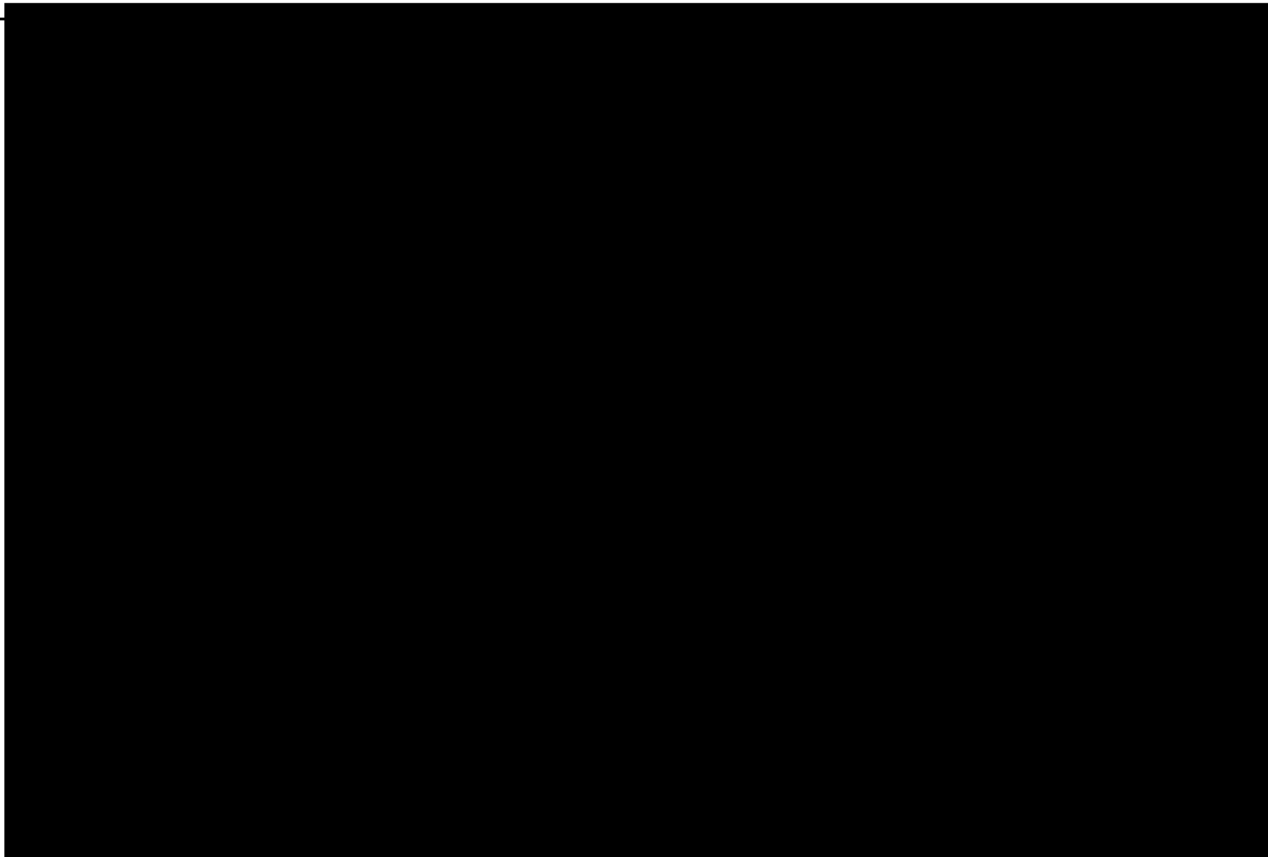


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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9 (annotated).

For another example.

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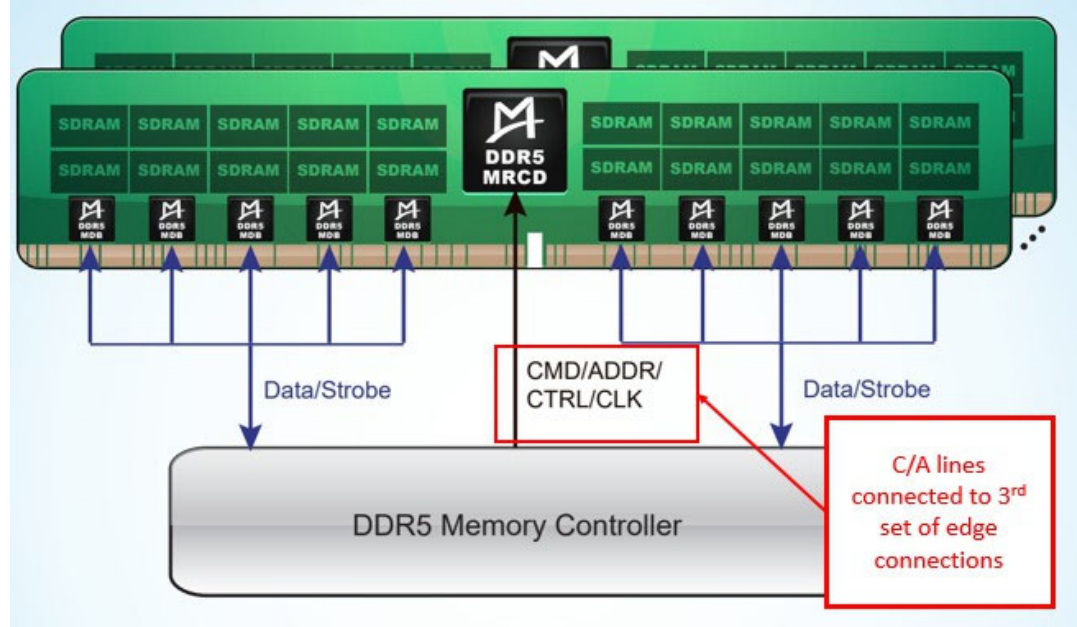


Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10 (annotated)

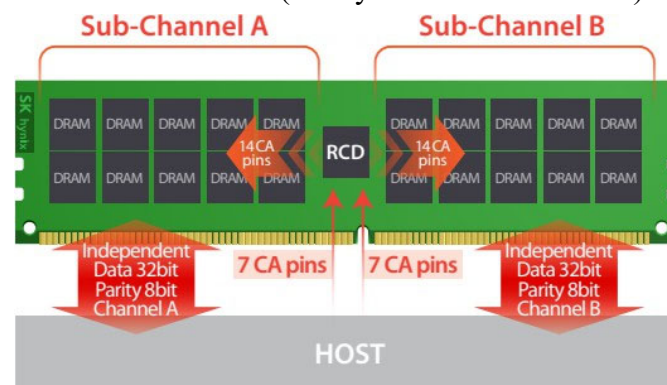


As another example, Montage DDR5 MRCD flyer describes and depicts address, command, clock and control signals being sent to the MRCD, which are then buffered and re-driven to the DRAMs in operation with the MDBs. *See* Ex. CC-12 (M88MR5RCD01 Flyer English) at 1–2 (annotated):

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As yet another example, SK hynix describes and depicts C/A lines from the edge connections to the RCD. *See* Ex. CC-40 (SK hynix A New Standard) at 5:

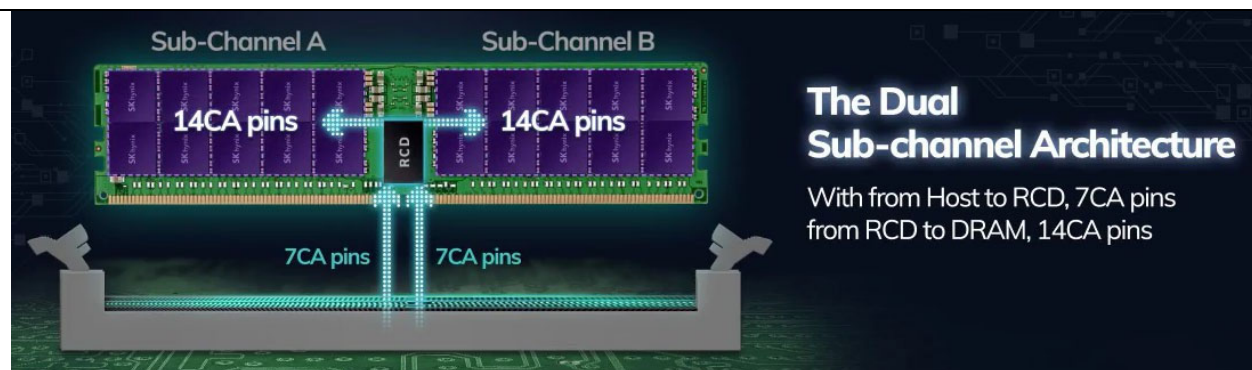


| New Sub-channel DRAM Architecture

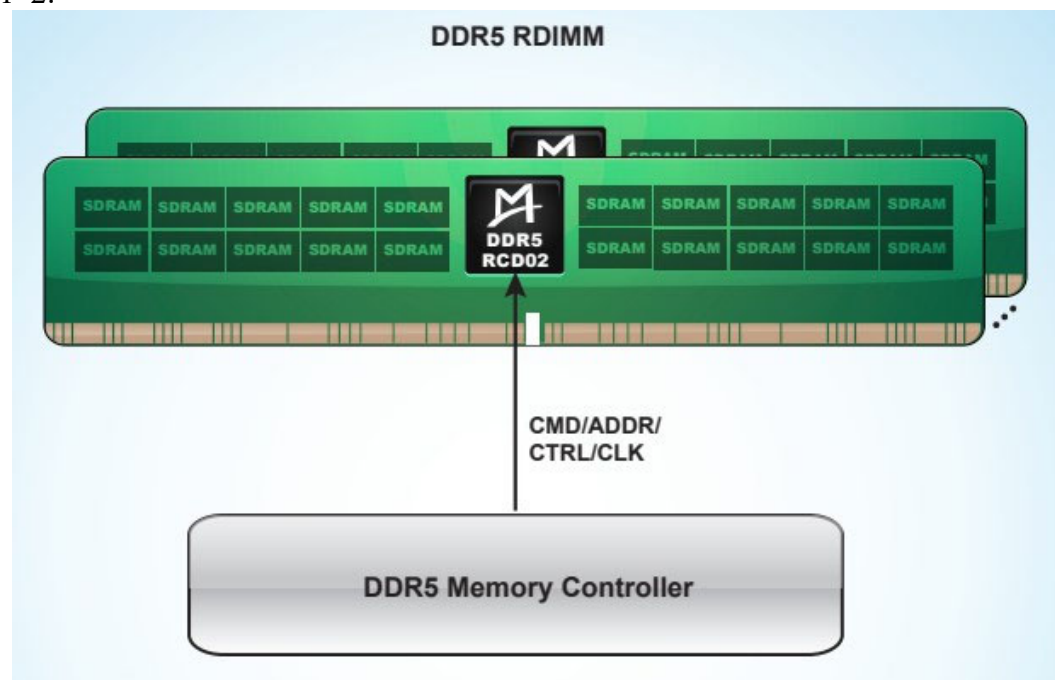
The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

See also Ex. CC- 41 (RDIMM SK hynix) at 4:

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Furthermore, Montage's M88DR5RCD02 RCD, included in both Netlist's DDR5 RDIMM and Overclock RDIMM, is described and shown as receiving CA pins. *See* Ex. CC-43 (M88DR5RCD02 Flyer) at 1–2:



These edge connections are configured to be electrically connected to the corresponding connectors of

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the control/address conduits in the memory slot.

See also Ex. CC-78C (0887-06REVA-0114) at 2–3 (annotated).

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See also Ex. CC-79C (0888-06REVA_0722) at 2–3 (annotated)

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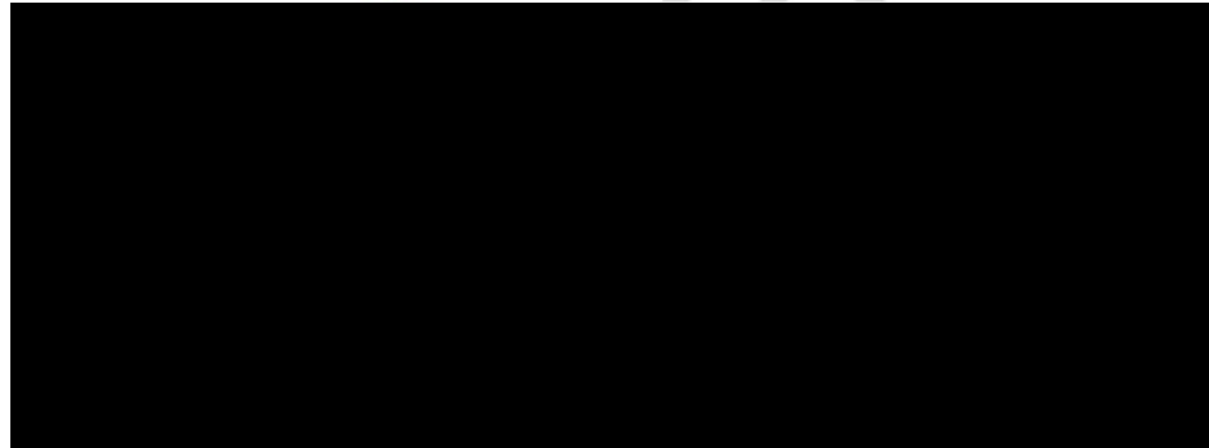
U.S. Patent No. 12,373,366 Domestic Industry Claim Chart

[1.b] a controller including a voltage monitor circuit and nonvolatile memory,

Each of the Practicing '366 Products includes a controller (such as the non-voltage-converter/regulator portions of the power management IC or PMIC) that includes a voltage monitor circuit and nonvolatile memory.

For example, Netlist's DDR5 Overclock RDIMM Datasheet [REDACTED]

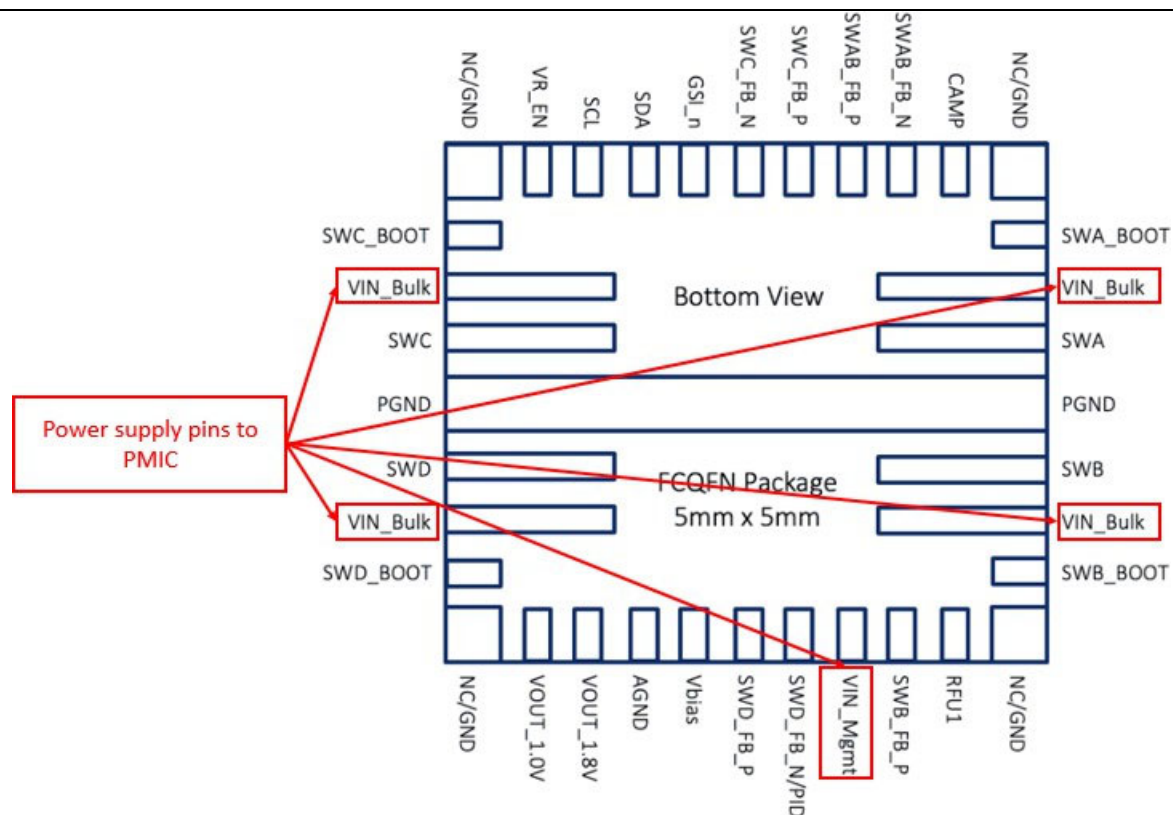
x. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 13:



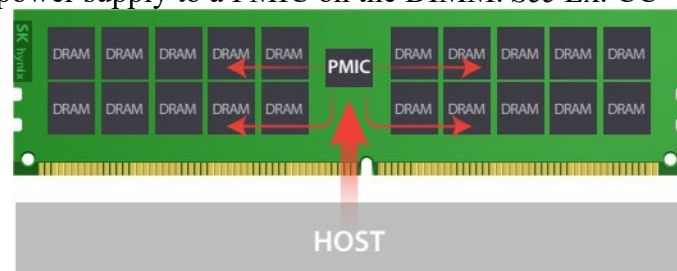
See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–6 [REDACTED]

As another example, Netlist's DDR5 MRDIMM includes Rambus PMIC 5020 with pins VIN_Bulk and VIN_MGMT the supply power to the PMIC. See Ex. CC-76 (Rambus DDR5 PMICs) at 2 (annotated):

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As yet another example, SK hynix illustrates and explains that power is supplied from an external power supply to a PMIC on the DIMM. See Ex. CC-40 (SK hynix A New Standard) at 5:



Stable Power Management

On-board power management IC (PMIC) eliminates the need for external voltage regulation and ensures more stable power provision to the DRAM.

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As yet another example, Netlist's DDR5 VLP RDIMM Datasheet describes

Ex. CC-38C (Netlist DDR5 VLP

Furthermore, illustrated above.

x. CC-38C (Netlist DDR5 VLP
RDIMM Datasheet Version 1.1) at 4; *see also* Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet

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Version 1.1) at 8.

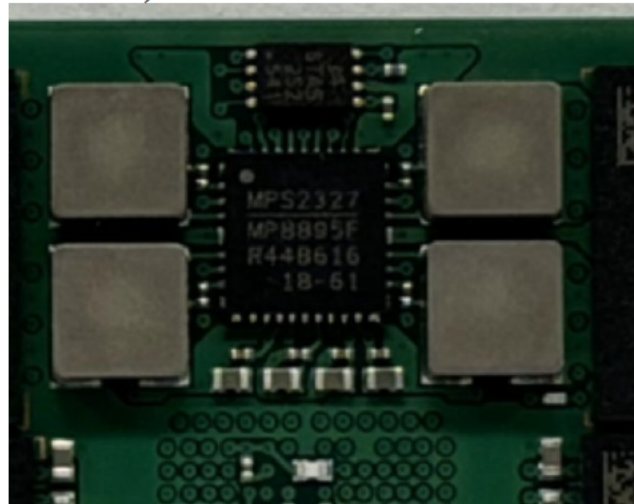
According to Netlist's web site, the "DDR5 RDIMM includes a Power Management Integrated Circuit (PMIC) to manage power consumption, voltage regulation, and other aspects of power delivery." Ex. CC-33 (Netlist DDR5 DIMMs) at 1.

Furthermore, Netlist

Ex. CC-53C (0887-99A-1M6060 Prelim 02-08-22) at 3

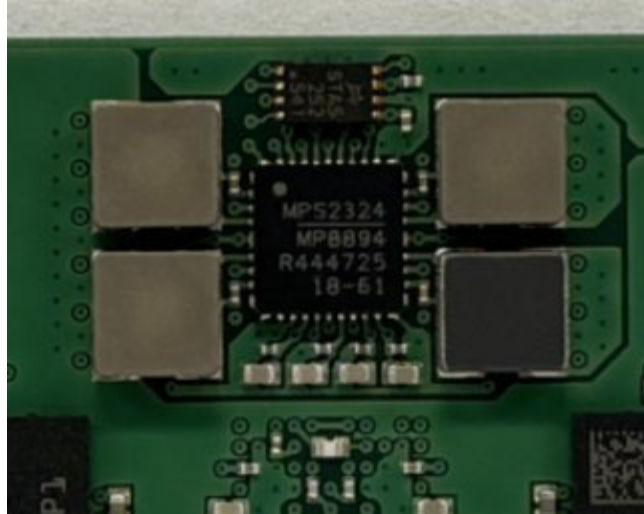
Ex. CC-54C (0888-97A-1A6B61-X00) at 3

Ex. CC-36 (Product Images) at 3 (depicting an MPS MP8895F PMIC on Netlist's DDR5 Overclock RDIMM); Ex. CC-36 (Product Images) at 10 (depicting an MPS MP8894 PMIC on Netlist's DDR5 RDIMM); Ex. CC-36 (Product Images) at 16 (depicting a Rambus P1947XXGA112 PMIC on Netlist's DDR5 MRDIMM).

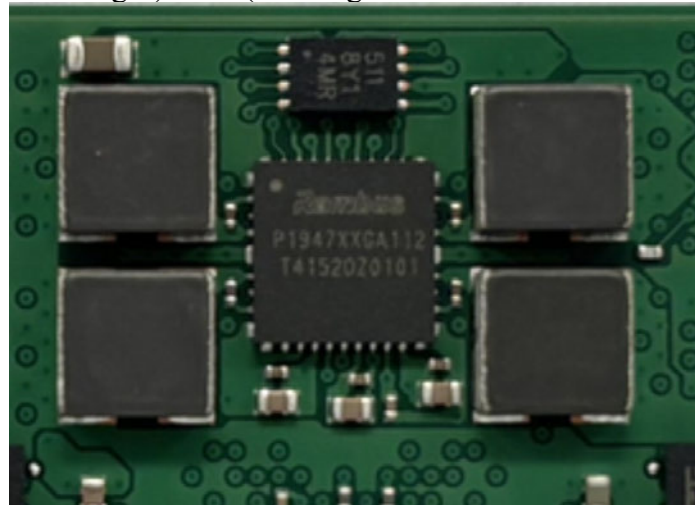


Ex. CC-36 (Product Images) at 3 (showing MP8895F on Netlist DDR5 Overclock RDIMM).

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Ex. CC-36 (Product Images) at 10 (showing MP8894 PMIC on Netlist DDR5 RDIMM).



Ex. CC-36 (Product Images) at 16 (showing P1947XXGA112 on Netlist DDR5 MRDIMM).

Each PMIC includes a voltage monitoring circuit. For example, each datasheet of the PMICs included in Netlist's RDIMMs details that the respective PMIC is specified to operate with certain operating voltages of VIN_BULK and VIN_MGMT. *See* Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1; Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1; Ex. CC-28 (MPQ8895F); Ex. CC-29

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(MPQ8894); Ex. CC-76 (Rambus DDR5 PMICs).

This supports that there is a voltage monitoring circuit so that the PMIC can react when the input voltage is outside the specified operating range.

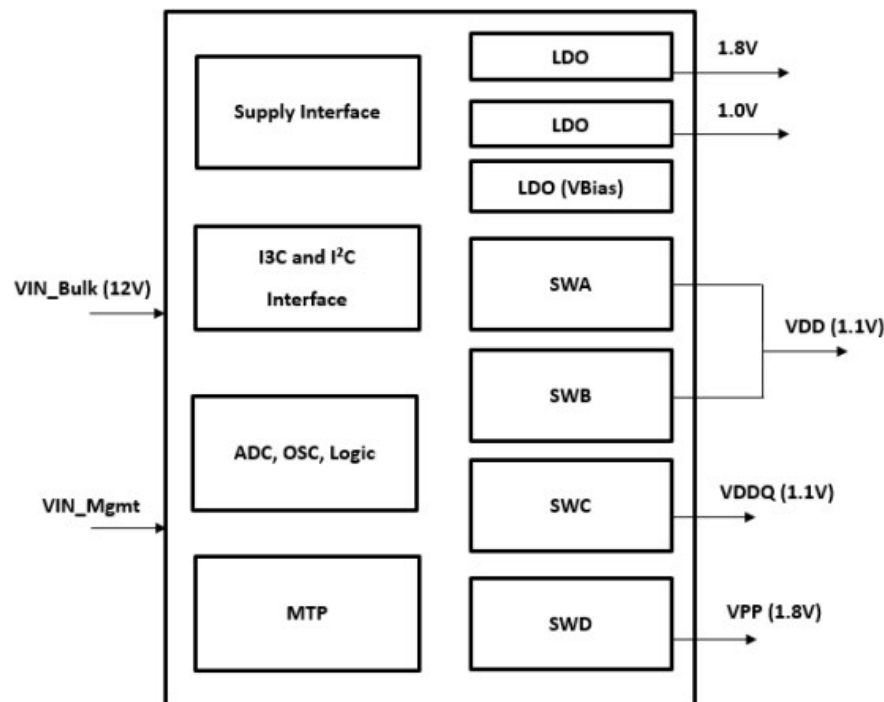
As another example, Renesas, supplying both P8900 and P8910, notes that each of P8900 and P8910 includes “VIN_Bulk input supply protection feature: Input over voltage.” *See* Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1; Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 (also including input undervoltage protection). This again supports the presence of an input voltage monitor circuit in the PMIC.

Furthermore, MPS, supplying both MP8895F and MP8894, notes that each of MP8895F and MP8894 includes Over-Current Protection (OCP), Under-Voltage Protection (UVP), Under-Voltage Lockout (UVLO), Thermal Shutdown, and Over-Voltage Protection (OVP). *See* Ex. CC-28 (MPQ8895F); Ex. CC-29 (MPQ8894).

Concerning Rambus P1947XXGA112 PMIC 5020, discovery will show that the Rambus PMIC 5020 includes the same features present in the examples above (*e.g.*, Over-Voltage Protection (OVP), etc.). *See* Ex. CC-77 (Rambus DDR5 Server PMICs Product Brief) at 1 (advertising support for “[a]dvanced protection and reliability features”).

The PMICs on Netlist’s DDR5 RDIMMs also each include non-volatile memory, such as in the form of MTP (multi-time programmable non-volatile memory). *E.g.*, Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1:

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Block Diagram

▪ **Multi Time Programmable Non-Volatile Memory**

See also Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 (showing and advertising the same non-volatile memory).

Ex. CC-29 (MPQ8894) at 2:

✓ Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory

See also Ex. CC-28 (MPQ8895F) at 2 (“Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory”).

Additionally, based on the above datasheets, each of the Practicing ’366 Products includes a respective PMIC that complies with corresponding JEDEC standards. See Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1 (“The P8900 is a power management IC (PMIC) designed for typical DDR5 RDIMM,

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DDR5 LRDIMM, and DDR5 NVDIMM applications.”); Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 (“The PMIC is fully compliant to the JEDEC DDR5 PMIC (PMIC5010; Low Current PMIC) specification.”); Ex. CC-29 (MPQ8894) at 1 (“JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev. 1.8 Compliant”); Ex. CC-28 (MPQ8895F) at 1 (“JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev 1.8 Compliant”); Ex. CC-76 (Rambus DDR5 PMICs) at 1 (“Meets or exceeds all JESD300-5 performance specifications”); Ex. CC-80 (Rambus Expands Chipset) at 3 (“The Rambus DDR5 server PMIC family includes products for the JEDEC extreme current (PMIC5020) . . . specifications.”).

According to the JEDEC Power Management IC specifications, the PMICs are each to include an input voltage monitoring circuit and nonvolatile memory. For example, the JESD301-1 standard for PMIC5000/PMIC5010 contains the following relevant description.

2.2 Common Features summary

- VIN_Bulk input supply protection feature: Input over voltage
- Multi Time Programmable Non-Volatile Memory

Ex. CC-85 (JESD301-1) at 2; Ex. CC-81 (JESD301-1A) at 2.

8.15 Input Over Voltage Protection

An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail.

Ex. CC-85 (JESD301-1) at 40; Ex. CC-81 (JESD301-1A) at 38.

See also Ex. CC-82 (JESD301-2) at 2, 55; Ex. CC-83 (JESD301-4A) at 2, 38; Ex. CC-84 (JESD301-6) at 2, 59.

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[1.b.1] the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines,

Each of the Practicing '366 Products includes a controller (PMIC) that is coupled to the PCB and to an input voltage supply line (*e.g.*, to the voltage supply line connected to VIN_BULK and/or to the voltage supply line connected to the VIN_MGMT) of the set of voltage supply lines (*e.g.*, VIN_BULK, VIN_MGMT and VSS).

For example, Netlist's DDR5 Overclock RDIMM Datasheet illustrates

[REDACTED]

[REDACTED]

See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10 ([REDACTED])

[REDACTED]:

[REDACTED]

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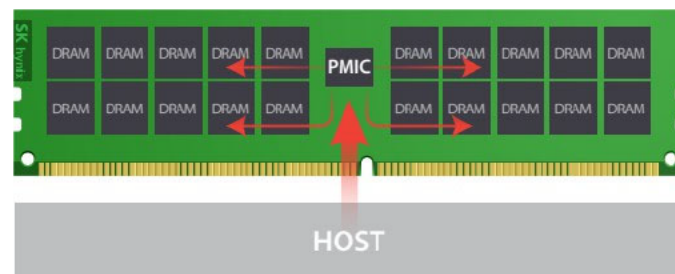
Additionally, [REDACTED]

See Ex. CC-78C (0887-06REVA-0114) at 4 [REDACTED]

Ex. CC-79C (0888-06REVA_0722)

at 4 [REDACTED]

As another example, SK hynix illustrates and explains that power is supplied from an external power supply to a PMIC on the DIMM, therefore the host provides an input voltage supply line, coupled to the PMIC. See Ex. CC-40 (SK hynix A New Standard) at 5:



| Stable Power Management

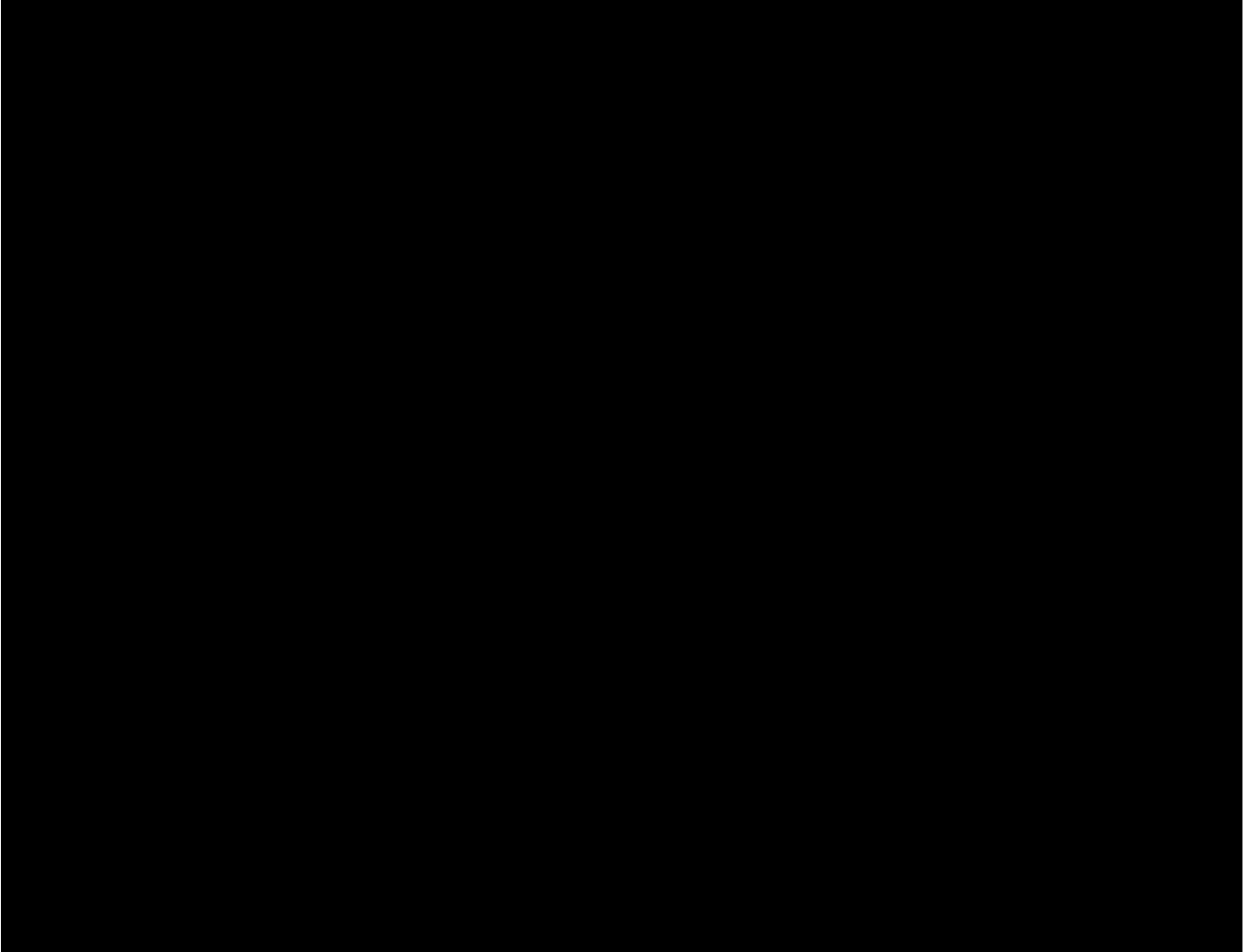
On-board power management IC (PMIC) eliminates the need for external voltage regulation and ensures more stable power provision to the DRAM.

As yet another example, Netlist's DDR5 VLP RDIMM Datasheet describes [REDACTED]

[REDACTED]. Ex. CC-38C (Netlist DDR5 VLP

RDIMM Datasheet Version 1.1) at 4 (annotated):

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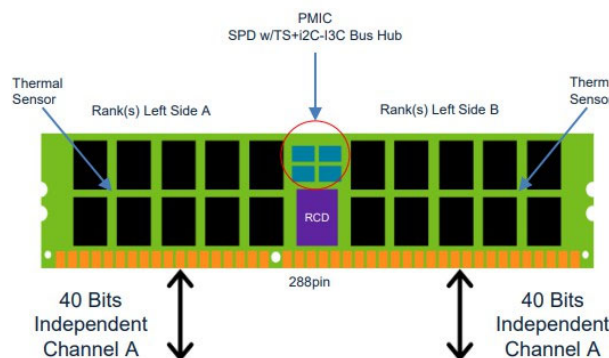
	 <i>See also</i> Ex. CC-38C (Netlist DDR5 VLP RDIMMs Datasheet) at 8 (annotated) ( :
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Furthermore, according to Netlist's web site, the "DDR5 RDIMM includes a Power Management Integrated Circuit (PMIC) to manage power consumption, voltage regulation, and other aspects of power delivery." Ex. CC-33 (Netlist DDR5 DIMMs) at 1. The PMIC on Netlist's DDR5 RDIMM is therefore coupled to the PCB.

See also Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1 (showing, as a change from DDR4 to DDR5, the PMIC is on the DIMM board):

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DIMM Changes from DDR4 to DDR5

Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data+ 8 ECC 2 channel per DIMM
DFE Support	DB Only	RCD, DB, and DRAM

Additionally, Rambus, a supplier of PMIC 5020 for Netlist's DDR5 MRDIMM, states that a change from DDR4 to DDR5 was "to move the power management ICs from the motherboard to the individual DDR5 memory modules," and that "[t]he DDR5 on-module PMIC receives a 12V input and generates the distinct voltage levels needed by the various components on the DDR5 module." *See* Ex. CC-76 (Rambus DDR5 PMICs) at 1.

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[1.b.2] the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;	As the evidence cited for elements [1.b] and [1.b.1] demonstrates, each of the Practicing '366 Products includes a PMIC having a voltage monitor that is configured to monitor an input voltage supplied via the input voltage supply line where the trigger occurs when the input voltage exceeds a first threshold voltage (<i>i.e.</i>, when there is an input over-voltage condition) because the voltage monitor circuit for each of the Practicing '366 Products provides input overvoltage lockout (OVLO) or overvoltage protection. <i>See</i>, <i>e.g.</i>, Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1 (showing the PMIC on Netlist's 2Rx8 DDR5 VLP RDIMM supports "VIN_Bulk input supply protection feature: Input over voltage"); Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 (showing the PMIC on Netlist's 1Rx8 DDR5 VLP RDIMM supports "VIN_Bulk input supply protection feature: Input overvoltage and input undervoltage"); Ex. CC-28 (MPQ8895F) (showing the PMIC on Netlist's DDR5 Overclock RDIMM supports "Over-Voltage Protection (OVP)"); Ex. CC-29 (MPQ8894) (showing the PMIC on Netlist's DDR5 RDIMM supports "Over-Voltage Protection (OVP)"); Ex. CC-77 (Rambus DDR5 Server PMICs Product Brief) at 1 (advertising support for "[a]dvanced protection and reliability features"). Additionally, the voltage monitor circuit is configured to: generate a trigger signal upon detecting a trigger condition; and transmit the trigger signal to at least one other portion of the controller. For example, as noted in connection with element 1b, each PMIC of the PMICs included on the Practicing '366 Products is compliant with JEDEC's PMIC standards. <i>See</i> Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1 ("The P8900 is a power management IC (PMIC) designed for typical DDR5 RDIMM, DDR5 LRDIMM, and DDR5 NVDIMM applications."); Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 ("The PMIC is fully compliant to the JEDEC DDR5 PMIC (PMIC5010; Low Current PMIC) specification."); Ex. CC-29 (MPQ8894) ("JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev. 1.8 Compliant"); Ex. CC-28 (MPQ8895F) ("JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev 1.8 Compliant"); Ex. CC-76 (Rambus DDR5 PMICs) at 1 ("Meets or exceeds all JESD300-5 performance specifications"); Ex. CC-80 (Rambus Expands Chipset) at 3s ("The Rambus DDR5 server PMIC family includes products for the JEDEC extreme current (PMIC5020) . . . specifications."). According to the JEDEC Power Management IC specifications, an over-voltage condition would trigger a VR disable command. <i>See</i> Ex. CC-85 (JESD301-1) at 36 and 37, Table 30.
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8.13 Function Interrupt - PWR_GOOD and GSI_n Output Signals

This section defines the output functionality of GSI_n pin and PWR_GOOD (Table 145, "Register 0x32" [5] = '0') pin.

When mask register bits are not set, the PMIC asserts its GSI_n output and PWR_GOOD output signals as shown in Table 30 when any event occurs. The table also highlights 11 events that causes PMIC to generate internally VR Disable command. For remaining events that does not trigger internal VR Disable command, the PMIC continues to operate as normal.

Ex. CC-85 (JESD301-1) at 36.

Table 30 — Events Interrupt Summary

Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	PWR_GOOD Output	GSI_n Output
VIN Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low
VIN Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low
VIN Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low

Ex. CC-85 (JESD301-1) at 37 (showing that input voltage monitoring is performed for both VIN_BULK and VIN_MGMT).

The VR Disable Command is generated in response to a trigger signal—such as an input over-voltage detection or alert signals output by the voltage monitor circuit (*e.g.*, as a result of a comparison by a comparator of the detected input voltage to a reference voltage corresponding to a threshold voltage set by an overvoltage threshold register).

Table 122 — Register 0x1B

R1B			
Bits	Attribute	Default	Description
7	RW	0	R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD VIN Bulk Input Over Voltage Threshold Setting For GSI_n Assertion 0 = 14.5 V 1 = 16.0 V

Ex. CC-85 (JESD301-1) at 136 (annotated).

The trigger signal generated by the voltage monitor circuit is configured to be transmitted by the voltage monitor circuit to other circuitry within the PMIC, *e.g.*, to logic or state machines to determine whether VR Disable Command should issue, to circuitry that is configured to execute a power down

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sequence in response to the generated VR Disable command, and/or to circuitry that would trigger writing to real time status registers (*see* Table 103, “Register 0x08”) and/or error log registers (*see* Table 100, “Register 0x05”).

The PMIC is configured to perform, in response to the trigger signal signifying detection of an input overvoltage condition, a write operation to write data into the nonvolatile memory, such as copying status register from volatile memory to MTP and/or updating the error log register in MTP non-volatile memory.

Regardless of the either mode of the operation, the PMIC, on its own, can generate VR Disable command at any time due to one or more events listed in Table 30 under column “Trigger VR Disable”. The PMIC executes power off sequence config0 (Table 174, “Register 0x58”) to power off sequence config3 (Table 177, “Register 0x5B,”) to preserve the appropriate voltage relationship as configured by the DIMM vendors. The PMIC may also reset its Vbias LDO regulator.

The Figure 16 below shows the timing relationship once the PMIC registers VR Disable command internally due to fault. The waveform shows each buck regulator output soft stop time and delay time once the soft stop time expires from each power off sequence config0 to power off sequence config3 registers. Note that if more than one regulators

Ex. CC-85 (JESD301-1) at 28 (annotated).

8.7 Power Down Output Regulators During Power On Sequence

During power on as described in Section 8.2 it is possible that PMIC can trigger VR Disable command on its own as described in Table 30, “Events Interrupt Summary” when one or more regulators are already turned on even while other remaining output regulators are not yet turned on because PMIC has not completed the power on sequence config registers. For these type of cases, the PMIC will not execute the remaining power on sequence config registers and will immediately jump to executing the power off sequence config0 to power off sequence config3 registers. The PMIC will update the status registers and error log registers appropriately as normal because it generated VR Disable command on its own. The PWR_GOOD output signal would remain low.

Ex. CC-85 (JESD301-1) at 29 (annotated).

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8.15 Input Over Voltage Protection

An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail.

There are two possibilities where PMIC recognizes the input over voltage event.

1. VIN_Mgmt input goes above the threshold set in register Table 122, "Register 0x1B" [5].
2. VIN_Bulk input goes above the threshold set in register Table 122, "Register 0x1B" [7].

When either one or both event occurs for a period longer than tInput_OV_GSI Assertion time then PMIC sets the register Table 103, "Register 0x08" [1:0] accordingly and drives GSI_n output signal as shown in Table 30 at the same time. Note that at this point, the PMIC does not assert PWR_GOOD output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing '1' to register Table 111, "Register 0x10" [1:0] appropriately or by writing '1' to global status clear register Table 115, "Register 0x14" [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 103, "Register 0x08" [1:0] will remain at '1'.

In programmable mode (i.e., Table 142, "Register 0x2F" [2] = '1'), if VIN_Bulk input supply over voltage condition persists greater than tInput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators and asserts PWR_GOOD signal. The PMIC keeps its VOUT_1.8V and

Ex. CC-85 (JESD301-1) at 40 (annotated).